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Questions

Q1.

Patients with cystic fibrosis may develop serious bacterial infections.

A patient with cystic fibrosis developed bacterial infections including *Mycobacterium*.

(a) (i) This patient was given a combination of antibiotics for several months.

Explain why a combination of antibiotics had to be given for several months.

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(ii) This patient did not respond to the combination of antibiotics, and later needed a lung transplant.

Suggest why this patient needed a lung transplant.

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(iii) Following the transplant, the patient was given immunosuppressive drugs.

Immunosuppressive drugs weaken the immune system. Some of these drugs work by preventing DNA synthesis in the patient.

As a result of the immunosuppressive drug treatment, the infection with *Mycobacterium* developed faster.

Explain why the infection with *Mycobacterium* developed faster when the patient was taking immunosuppressive drugs.

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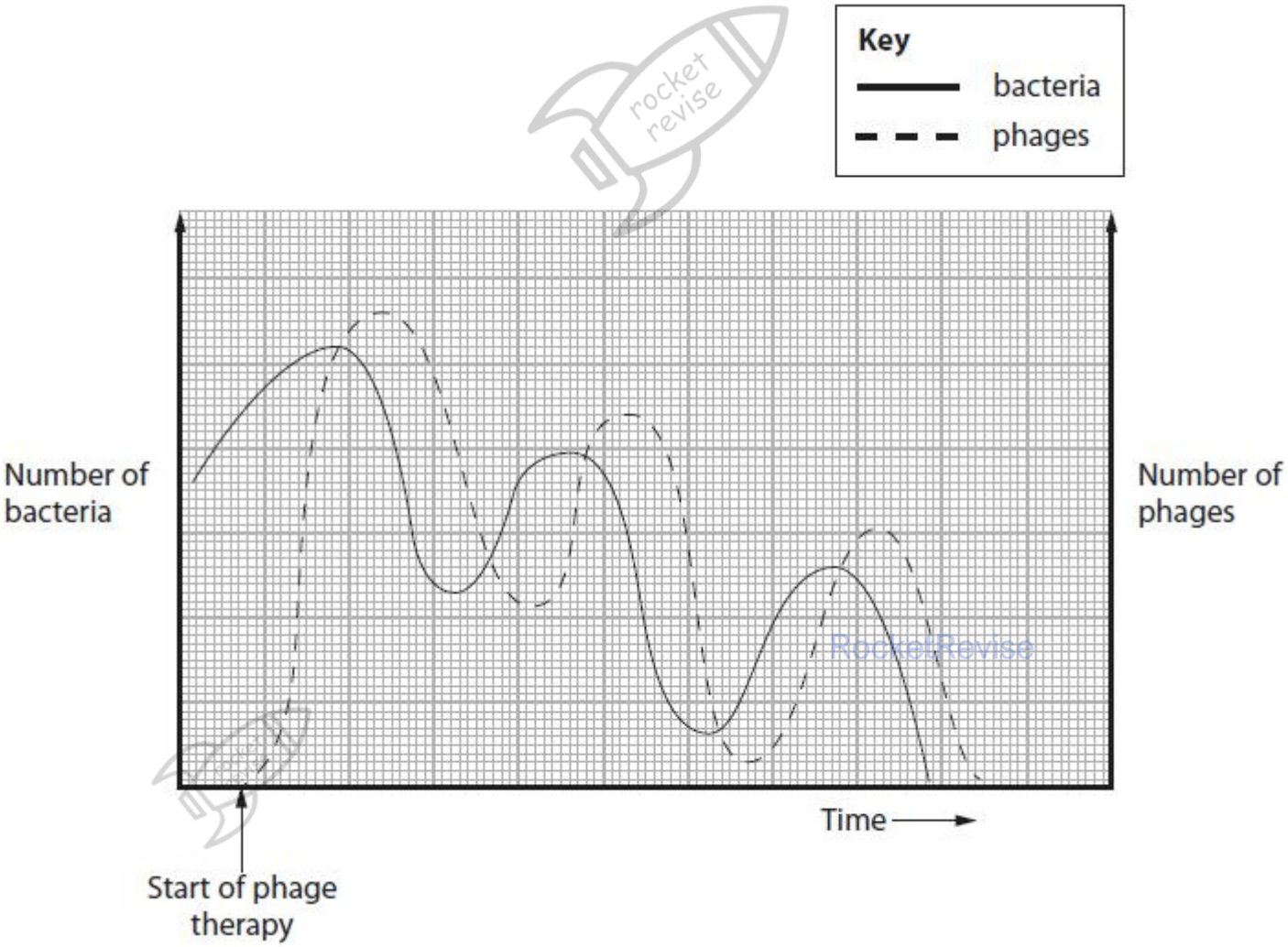
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*(b) Phage therapy was used to treat the *Mycobacterium* infection in this patient.

Phages are viruses that target bacterial cells.
The graph shows how the number of *Mycobacterium* and the number of phages changed following the start of phage therapy.



Explain the changes in the number of bacteria and phages following the start of the phage therapy.
Use the information in the graph to support your answer.

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(Total for question = 15 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

There are four types of immunity.

(a) Artificial immunity develops when a person is immunised by an injection.

The table gives some statements about artificial immunity.
For each statement, put **one** cross ☒ in the appropriate box, in each row, to show which statements are correct for the types of artificial immunity.

(3)

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Statement	Type of artificial immunity			
	both active and passive	active only	passive only	neither active nor passive
Antibodies are injected into the person	☒	☒	☒	☒
B cells differentiate into plasma cells	☒	☒	☒	☒
Memory cells are formed	☒	☒	☒	☒

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(b) Natural active immunity can develop when a person is infected with a virus.

(i) Describe the role of macrophages in the development of natural active immunity to a virus, following infection.

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(ii) Explain why both T helper cells and T killer cells are needed in the immune response to a virus.

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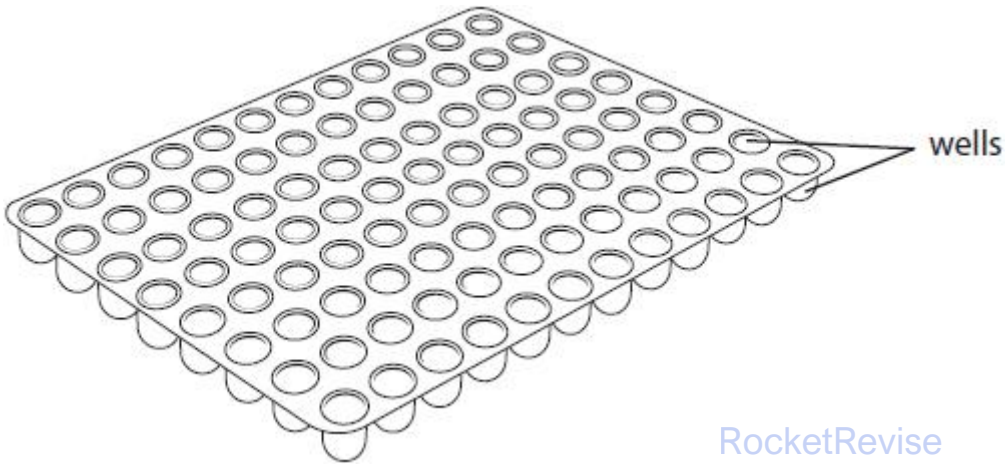
Q3.

Antimicrobial substances can be tested using a Minimum Inhibitory Concentration (MIC) assay.

This assay determines the lowest concentration of an antimicrobial substance that prevents visible growth of bacteria.

A microdilution plate is used in these assays. It is made of plastic and contains small wells that the antimicrobial substance and the bacteria can be added to.

The diagram shows a microdilution plate.



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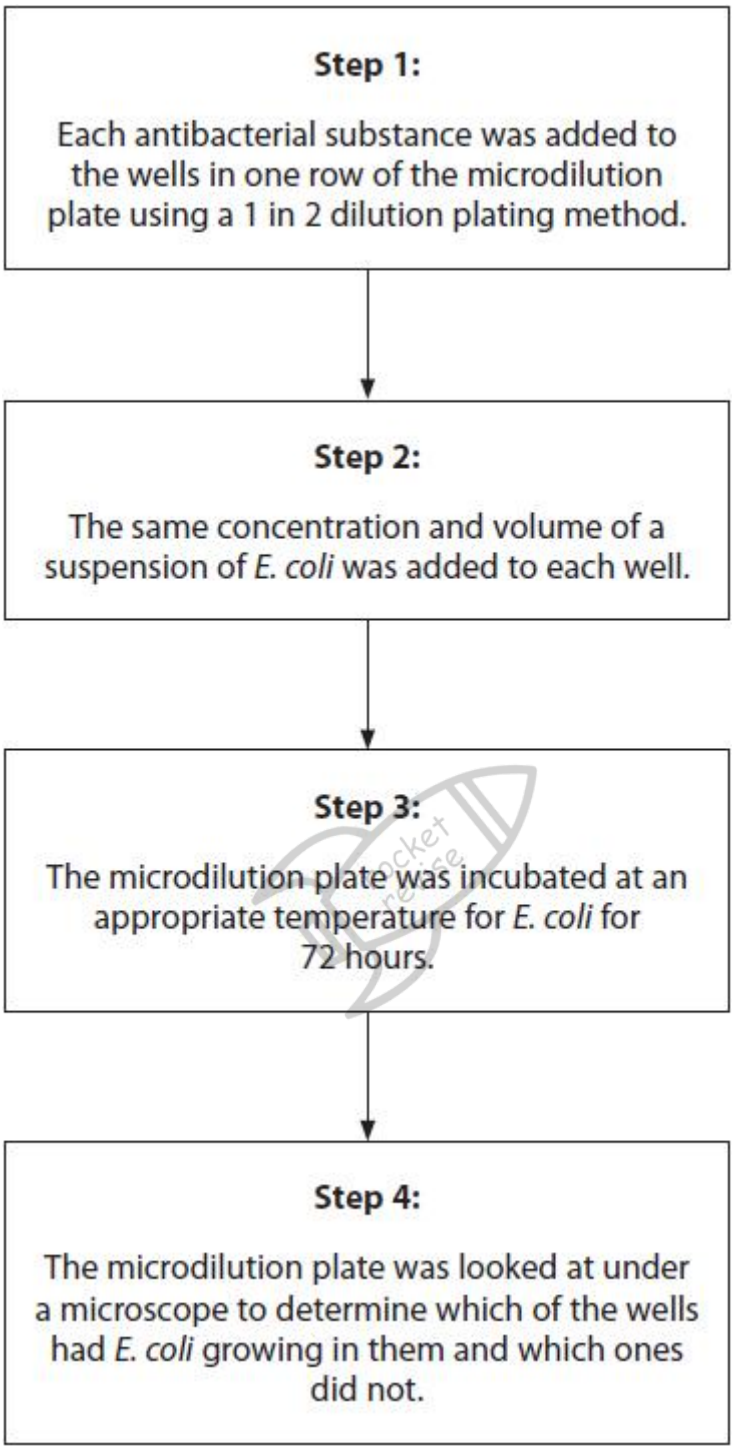
An investigation tested eight antimicrobial substances on one type of bacteria, *E. coli*.

Appropriate controls were included in this investigation.

The diagram shows the steps involved.

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(a) All steps in this investigation had to be carried out using aseptic technique.

(i) State the meaning of the term **aseptic technique**.

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(ii) Describe two aseptic techniques that could be used in this investigation.

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(iii) Explain why using aseptic technique in this investigation is important.

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(b) Explain why the microdilution plate had to be incubated at an appropriate temperature for *E. coli* for 72 hours.

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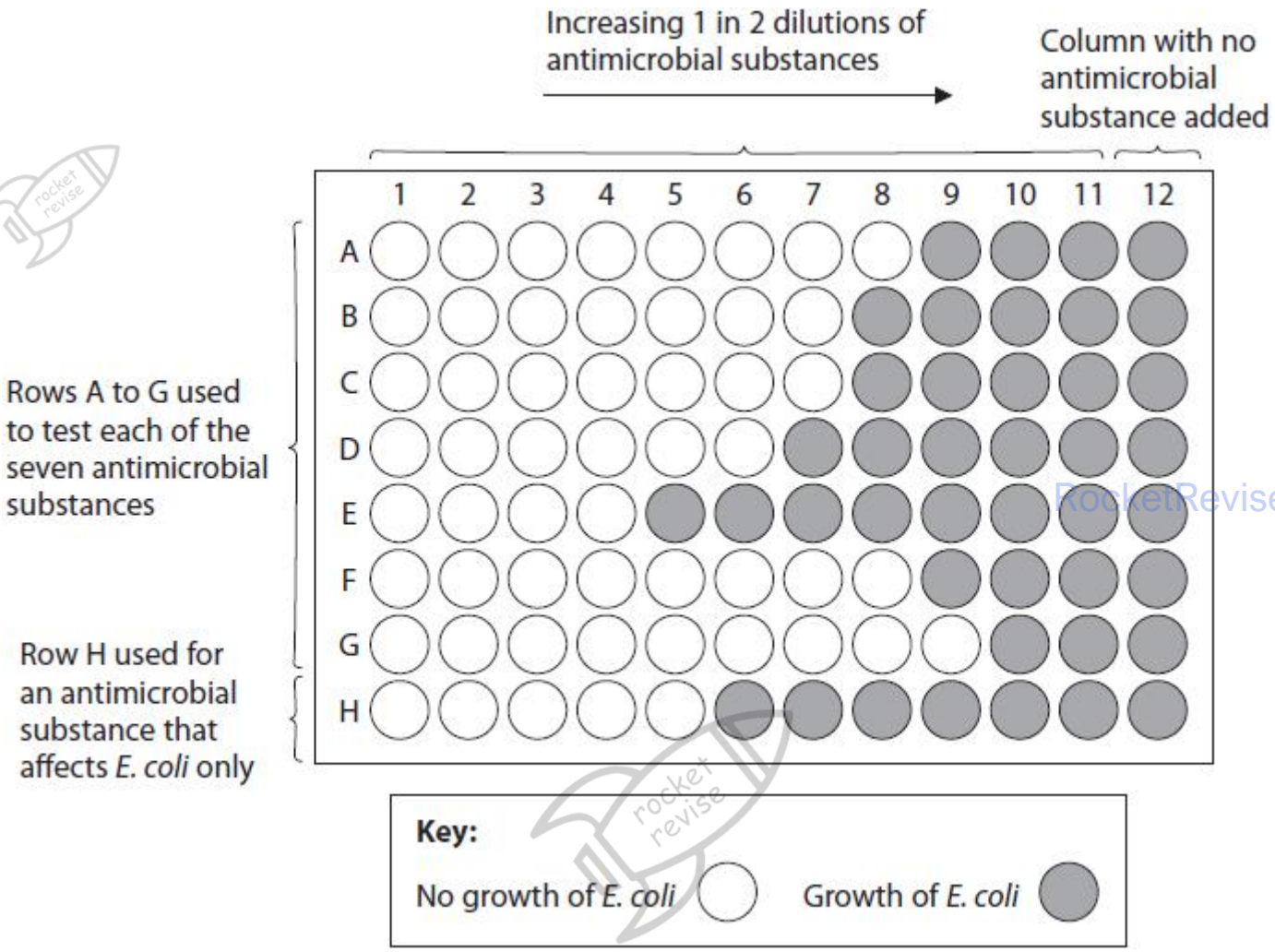
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(c) The diagram shows the results of the MIC assay from this investigation.

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(i) Explain why an antimicrobial substance that affects *E. coli* only was included in this assay (row **H**).

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(ii) Explain why there was one column that had no antimicrobial substance added to it (column 12).

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(iii) The MIC for the antimicrobial substance used in row **E** was in column 4.

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Describe how a 1 in 2 dilution plating method would have been carried out to achieve the dilution in this well.

(2)

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(iv) Calculate how many times more effective the antimicrobial substance used in row **G** is than the antimicrobial substance used in row **E**.

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(2)

Answer .....

(Total for question = 15 marks)

Q4.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student grew a bacterial culture in a liquid medium for 20 hours.

Every hour, a sample of the culture was removed.

This sample was added to some fresh medium and mixed thoroughly.

This mixture was spread onto agar in a Petri dish, so that the number of bacteria in the culture could be determined.

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(a) Which method is being used to determine the number of bacteria in this culture?

(1)

- ☐ **A** dilution plating
- ☐ **B** latency
- ☐ **C** mass method
- ☐ **D** optical method (turbidity)

(b) Explain how the Petri dish should have been sealed after the mixture was spread onto the agar.

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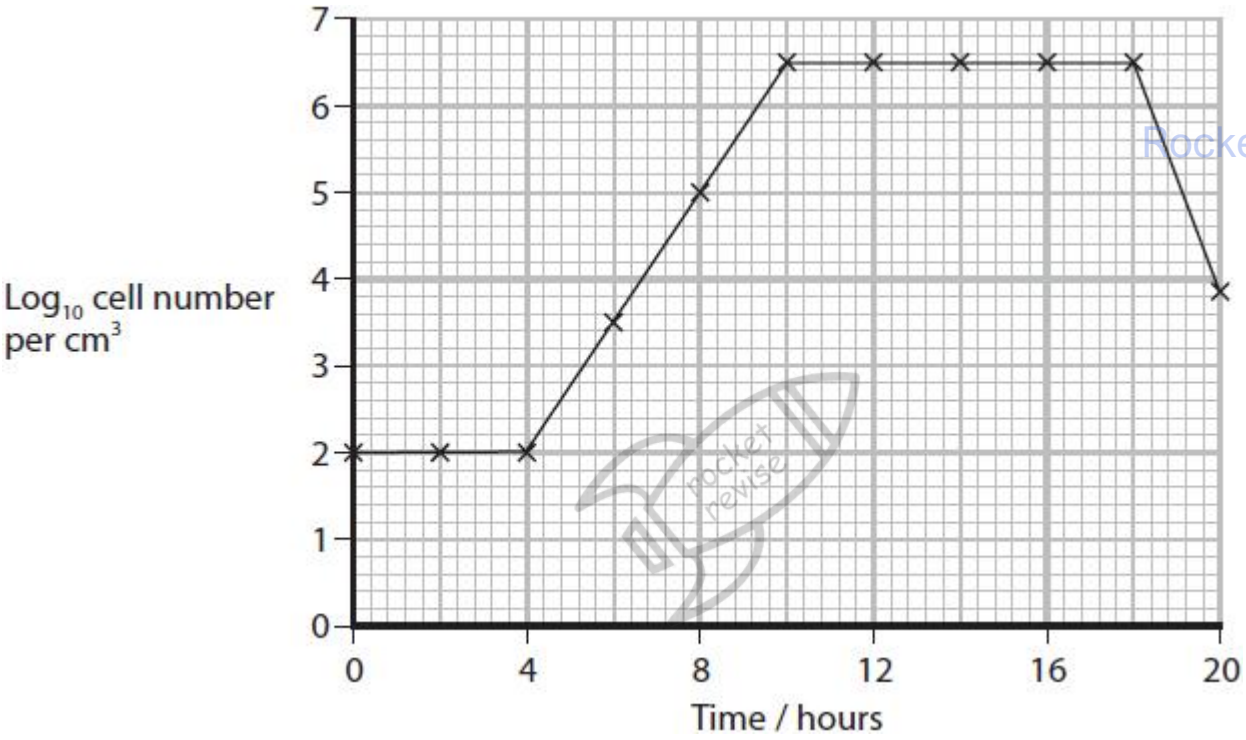
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(c) The graph shows the bacterial growth curve for this culture.



(i) Which shows the order of the phases of a bacterial growth curve?

(1)

- ☐ **A** exponential → lag → death → stationary
- ☐ **B** lag → stationary → death → exponential
- ☐ **C** lag → exponential → stationary → death
- ☐ **D** stationary → exponential → lag → death

(ii) The bacteria were not growing at their fastest rate in the conditions the student used.
How many of the following changes could result in a faster growth rate?

- decrease in pH
- decrease in temperature
- increase in pH
- increase in temperature

(1)

- ☐ **A** 1
- ☐ **B** 2
- ☐ **C** 3
- ☐ **D** 4

(iii) Calculate the growth rate constant (k) of these bacteria between 4 hours and 10 hours.

(2)

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Use the equation: $$k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$$

Give your answer to two decimal places.

Answer .....

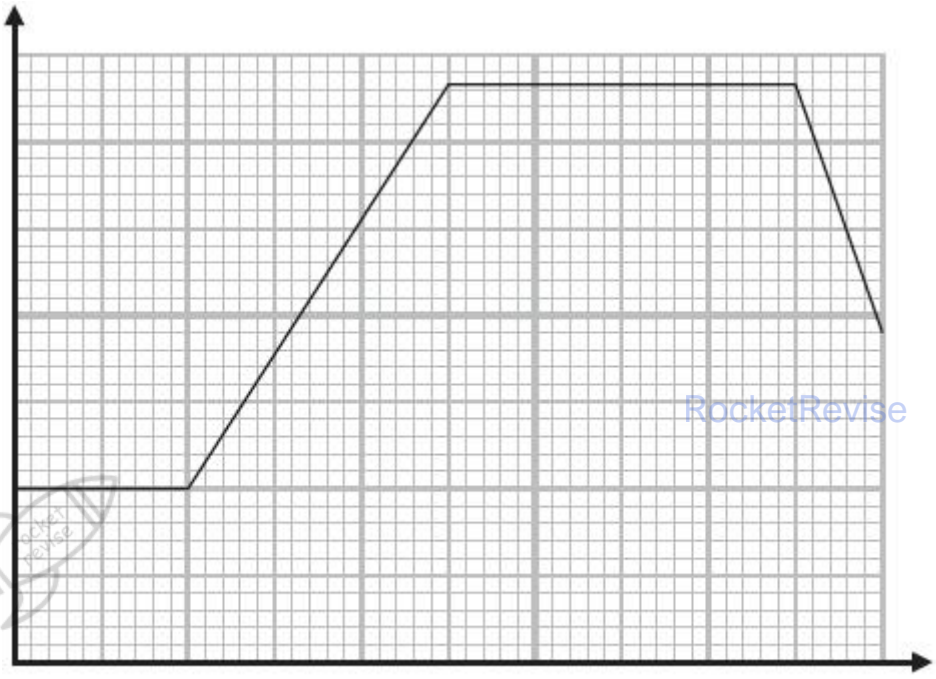
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(Total for question = 7 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graph shows a bacterial growth curve, produced by counting the number of bacterial colonies using the dilution plating technique.



(a) Which label should be used for the y-axis of this graph?

- ☒ **A** $\log_{10}$ number of living bacterial cells
- ☒ **B** $\log_{10}$ total number of bacterial cells

(1)

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☒ **C** number of living bacterial cells
☒ **D** total number of bacterial cells

(b) When culturing microorganisms, it is important that an aseptic technique is used.

(i) Explain the importance of using an aseptic technique.

(2)

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(ii) Explain **two** aseptic techniques that should be used in dilution plating.

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(Total for question = 5 marks)

Q6.

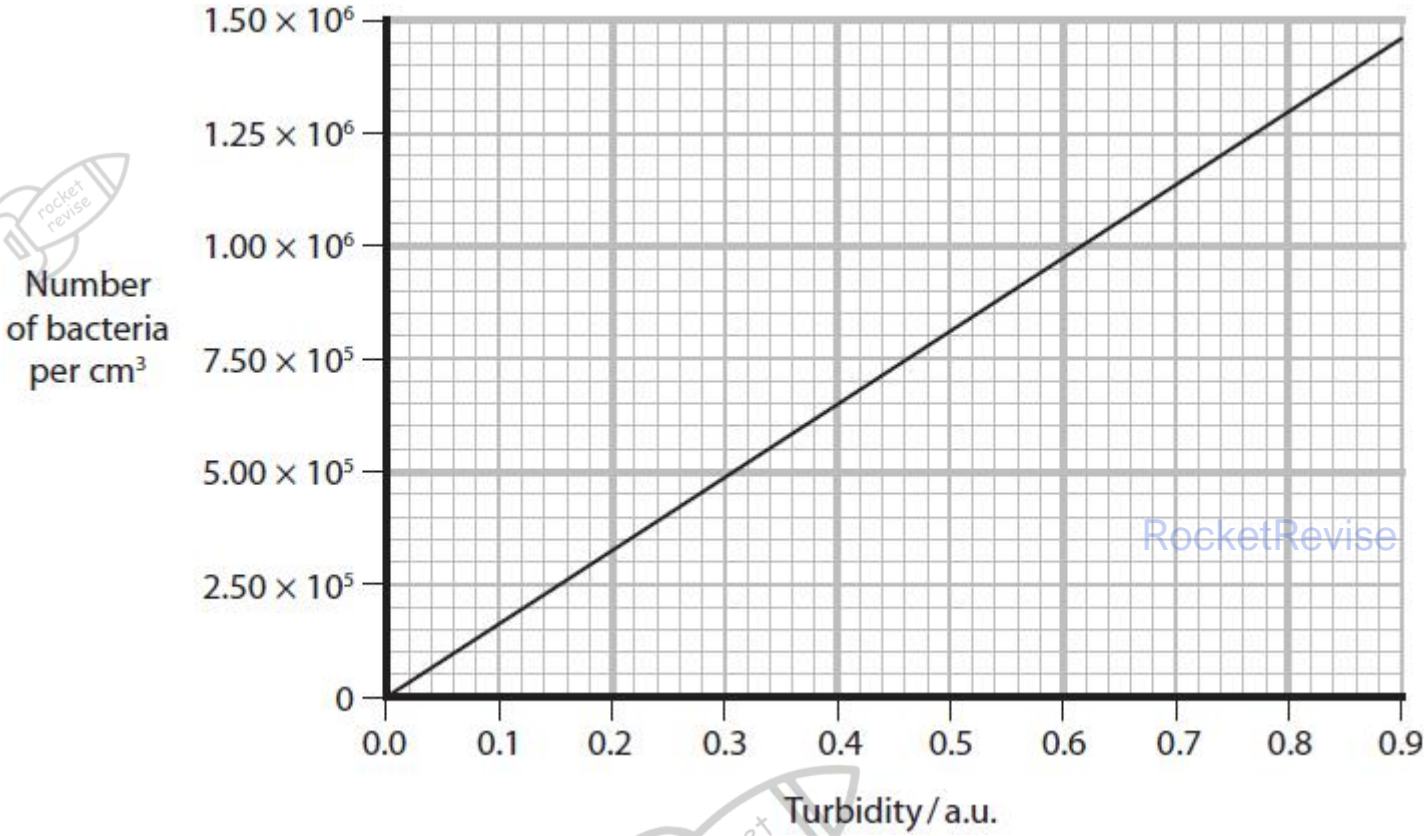
The growth of bacteria in a liquid culture was investigated over a 14 hour period, using an optical method (turbidity).

A calibration curve was used to convert the turbidity readings into the number of bacteria per cm^3 of liquid culture.

The graph shows the calibration curve used.

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The table shows the results for the first 10 hours of culture.

Time of culture / hours	Turbidity / a.u.	Increase in cell number in each two hour period
0	0.01	
2	0.01	0
4	0.04	5.0×10^4
6	0.56	
8	0.80	4.0×10^5
10	0.80	0

(a) Complete the table to show the increase in cell number between 4 and 6 hours of culture.

(1)

(b) Explain the changes during the first six hours of this culture.

(2)

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(c) The number of cells remained constant after 8 hours to the end of the 14 hour period of this culture.

Bacterial growth was also investigated using dilution plating and colony counts.

(i) Describe the method used in dilution plating.

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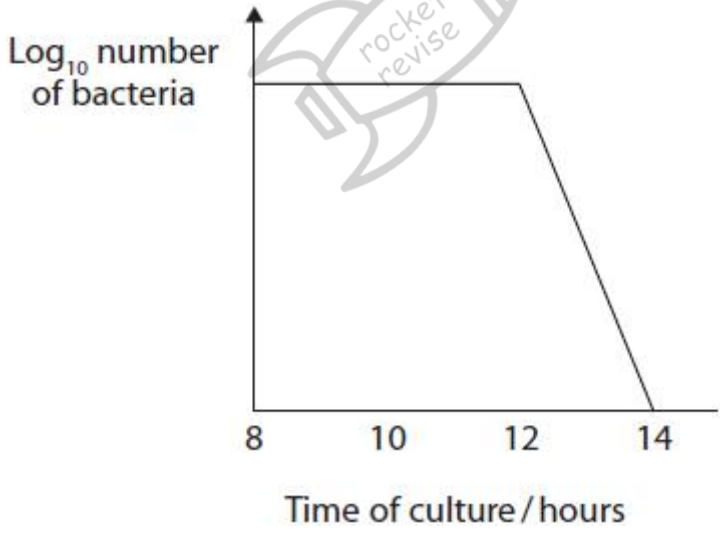
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(ii) The sketch shows the shape of the growth curve produced using dilution plating and colony counts between 8 and 14 hours of this culture.



There are differences in the numbers of bacteria between 8 and 14 hours measured using optical methods (turbidity) and measured using dilution plating and colony counts.

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Explain the differences obtained by these two methods.

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(Total for question = 8 marks)

Q7.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Bacteriostatic and bactericidal antibiotics are used to treat bacterial infections.

(a) Give the meaning of the term **bactericidal antibiotic**.

(1)

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(b) State why antibiotics affect bacteria and usually not the person taking them.

(1)

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(c) Codes of practice advise doctors and patients about the use of antibiotics.

Describe **two** pieces of advice given to patients.

(2)

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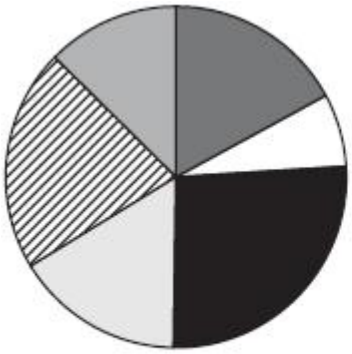
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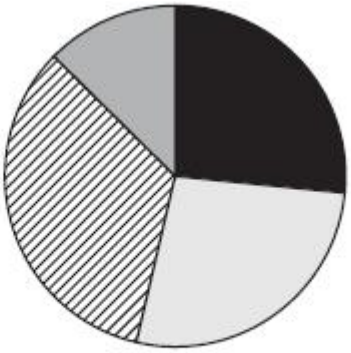
(d) The charts show the types of bacteria present in the throat of a patient, before and after treatment with an antibiotic.

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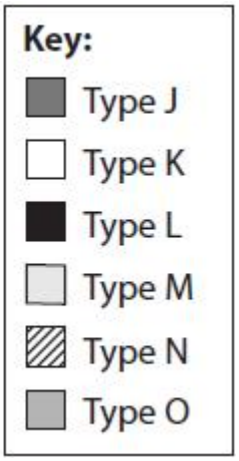
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Before treatment with antibiotic



After treatment with antibiotic



Describe the effects of this antibiotic on the bacteria present in the throat.

(3)

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(Total for question = 7 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Tetracycline is a bacteriostatic antibiotic and vancomycin is a bactericidal antibiotic.

(a) Compare and contrast bacteriostatic antibiotics with bactericidal antibiotics.

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(b) Antibiotics can be identified using Rf values obtained from chromatography.

The table shows the Rf values for tetracycline and vancomycin, obtained using two different

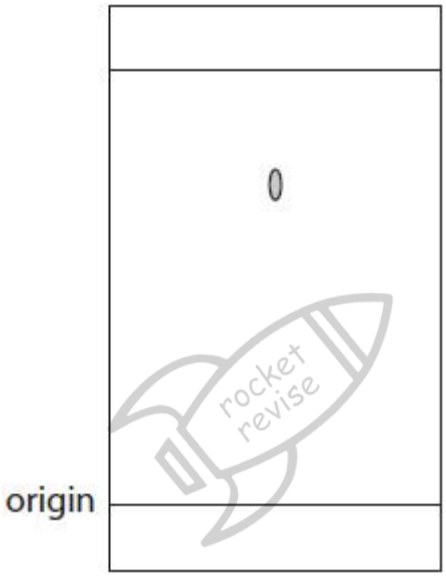
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chromatography methods.

Antibiotic	Rf value	
	Method 1	Method 2
Tetracycline	0.51	0.75
Vancomycin	0.09	0.29

The diagram shows a chromatogram for one of these antibiotics using one of the methods.



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(i) Describe how this chromatogram could be produced.

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(ii) Which identifies the antibiotic in the diagram and the method used?

(1)

- ☒ **A** tetracycline, using method 1
- ☒ **B** tetracycline, using method 2
- ☒ **C** vancomycin, using method 1
- ☒ **D** vancomycin, using method 2

(iii) Explain why the four values in the table are all different from each other.

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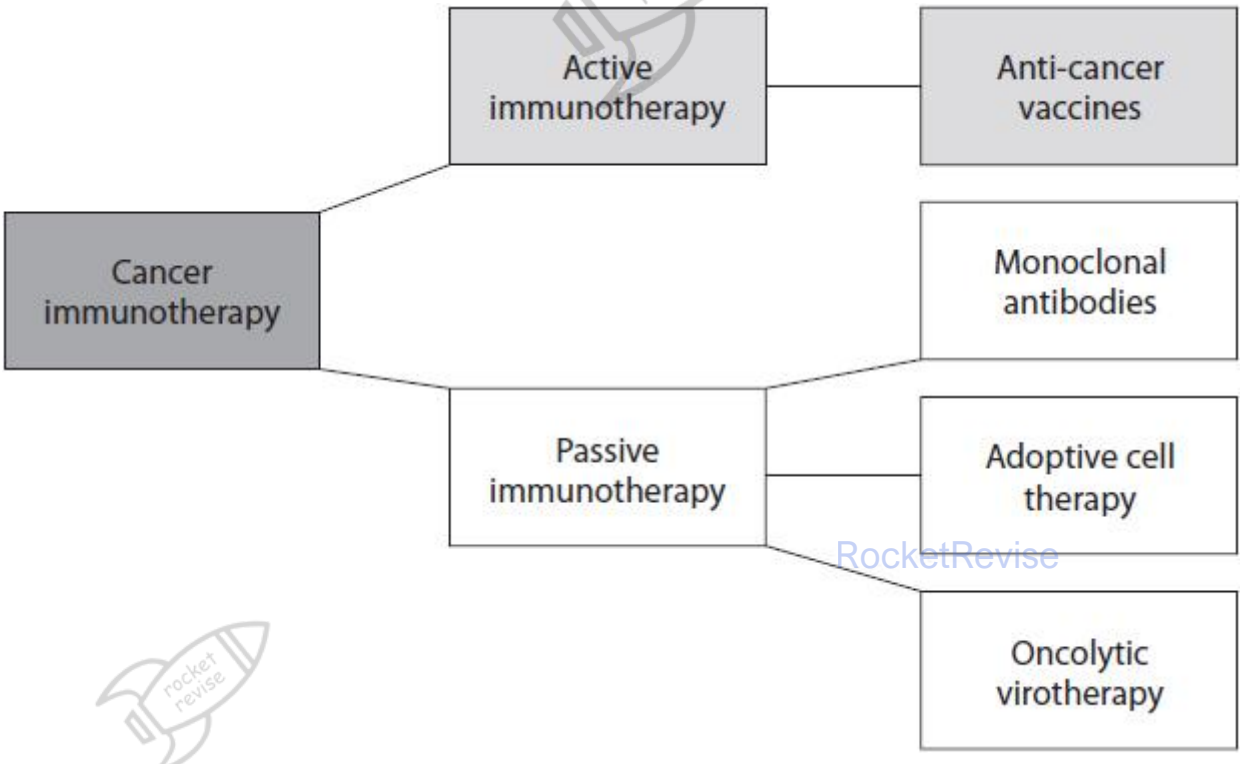
(Total for question = 8 marks)

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Q9.

Cancer immunotherapy stimulates and strengthens an individual's active and passive immune systems to destroy cancer cells.

The diagram shows how different types of cancer immunotherapy can be classified.



(a) Suggest what the difference is between active immunotherapy and passive immunotherapy.

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(b) Cancer cells result from mutations.

Some of these mutations can result in the formation of altered membrane proteins called neoantigens. The neoantigens are not present on the surface of normal cells.

(i) Explain how a mutation can result in the formation of a neoantigen.

(2)

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(ii) Some active anti–cancer vaccines can contain the mRNA that codes for the neoantigens.
Explain how vaccines containing the mRNA coding for neoantigens stimulate the immune response.

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(c) Monoclonal antibodies are made in a laboratory. Large numbers of identical antibodies specific to one type of antigen can be made.

Explain how these antibodies help the immune system to destroy cancer cells.

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(d) In adoptive cell therapy, T cells are taken from the patient and large numbers of genetically identical cells are made in the laboratory.

These cells are then injected back into the patient.
Explain why it is necessary to produce large numbers of genetically identical cells to make this treatment effective.

(2)

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(e) In oncolytic virotherapy, viruses that destroy cancer cells are injected into the patient.

Not all types of virus are suitable for this therapy.
Suggest **two** features of a virus that make it suitable for use in this type of therapy.

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(Total for question = 14 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a young giant panda feeding on bamboo.

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(Source: © Cheryl Schneider/Alamy Stock Photo)

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Giant pandas have evolved the ability to eat bamboo.

Genes that code for enzymes that digest the cellulose in the bamboo are not present in the genome of the giant panda.

The giant pandas depend on microorganisms in their intestines to produce enzymes that can digest the cellulose.

(a) Which type of glycosidic bonds are broken in the digestion of cellulose?

(1)

- ☒ **A** ɑ 1-4
- ☒ **B** ɑ 1-6
- ☒ **C** β 1-4
- ☒ **D** β 1-6

(b) The microorganisms in the intestines of giant pandas include both bacteria and fungi.

Fungi are eukaryotic organisms.
Put one cross [x] in the appropriate box, in each row, to show if genes are found in each of the structures in these microorganisms.

(3)

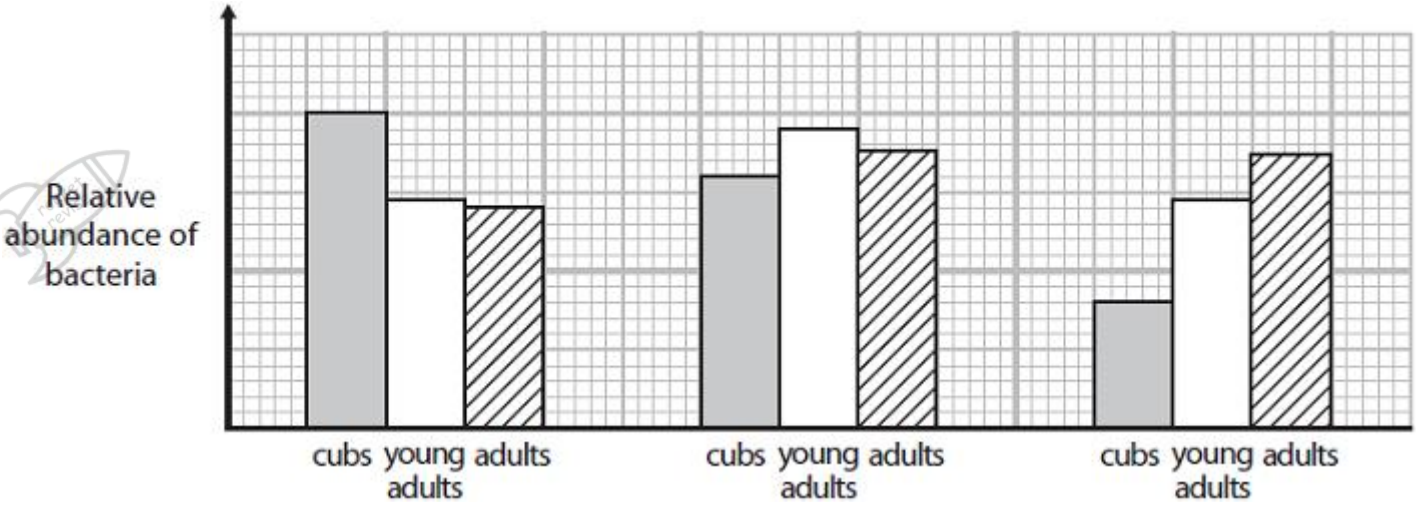
Structure	Microorganism			
	Bacteria only	Fungi only	Both bacteria and fungi	Neither bacteria nor fungi
mitochondria	☒	☒	☒	☒
nuclei	☒	☒	☒	☒
70S (small) ribosomes	☒	☒	☒	☒

(c) In one study, the succession of bacteria in the intestines of developing giant pandas was determined.

Three stages of development were studied: cubs, young adults and adults.
The graph shows the relative abundance of four types of bacteria in the three developmental stages of the giant panda.

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(i) Which sequence shows the succession of bacteria during the development of a giant panda?

(1)

- ☒ **A** Clostridium → Streptococcus → Escherichia and Shigella
- ☒ **B** Escherichia and Shigella → Clostridium → Streptococcus
- ☒ **C** Escherichia and Shigella → Streptococcus → Clostridium
- ☒ **D** Streptococcus → Escherichia and Shigella → Clostridium

(ii) Describe one method that could be used in this study to determine the relative abundance of Streptococcus in a giant panda.

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(Total for question = 9 marks)

Q11.

Alzheimer's disease is a cause of dementia in older people.

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The neurones in the brain are damaged and eventually destroyed.

Abnormal plaques are seen in the brains of people with Alzheimer's disease. These plaques are made of a protein called ß-amyloid.

Drug ATD is a new drug used to treat people with Alzheimer's disease.

This drug is an antibody that is specific for ß-amyloid.

(a) Drug ATD was recently approved for the treatment of Alzheimer's disease.

Describe the information that had to be collected by scientists before this drug could be approved.

(3)

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(b) (i) Explain the phrase: 'an antibody that is specific for ß-amyloid'.

(2)

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(ii) Drug ATD reduces the build-up of ß-amyloid in the brains of people with Alzheimer's disease.

Suggest how drug ATD could result in the reduction of the build-up of ß-amyloid.

(3)

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(c) Drug ATD was derived from a B memory cell taken from an elderly person who showed no signs of Alzheimer's disease.

(i) Suggest why this person did not have Alzheimer's disease.

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(ii) Explain why this drug could be produced only from B memory cells taken from a person who did not have Alzheimer's disease.

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(Total for question = 12 marks)

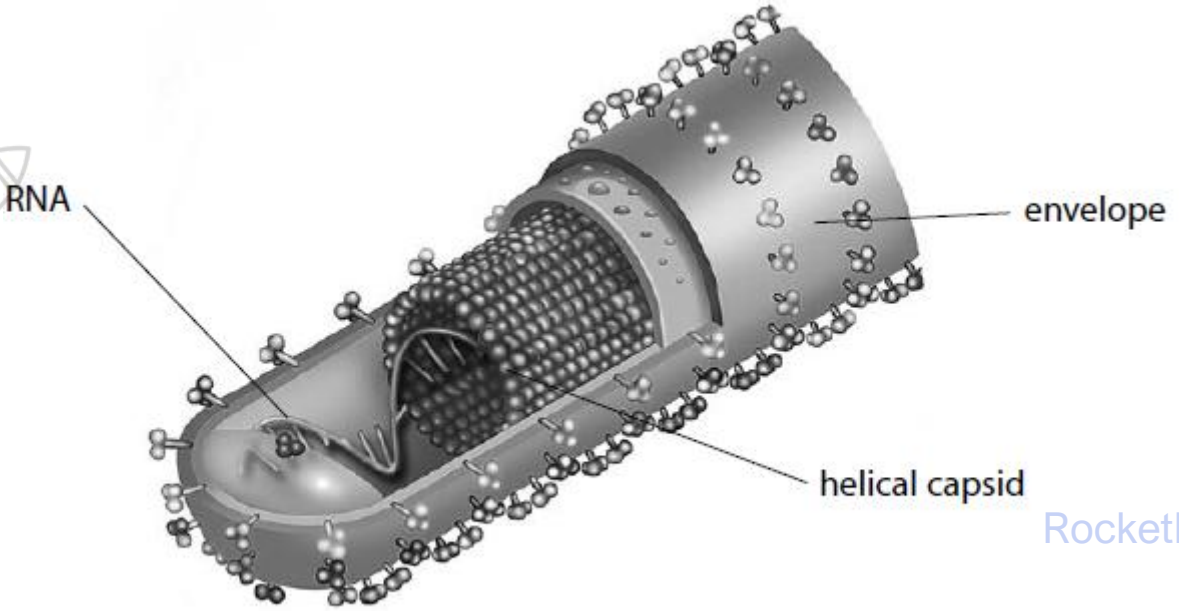
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Q12.

(a) The diagram shows the structure of an Ebola virus.

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© Science History Images / Alamy Stock Photo

(i) Which of the following viruses contain the same type of nucleic acid as the Ebola virus?

(1)

- ☒ **A** human immunodeficiency virus (HIV) only
- ☒ **B** human immunodeficiency virus (HIV) and tobacco mosaic virus (TMV)
- ☒ **C** lambda phage (λ phage) only
- ☒ **D** lambda phage (λ phage) and tobacco mosaic virus (TMV)

(ii) Which of the following viruses have a helical capsid?

(1)

- ☒ **A** human immunodeficiency virus (HIV) and lambda phage (λ phage)
- ☒ **B** human immunodeficiency virus (HIV) only
- ☒ **C** lambda phage (λ phage) and tobacco mosaic virus (TMV)
- ☒ **D** tobacco mosaic virus (TMV) only

(iii) What type of molecule makes up the capsid of a virus?

(1)

- ☒ **A** carbohydrate
- ☒ **B** lipid
- ☒ **C** nucleic acid
- ☒ **D** protein

(iv) The volume of an Ebola virus is approximately 7.76×10^4 nm^3.
Tobacco mosaic virus (TMV) is approximately 300 nm long and 80 nm in diameter.
Calculate how many times larger Ebola virus is than TMV.
Assume that TMV is a cylinder in shape.
The volume of a cylinder is calculated using the formula

$$V = \pi r^2 l$$

(2)

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Answer .....

(b) Human immunodeficiency virus (HIV) contains two enzymes that are **not** found in most other types of virus.

(i) Name these two enzymes.

(1)

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(ii) Explain why HIV contains these two enzymes.

(3)

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(Total for question = 9 marks)

Q13.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Changes in temperature affect the rate of enzyme activity and the survival of living organisms.

(a) Which describes an enzyme in a living organism?

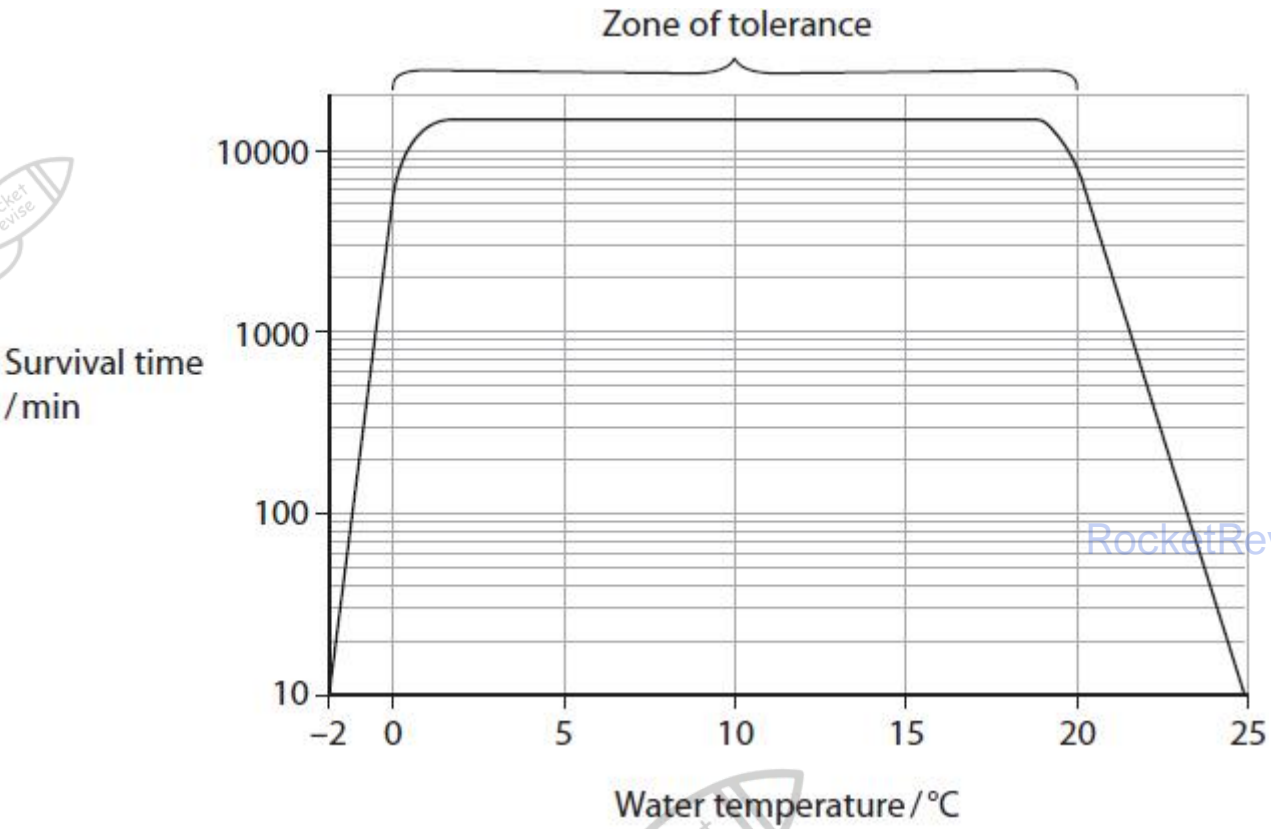
(1)

- ☒ **A** inorganic substance that decreases activation energy
- ☒ **B** inorganic substance that increases activation energy
- ☒ **C** organic substance that decreases activation energy
- ☒ **D** organic substance that increases activation energy

(b) The graph shows the survival time of one species of fish in water at different temperatures.

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(i) Name the type of scale that has been used on the y axis.

(1)

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(ii) The zone of tolerance is the range of temperatures that a fish could survive at for longer than 5000 minutes.

Which **two** biotic factors could cause the fish to die in this range of temperatures?

(1)

- ☒ **A** level of salt in the water and wave action
- ☒ **B** pathogens and wave action
- ☒ **C** predators and pathogens
- ☒ **D** predators and level of salt in the water

(iii) The body temperature of a fish changes with the temperature of its surroundings.

Explain why the survival time of fish is reduced at temperatures lower and higher than those in the zone of tolerance.

(3)

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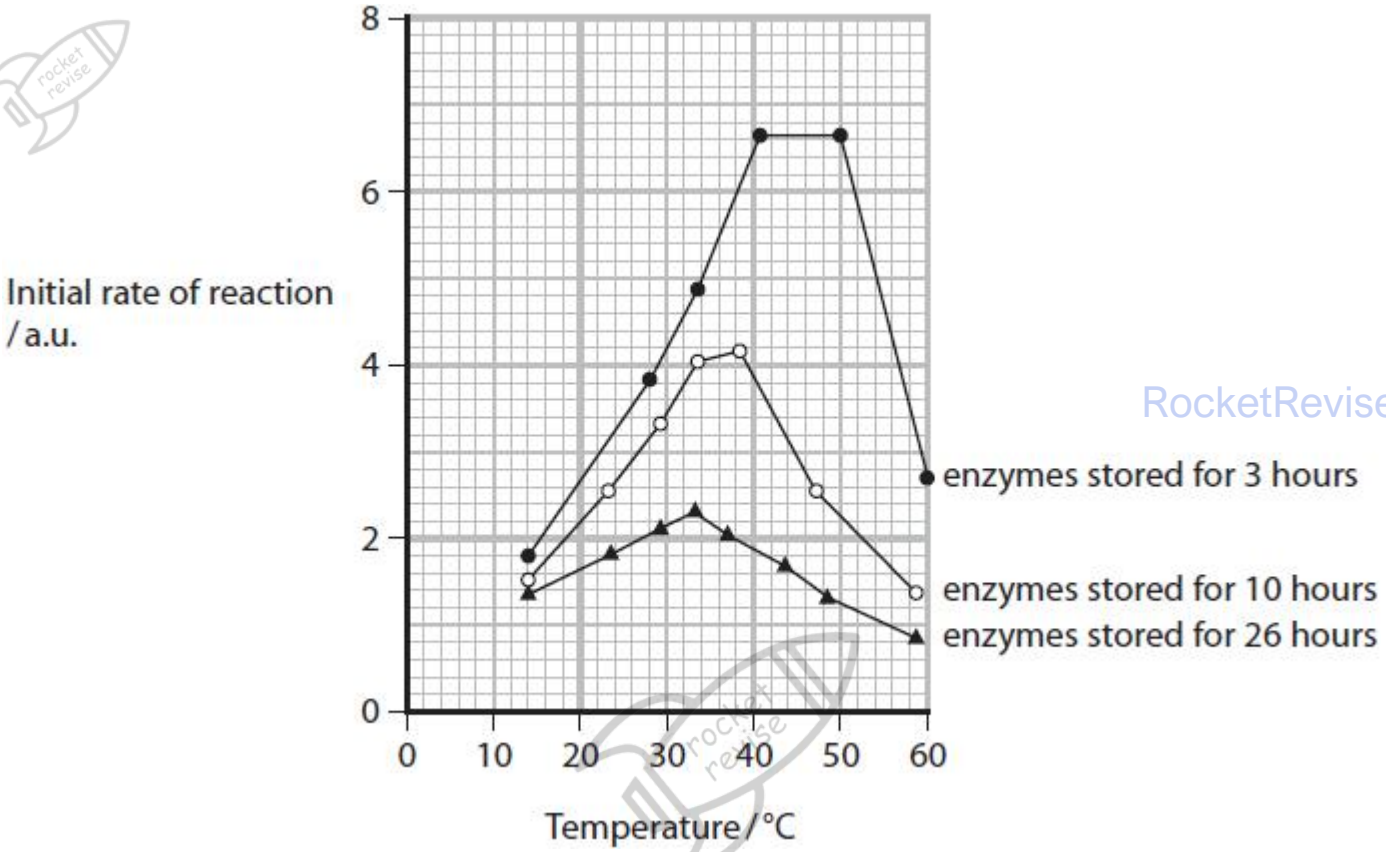
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(c) Enzymes were stored for different lengths of time.

The effect on the initial rate of reaction was investigated.
The graph shows the results of this investigation.



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(i) Calculate the Q_{10} for this enzyme stored for 3 hours.
Use the formula

$$Q_{10} = \frac{R_{(t+10)}}{R_t}$$

where R_t is the initial rate of reaction at 20 °C.

(3)

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Answer .....

(ii) Explain why the optimum temperature for this enzyme is affected by storage time.

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(Total for question = 12 marks)

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Q14.

Forensic entomology is a method for estimating the time of death of a mammal.

(a) Seventy-two hours or more after death, forensic entomology is the most accurate method for estimating the time of death.

Forensic entomology can also provide information about the place of death and indicate if the body has been moved.

(i) Explain why forensic entomology is the most accurate method for estimating the time of death, if this is greater than 72 hours.

(2)

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(ii) Explain how forensic entomology can indicate if a body has been moved from the place of death.

(2)

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(b) A study looked at the succession of insects associated with the decomposition of a dead mammal in the Andean Coffee region.

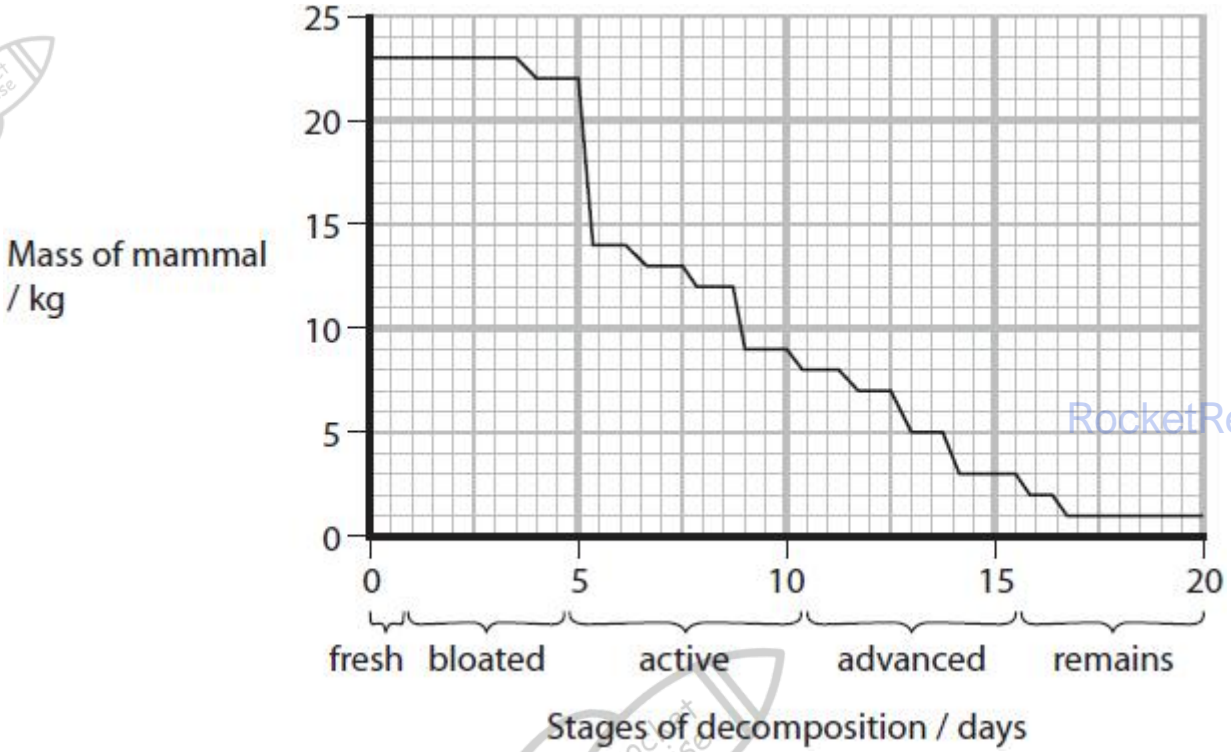
In this study, a mammal was killed, placed inside a metal cage in this region and left until it had completely decomposed.

The body of this mammal was monitored regularly.

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The stages of decomposition were identified and various measurements were taken and recorded.
Insects in the different stages of their lifecycle were collected and identified.
The graph shows the changes in mass of this mammal during the decomposition period.



(i) Suggest why the mammal was placed inside a metal cage to decompose.

(1)

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(ii) Calculate the rate of change in mass between day 5 and day 15 of decomposition.
Express your answer in kg hr^{-1}.
Give your answer to two decimal places.

(2)

Answer ..... Kg hr^{-1}

(iii) Explain the changes in mass of this mammal during decomposition.

(4)

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(iv) Suggest why some of the insect eggs, collected from the decomposing mammal, were taken back to the laboratory and kept for a few days.

(1)

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(v) The table shows data on some of the insects collected from this decomposing mammal.

Type of insect	Percentage of some of the insect found at each stage of decomposition (%)				
	Fresh	Bloated	Active	Advanced	Remains
Lucilia	100.0	61.4	1.8	3.7	0.0
Cochliomyia	0.0	15.5	68.4	35.1	0.0
Chrysomya	0.0	6.4	9.8	1.2	0.0
Ophyra	0.0	0.0	0.6	30.9	83.9
Fannia	0.0	0.0	0.0	0.1	0.0
Other types of insects	0.0	16.7	19.4	29.0	16.1

Explain how these results illustrate succession.
Use the information in the table to support your answer.

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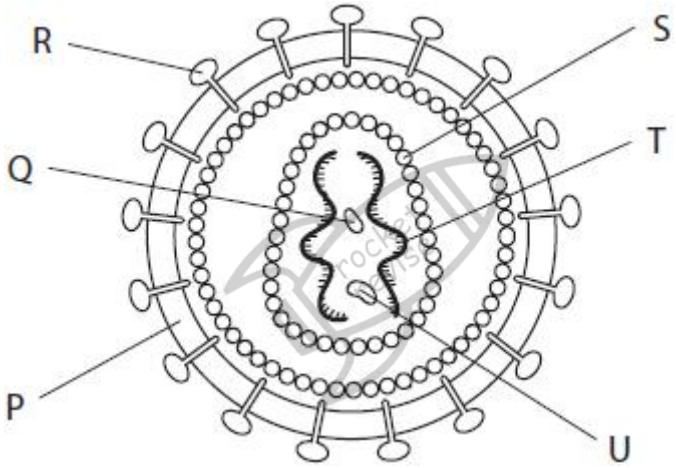
(Total for question = 15 marks)

Q15.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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The diagram shows a human immunodeficiency virus (HIV).



(a) (i) Which structure is RNA?

(1)

- ☒ **A** R
- ☒ **B** Q
- ☒ **C** S
- ☒ **D** T

(ii) How many of the following structures contain protein?

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- structure Q
- structure R
- structure S

(1)

- ☒ **A** 0
- ☒ **B** 1
- ☒ **C** 2
- ☒ **D** 3

(iii) Which virus has structure P and contains RNA?

(1)

- ☒ **A** Ebola virus only
- ☒ **B** Ebola virus and tobacco mosaic virus (TMV)
- ☒ **C** Tobacco mosaic virus (TMV) only

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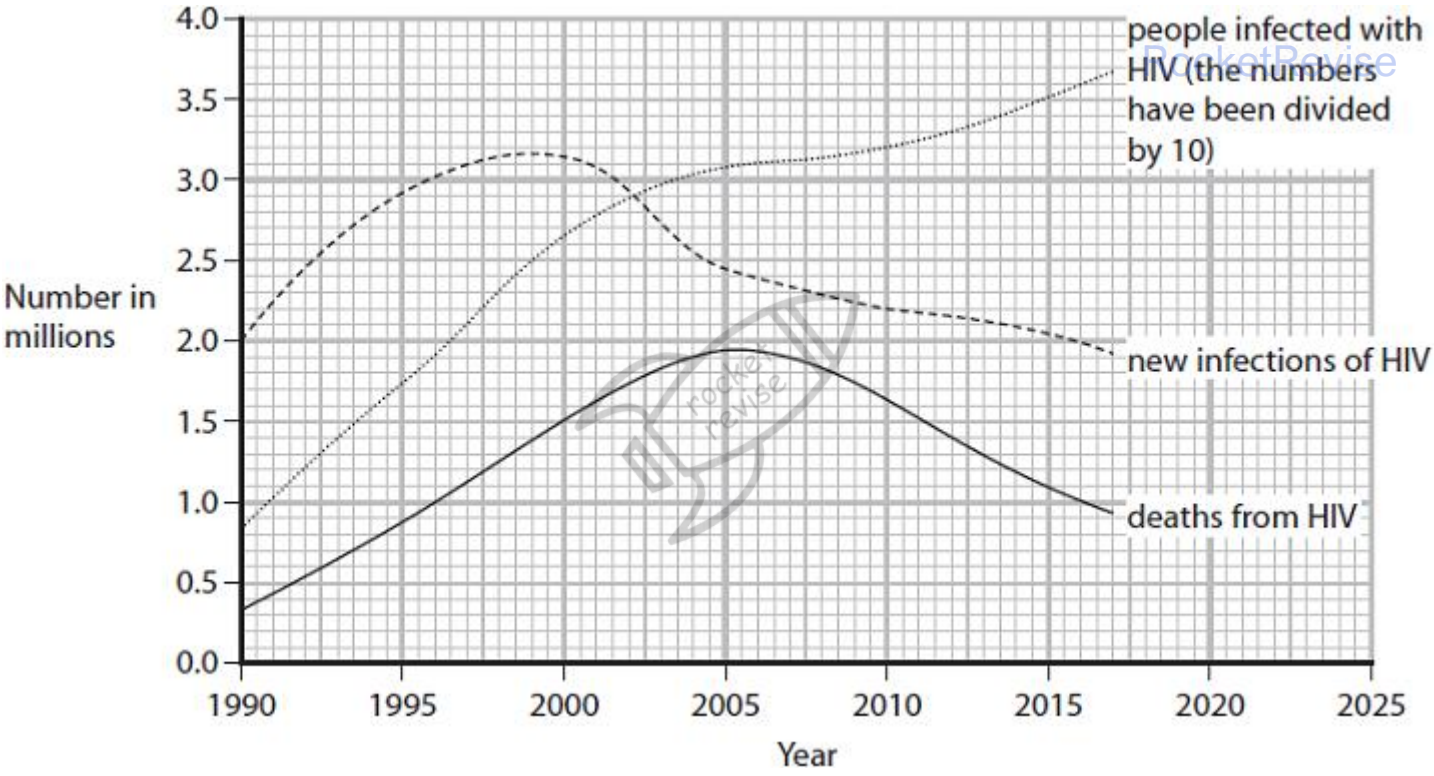
☒ **D** Neither Ebola virus nor tobacco mosaic virus (TMV)

(iv) The diameter of HIV is 120 nm.
Which is the approximate magnification of this drawing?

(1)

☒ **A** × 500
☒ **B** × 5 000
☒ **C** × 50 000
☒ **D** × 500 000

(b) The graph shows the number of HIV infections and the number of deaths from HIV in the world from 1990 to 2017.



(i) The data plotted for the number of people infected with HIV has been divided by 10.
Explain why these numbers were divided by 10.

(2)

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(ii) Calculate the mean rate of increase in the number of people infected with HIV from 2005 to 2010.
Express your answer in standard form to an appropriate number of significant figures.

(2)

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Answer ..... mean number of infections year^{-1}

(iii) Explain why the number of people infected with HIV is going up although the number of new infections and deaths is going down, after 2005.

(2)

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(iv) State how the graph can be used to estimate the number of deaths in 2025.

(1)

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(v) Explain why this estimate would be lower if a vaccine for HIV became available in 2023.

(3)

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(Total for question = 14 marks)

Q16.

The immune response to cancer involves both T helper and T killer cells.

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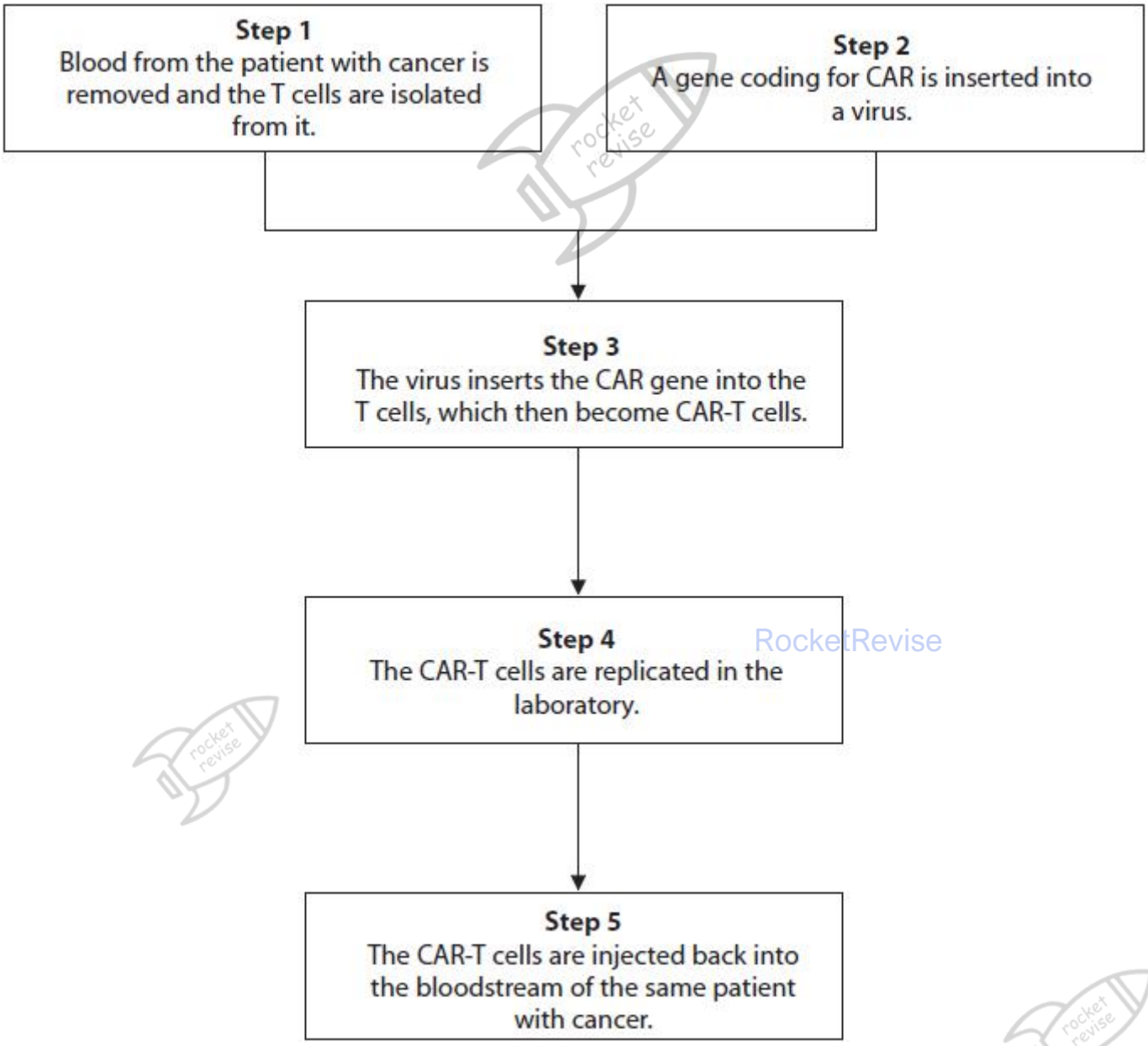
(a) Which row of the table shows the cells that present antigens to each of these T cells?

(1)

	T helper cells	T killer cells
☐ A	host-infected cells	host-infected cells
☐ B	host-infected cells	macrophages
☐ C	macrophages	host-infected cells
☐ D	macrophages	macrophages

(b) Immunotherapy, using genetically-modified T cells, is being developed to treat blood cell cancers.

Cancer cells have a specific antigen on their surface.
The T cells have to possess a receptor (CAR) on their surface that binds to this antigen.
A gene coding for CAR has to be inserted into the T cells. These genetically-modified T cells are called CAR-T cells.
The diagram shows some steps in this procedure.



(i) The virus used in this procedure is modified so that it is not pathogenic.

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Which virus would be suitable to use in **Step 2**?

(1)

- ☐ **A** Ebola virus
- ☐ **B** human immunodeficiency virus (HIV)
- ☐ **C** lambda phage (*λ* phage)
- ☐ **D** tobacco mosaic virus (TMV)

(ii) Explain why the CAR gene is inserted into the T cells in **Step 3**.

(2)

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(iii) Explain the importance of mitosis in the replication of the CAR-T cells in **Step 4**.

(3)

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(iv) In **Step 1**, T cells are removed from a patient with cancer.
Explain the advantages of injecting CAR-T cells into the same patient in **Step 5**.

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(3)

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(v) Currently, the use of CAR-T cells has been effective against blood cell cancers only.
Suggest why CAR-T cells are not effective against cancer cells in solid tumours growing in other tissues in the body.

(2)

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(Total for question = 12 marks)

Q17.

Human Immunodeficiency Virus (HIV) infects human cells, causing symptoms that may result in death.

(a) Explain why the virus is an immunodeficiency virus.

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(b) A number of antiretroviral drugs are given to patients with HIV.
These include reverse transcriptase inhibitors and integrase inhibitors.

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(i) Explain how these two drugs reduce the development of HIV.

(2)

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(ii) These two drugs are often given in combination with other antiretroviral drugs. Suggest why patients are given combinations of drugs.

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(iii) If a patient stops taking the antiretroviral drugs, the number of virus particles increases again. Suggest why this increase occurs.

(2)

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(Total for question = 10 marks)

Q18.

Antibodies play an important role in the immune response.

(a) Explain the importance of antibodies.

(4)

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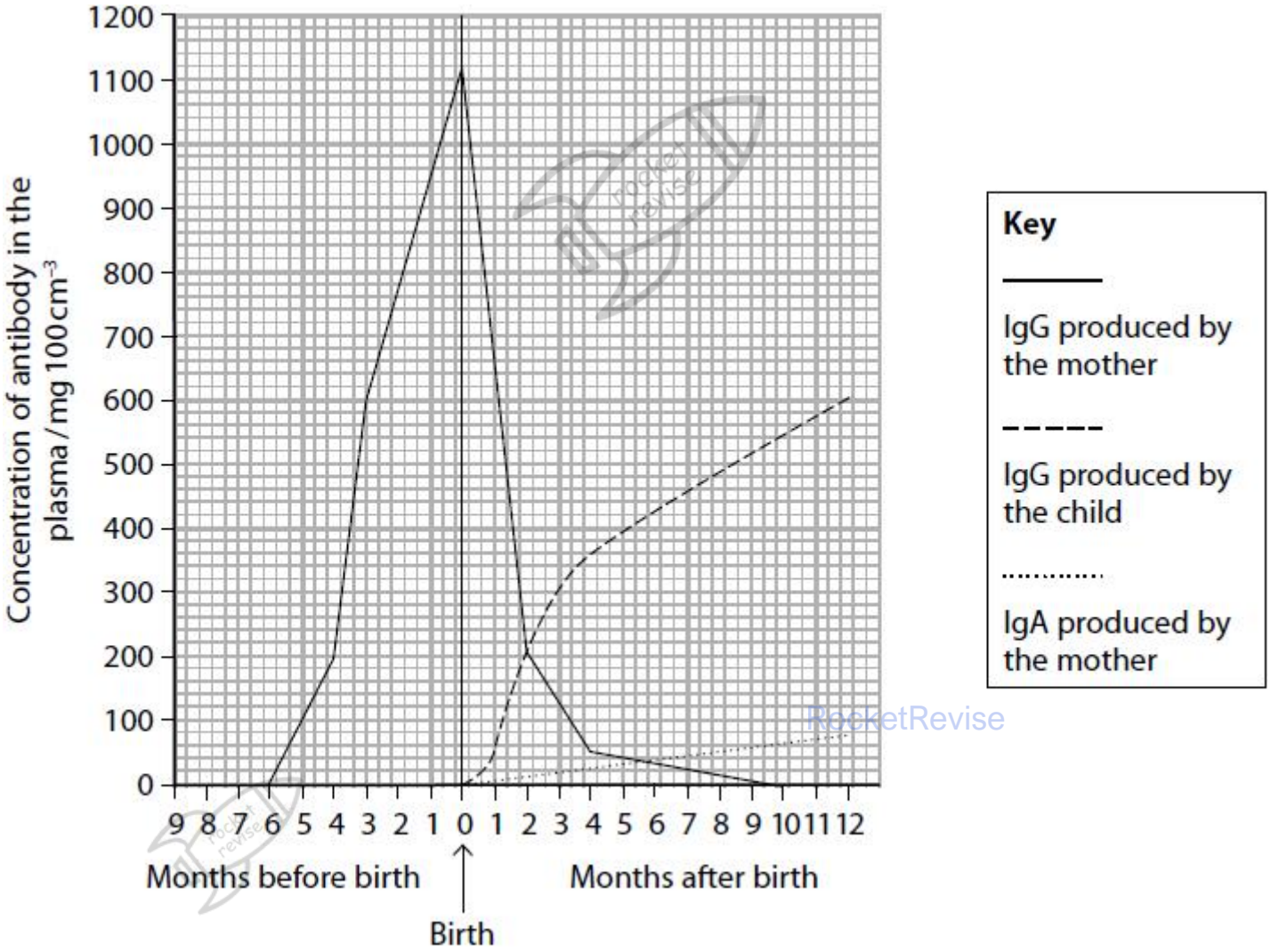
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(b) There are different classes of antibody, including IgA, IgG and IgM.

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The graph shows the changes in the concentration of IgA and IgG in the plasma of a fetus before birth and in a child after birth.



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(i) Describe the changes in the classes of antibody in a fetus before birth and in a child after birth. Use the information in the graph to support your answer.

(3)

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(ii) Which type of immunity explains the presence of IgA?

(1)

- ☒ **A** artificial active
- ☒ **B** artificial passive
- ☒ **C** natural active
- ☒ **D** natural passive

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(iii) Suggest why the concentration of antibodies is less in a child 12 months after birth than in an adult.

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(Total for question = 10 marks)

Q19.

Rheumatoid arthritis is an inflammatory condition caused by the overactivity of the immune system.

(a) The photographs show the hand of a person with rheumatoid arthritis and the hand of a person without rheumatoid arthritis.

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(Source: © Mediscan/Alamy Stock Photo)

Hand with rheumatoid arthritis

Magnification × 0.2

Hand without rheumatoid arthritis

Magnification × 0.2

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(i) Give **one** piece of evidence, shown in the photographs, that rheumatoid arthritis is an inflammatory condition.

(1)

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(ii) Give **two other** signs of inflammation that are experienced in the hand of a person with rheumatoid arthritis.

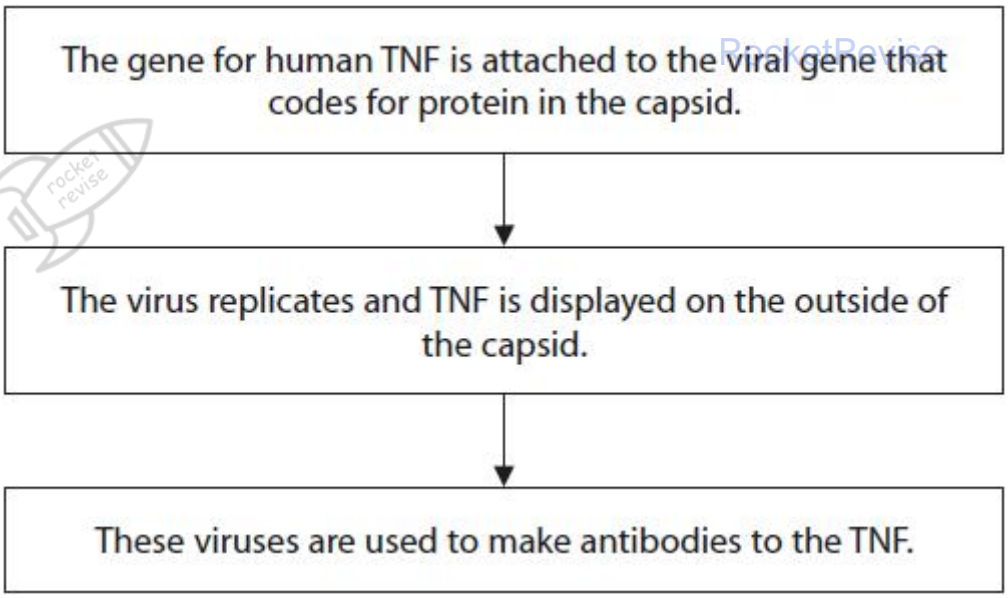
(1)

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(b) Tumour necrosis factor (TNF) is a protein that can be released by several types of cell. It plays an important part in the immune response.

This protein binds to specific receptors on the surface of cell membranes stimulating chemical reactions inside the cell. As a result, a number of responses may occur that include inflammation and the stimulation of phagocytosis by macrophages.
Antibodies to TNF are used to treat rheumatoid arthritis.
Scientists have used a virus that infects bacteria to produce antibodies to human TNF.
The diagram summarises this process.



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(i) Explain how attaching the gene for TNF to the viral gene results in TNF being displayed on the outside of the capsid.

(2)

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(ii) Explain why antibodies to TNF are used in the treatment of rheumatoid arthritis.

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(2)

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(iii) Patients being treated with antibodies to TNF are more susceptible to tuberculosis (TB), which can be fatal.

Explain why patients being treated with antibodies to TNF can die from TB.

(4)

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(Total for question = 10 marks)

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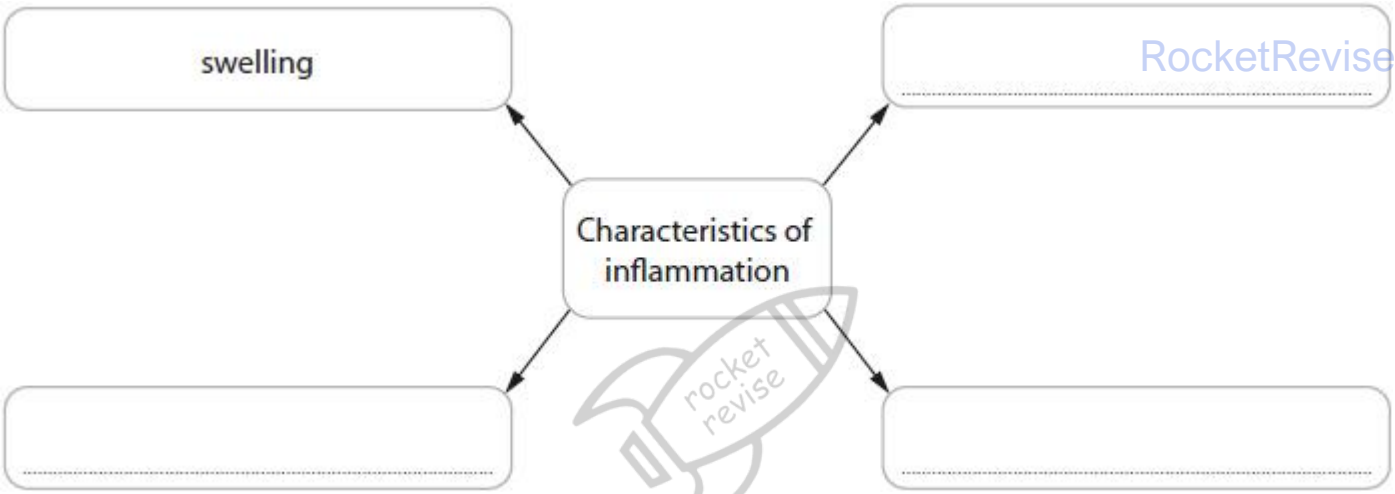
Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Inflammation is involved in the non-specific response of the body to infection and injury.

(a) (i) Complete the diagram to show the four characteristics of inflammation.

(2)



(ii) Inflammation occurs following a cut to the skin.
Describe the role of **two** of these characteristics of inflammation in response to a cut to the skin.

(2)

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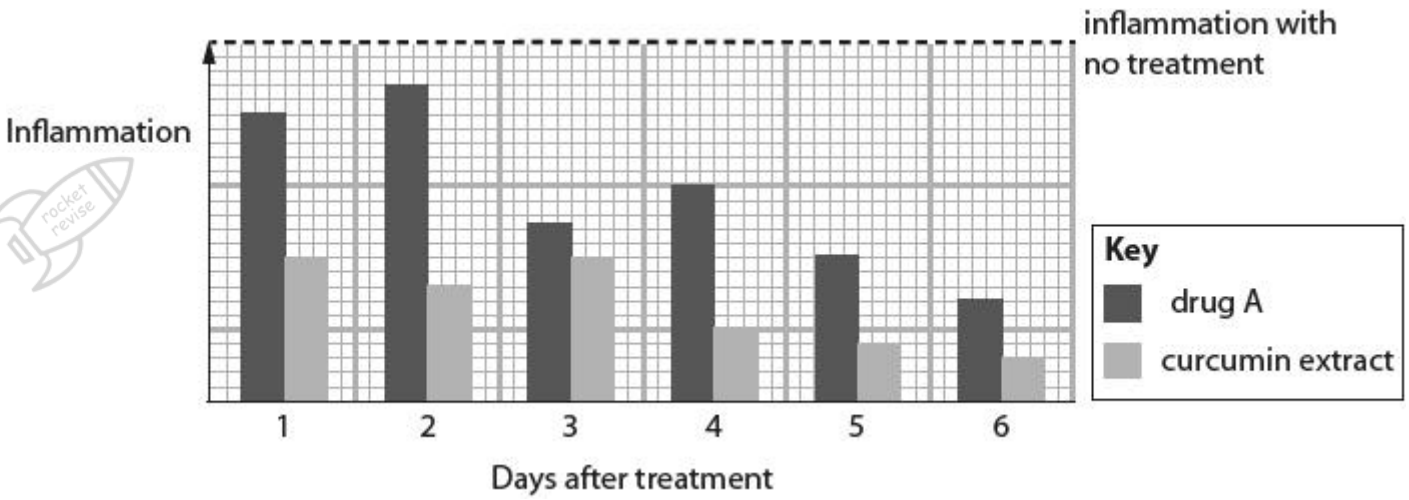
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(b) Anti-inflammatory drugs reduce inflammation.

Turmeric is a spice that is added to food.
Curcumin is a chemical in turmeric that has been shown to reduce inflammation.
(i) The graph shows the effect of treating inflammation with curcumin extract and an anti-inflammatory drug, drug A.

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Compare and contrast the effect of treating inflammation with curcumin extract and with drug A.

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(ii) Many treatments use 1 g curcumin extract.
A sample of turmeric contains 3 % curcumin.
Which is the mass of turmeric for one treatment?

(1)

- ☒ **A** 0.03 g
- ☒ **B** 3.33 g
- ☒ **C** 33.3 g
- ☒ **D** 33.4 g

(Total for question = 8 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Malaria is a life-threatening disease caused by the pathogen *Plasmodium*.

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Plasmodium is transmitted into the blood plasma of a person through the bite of an infected mosquito.

The *Plasmodium* is a sporozoite that travels to the liver cells.

This sporozoite produces merozoites that infect red blood cells.

(a) State the meaning of the term **infection.**

(1)

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(b) In 2019, there were an estimated 229 million cases of malaria worldwide.

The number of deaths was estimated at 4.09×10^5 per year.

(i) Which is the approximate death rate from malaria in 2019?

(1)

- ☒ **A** 1 in 180
- ☒ **B** 1 in 600
- ☒ **C** 1 in 1 800
- ☒ **D** 1 in 6 000

(ii) In 2019, 274 000 young children died from malaria.
Which is the ratio of deaths in young children to deaths in other age groups?

(1)

- ☒ **A** 0.7 : 1.0
- ☒ **B** 1.0 : 0.5
- ☒ **C** 1.0 : 1.5
- ☒ **D** 2.0 : 1.0

(c) Scientists are trying to develop vaccines to protect against malaria.

One vaccine uses living sporozoites.

(i) Suggest why sporozoites are used in a vaccine against malaria.

(3)

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(ii) In clinical trials of this vaccine, people were injected with the vaccine and two chemicals.
One chemical kills sporozoites and the other kills merozoites.
Deduce why these two chemicals were injected with the vaccine in these clinical trials.

(3)

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(iii) Later in the clinical trials, people were infected with one of two strains of sporozoites, three months after the vaccination.
One strain was identical to the one used in the original vaccine and the other strain was a different one.
The trials found that the vaccine was effective against both strains.
Explain these findings.

(3)

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(Total for question = 12 marks)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Antibiotics are medicines used to treat some medical conditions.

(a) The table shows some medical conditions and whether or not antibiotics are needed to treat the condition.

Medical condition	Are antibiotics needed to treat the condition?
Impetigo	yes
Whooping cough	yes
Middle ear infections	sometimes
Sinus infections	sometimes
Multiple sclerosis	no
Rheumatoid arthritis	no

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Explain the use of antibiotics to treat these conditions.

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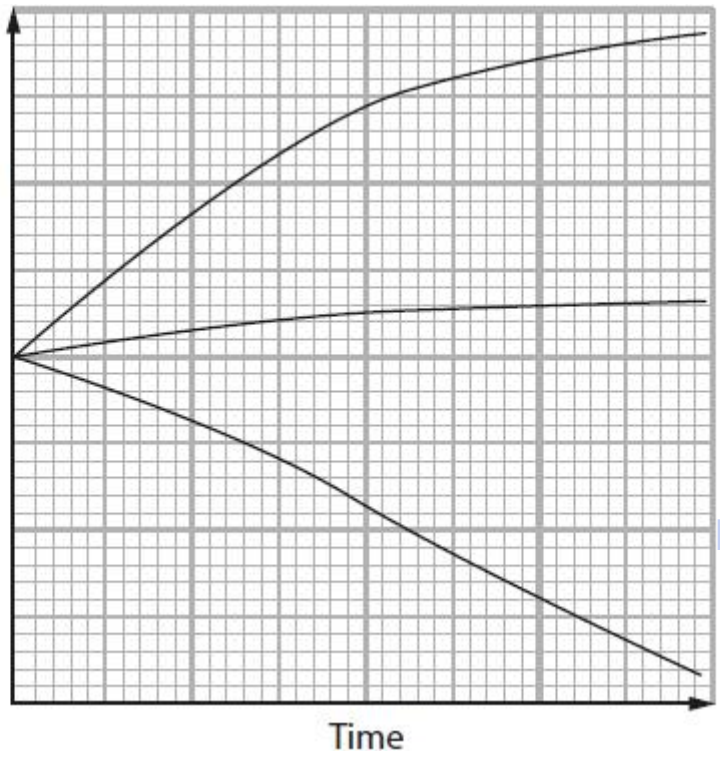
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(b) The graph shows the effect of three drugs, J, K and L, on the number of bacteria in a culture.

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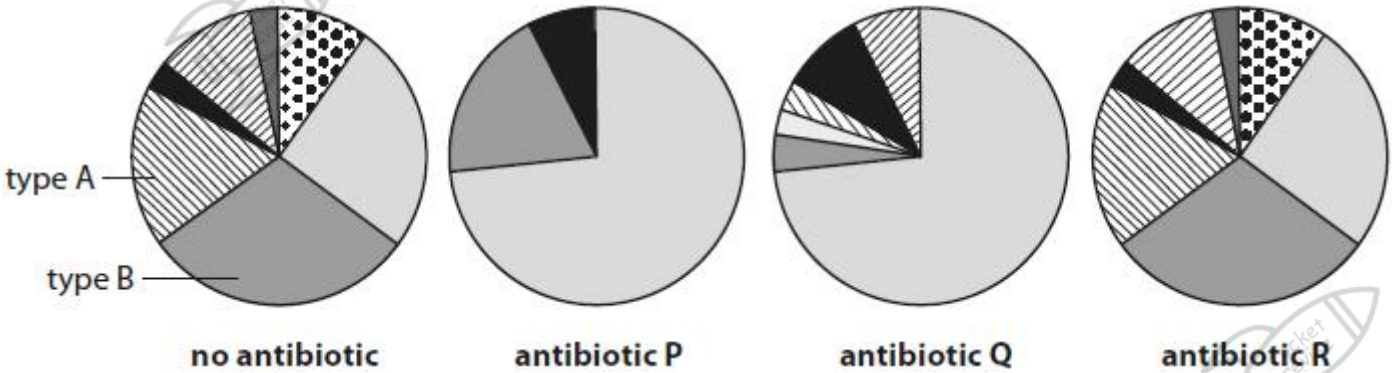
Which row of the table describes each of these three drugs?

(1)

		drug J	drug K	drug L
☐	A	not an antibiotic	bactericidal antibiotic	bacteriostatic antibiotic
☐	B	not an antibiotic	bacteriostatic antibiotic	bactericidal antibiotic
☐	C	bactericidal antibiotic	not an antibiotic	bacteriostatic antibiotic
☐	D	bacteriostatic antibiotic	not an antibiotic	bactericidal antibiotic

(c) One problem of taking antibiotics is their effect on gut flora.

The diagrams show the effects of three antibiotics, P, Q and R, on the proportion of different types of gut flora.
Each section in each of the diagrams represents a different type of gut flora.



(i) Explain the role of gut flora in protecting the body from infection.

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(2)

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(ii) Explain why antibiotics can affect gut flora.

(2)

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(iii) The number of type A in gut flora in the absence of antibiotics is 6 000 000.
Estimate the number of type B in gut flora in the absence of antibiotics.
Give your answer in standard form.

(1)

Answer .....

(iv) Deduce the effects of antibiotics P, Q and R on gut flora.

(3)

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(Total for question = 11 marks)

Q23.

Methicillin-resistant Staphylococcus aureus (MRSA) is a disease causing worldwide concerns.

Methicillin is one of the antibiotics to which this pathogen has developed resistance.

Castaneroxy A is a molecule extracted from the European chestnut tree, endemic to Southern Europe and Turkey.

The effects of Castaneroxy A have been investigated.

The investigations have found that Castaneroxy A:

- does not affect the natural skin flora
- prevents MRSA from producing toxins
- weakens the MRSA bacteria
- reduces the size of skin infections caused by MRSA.

(a) Explain why the development of this drug is an example of an 'evolutionary race' between humans and this pathogen.

(2)

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(b) Explain why it is important that Castaneroxy A does not affect skin flora.

(2)

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(c) Suggest how weakening the MRSA bacteria could help the recovery of a patient infected with this bacteria.

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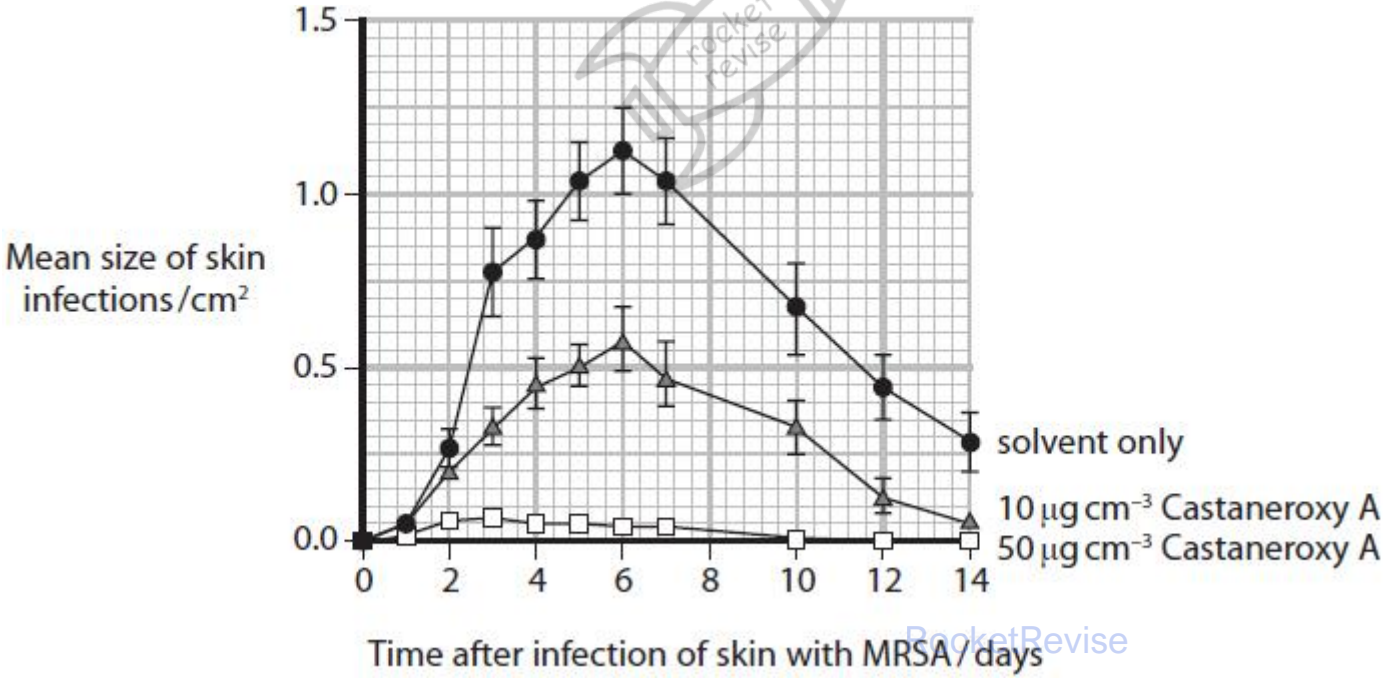
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(d) In an investigation, patients who had developed skin infections were divided into three groups.

Two groups were each treated with a different concentration of Castaneroxy A. The other group was treated with only the solvent that was used in the treatment.
The graph shows the results of these treatments on the mean size of the skin infections.



(i) Explain why this investigation included treatment with the solvent only.

(2)

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(ii) Determine the effect that Castaneroxy A has on the mean size of skin infections.

(3)

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(iii) The leaves of the chestnut tree contain 0.0019 %, by mass, of Castaneroxy A.
Calculate the mass of leaves, in kilograms, that would be needed to produce 1 dm^3 of Castaneroxy A at a concentration of 50 μg cm^{-3}.
Give your answer to the nearest kilogram.

(3)

Answer ..... kg

(Total for question = 14 marks)

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Q24.

The body responds to infection with non-specific responses and the immune response.

(a) Inflammation is characterised by heat, redness and two other characteristics.

(i) Name the **two** other characteristics of inflammation.

(1)

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(ii) Explain how heat helps the body respond to infection.

(2)

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(b) Describe how interferon helps the body respond to infection.

(2)

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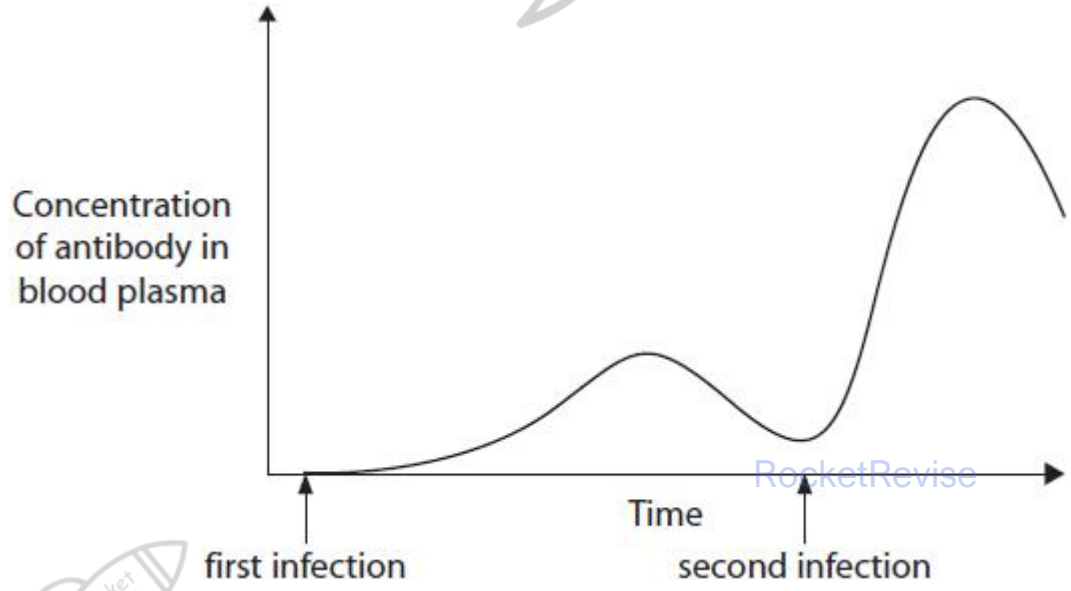
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(c) The immune response results in the production of antibodies following the first infection with a pathogen and on further infections with the same pathogen.

The graph shows the concentration of antibody in blood plasma following the first and second infection with the same pathogen.



Compare and contrast the changes in concentration of antibody in the blood plasma following the first infection and the second infection.

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(Total for question = 8 marks)

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Q25.

Human breast milk supports the growth and development of a baby.

One important component is a group of carbohydrates called oligosaccharides. These contain between three and ten monosaccharide units.

(a) During the first few days after birth, the milk produced is called colostrum. By two weeks, the composition of the milk has changed and it is called mature breast milk.

The charts show some of the nutritional components of colostrum and mature breast milk.



Estimate the percentage composition of oligosaccharides in the colostrum.

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(1)

Answer ..... %

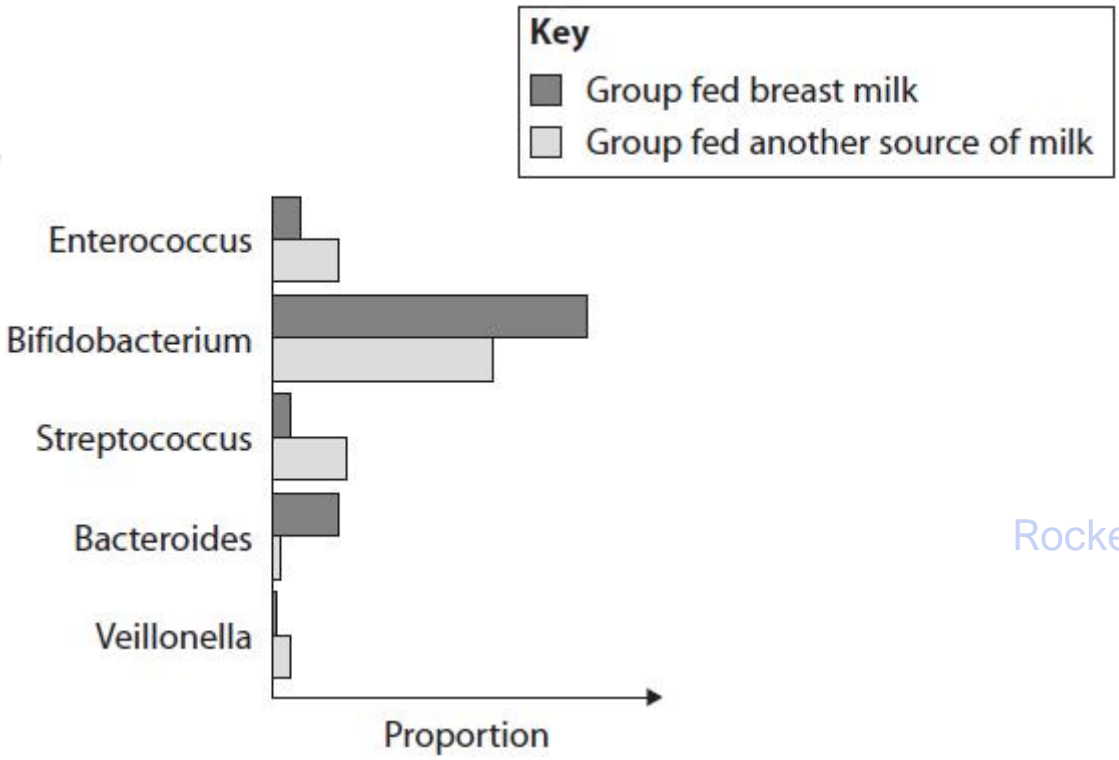
(b) Oligosaccharides in breast milk have been shown to be important in establishing a healthy gut flora in the developing baby.

(i) The effect on the types of bacteria in the guts of babies fed on breast milk or another source of milk was investigated.

The graph shows the proportion of five types of bacteria in the two groups of babies.

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Describe **two** conclusions that can be made about the gut flora of babies fed these two types of milk.

(2)

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(ii) Explain the importance of healthy gut flora in preventing infection in a developing baby.

(2)

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(c) Group B Streptococcus (GBS) bacteria are a common cause of infections in babies.

These bacteria are becoming increasingly resistant to antibiotics.
Scientists have shown that oligosaccharides isolated from breast milk have antimicrobial activity against GBS bacteria.
Explain the possible advantages of using these oligosaccharides, instead of antibiotics, to treat GBS bacterial infections.

(3)

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(Total for question = 8 marks)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Pathogens continuously adapt to invade their host cells.

Host cells continuously adapt to avoid infection by pathogens.

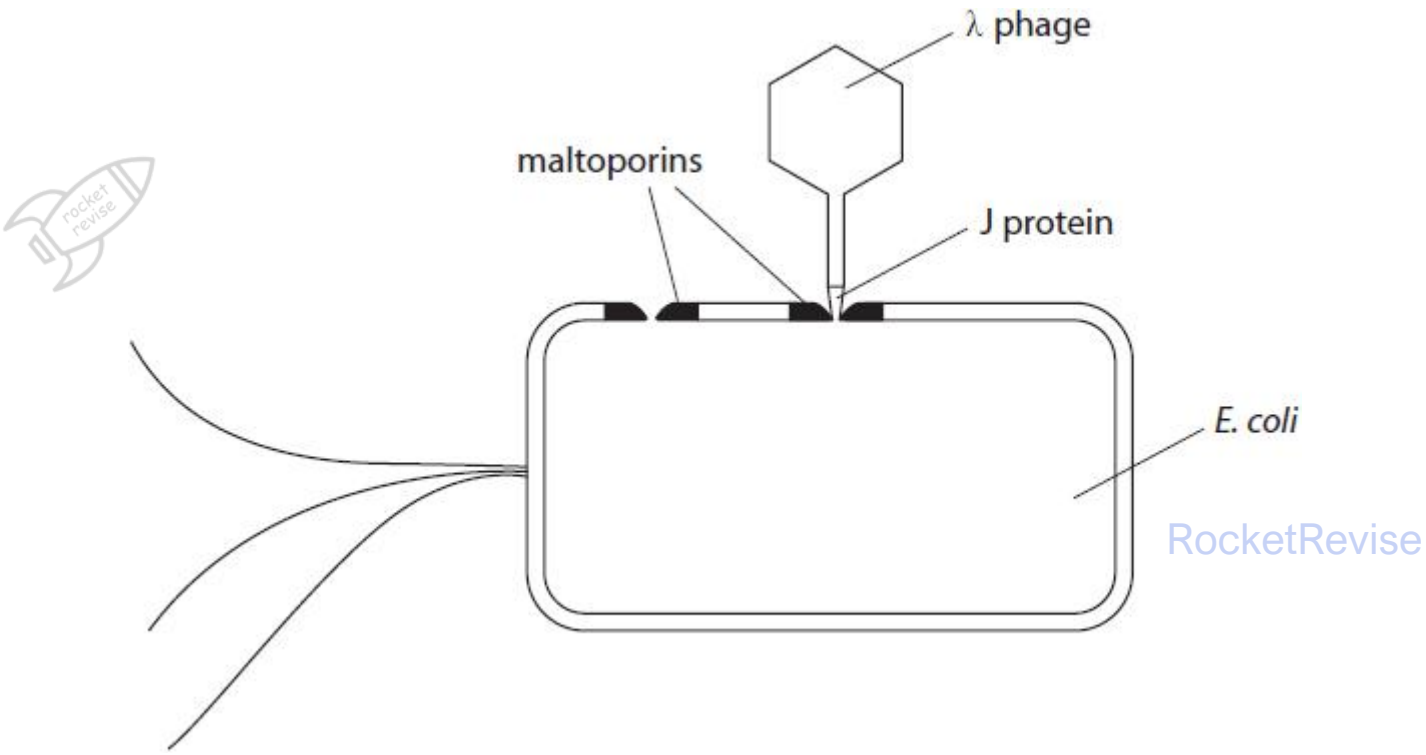
This is an example of an 'evolutionary race'.

The diagram shows the pathogen lambda phage (*λ* phage) attached to its host cell, *E. coli*.

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(a) How many of the following statements are correct for *λ* phage?

- the genetic material is DNA
- the capsid structure is described as complex
- the genetic material codes for the J protein

(1)

- ☒ **A** 0
- ☒ **B** 1
- ☒ **C** 2
- ☒ **D** 3

(b) Describe the role of the *λ* phage J protein.

Use the information in the diagram to support your answer.

(2)

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(c) Maltoporin is coded for by the *lamβ* gene of *E. coli*.

Maltoporin is involved in the transport of some sugars into *E. coli*.

(i) Give **one** reason why sugars are important to *E. coli*.

(1)

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*(ii) Discuss how E. coli and λ phage could interact and develop in their evolutionary race for existence.

Use the information in this question to support your answer.

(6)

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(Total for question = 10 marks)

Q27.

Dengue fever is caused by infection with the dengue virus. This virus is transmitted by mosquitoes.

The virus replicates inside the mosquito and is then passed onto a person in the saliva of the mosquito. The virus then replicates inside the infected person and causes the symptoms of the infection.

Symptoms include fever, headache and a rash. Complications include bleeding, low levels of platelets and low blood pressure.

It is estimated that globally, there are 390 million new cases of dengue infections per year. The infection rate is rising. This disease can be fatal.

Groups of scientists are working to develop vaccines, treatments and methods to reduce the spread of

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this virus.

(a) The dengue virus is an RNA virus. The RNA is translated on the endoplasmic reticulum of the host cell into one polyprotein.

(i) Describe how the viral RNA is translated.

(3)

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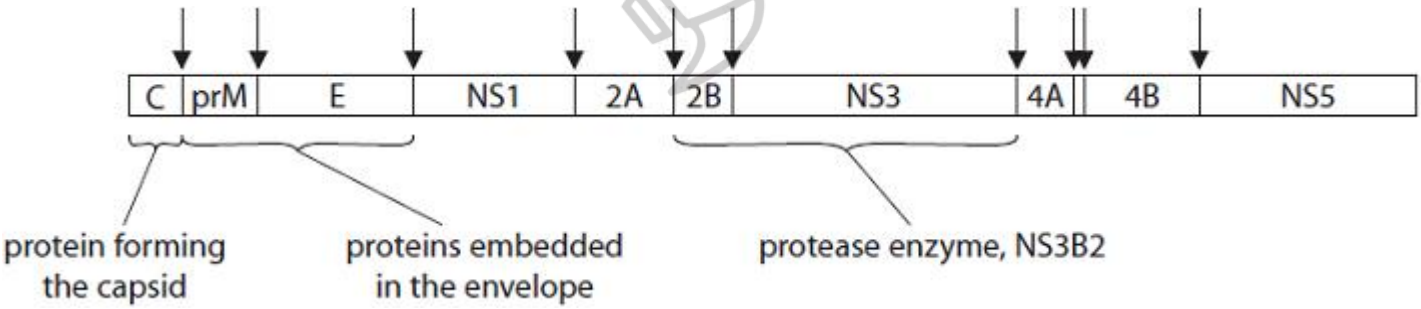
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(ii) The diagram shows this polyprotein and the function of some of the individual proteins it contains. The arrows indicate where the individual proteins need to be separated by the protease enzyme NS3B2.



One group of scientists has found that an inhibitor to this protease enzyme reduces the number of viruses produced in infected cells.
Suggest how this inhibitor could result in fewer viruses being produced.

(3)

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(b) Another group of scientists showed that an extract from the leaves of the papaya plant helped in recovery from dengue virus infection.

They showed that the platelet and white blood cell counts were higher in people treated with the extract than in people given a placebo.
Explain how these changes may have helped in recovery from dengue virus infection.

(4)

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(c) Another group of scientists infected the eggs of mosquitoes with *Wolbachia* bacteria.

The *Wolbachia* do not harm the mosquitoes but compete with the virus so that it is harder for it to replicate.
The bacteria can also be passed on to future generations of mosquitoes.
One study in Indonesia showed a 77% reduction in new cases of dengue infection after the introduction of *Wolbachia* into the mosquito population.
Describe **two** advantages of this method of disease control.

(2)

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(Total for question = 12 marks)

Q28.

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The presence of skin flora is one way that the skin protects the body from infection.

(a) (i) Describe how skin flora protect the body from infection.

(2)

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(ii) Give **one** other way in which the skin protects the body from infection.

(1)

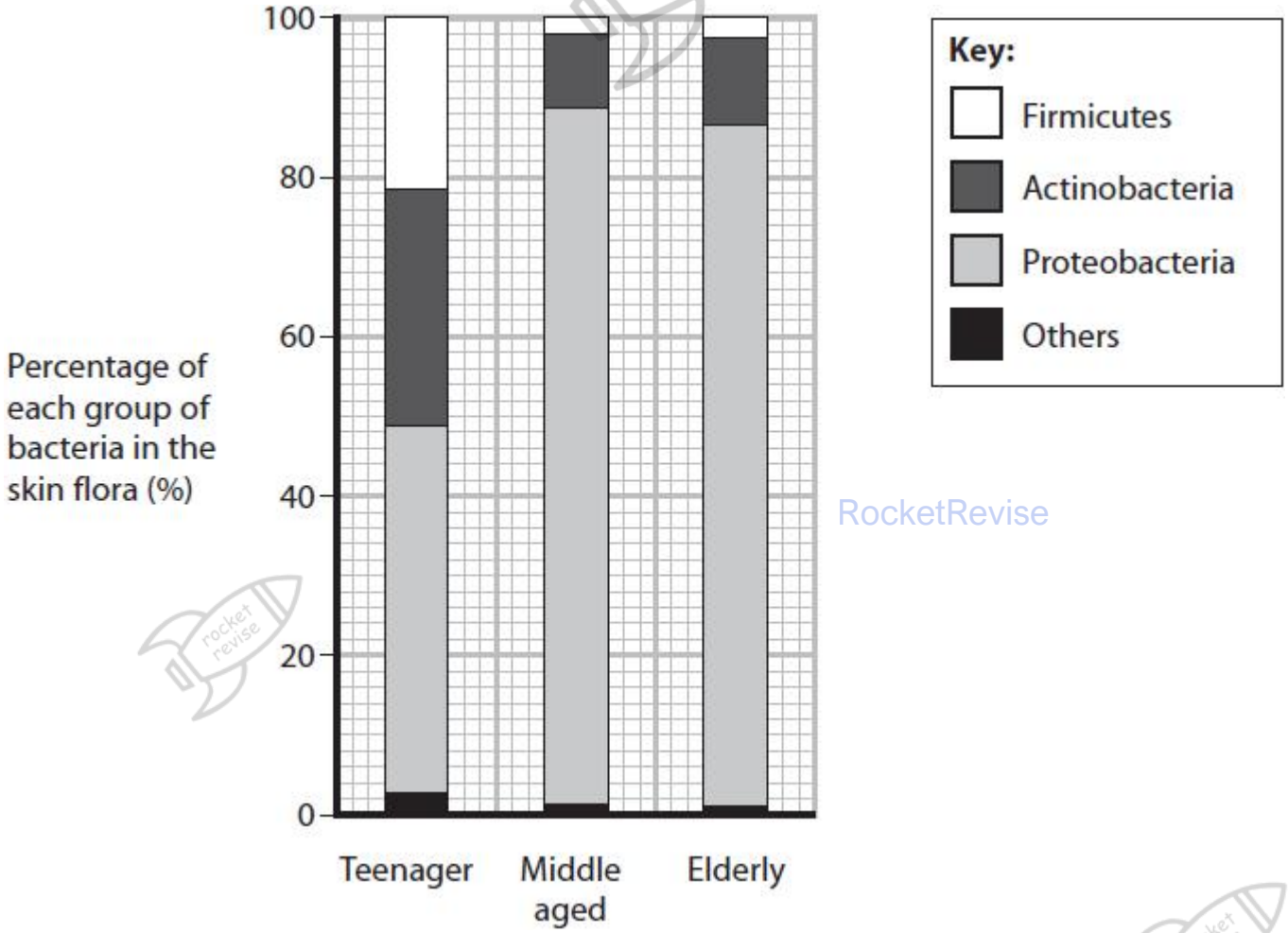
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(b) The groups of bacteria present on the skin of men of different ages was investigated.

The graph shows some of the results of this investigation.



(i) Describe **three** conclusions that could be made from this graph.

(3)

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(ii) Describe how this study should have been designed to collect valid data.

(2)

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(c) Some bacteria can get through the skin.

(i) Describe the role of macrophages in the non-specific response of the body to these bacteria.

(2)

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(ii) Describe the role of macrophages in the immune response of the body to these bacteria.

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(Total for question = 12 marks)

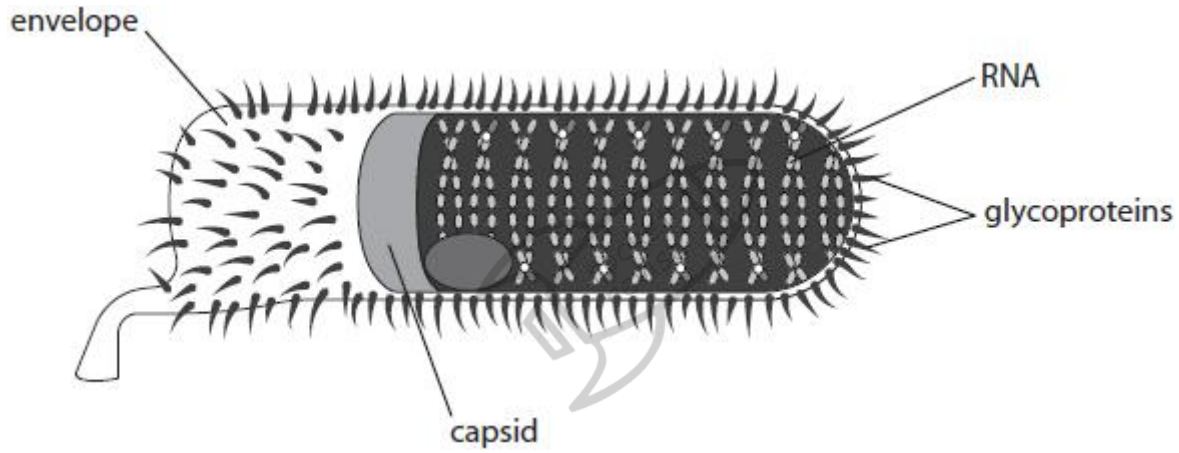
Q29.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Rabies is a disease caused by a virus.

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(a) The diagram shows the structure of the rabies virus.



(i) The rabies virus has an envelope.
Which of the following pairs of viruses have an envelope?

(1)

- ☒ **A** Ebola virus and human immunodeficiency virus (HIV)
- ☒ **B** human immunodeficiency virus (HIV) and tobacco mosaic virus (TMV)
- ☒ **C** lambda phage (λ phage) and Ebola virus
- ☒ **D** tobacco mosaic virus (TMV) and lambda phage (λ phage)

(ii) The structure of the rabies capsid is described as complex.
Which of the following has a complex capsid structure?

(1)

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- ☒ **A** Ebola virus
- ☒ **B** human immunodeficiency virus (HIV)
- ☒ **C** lambda phage (λ phage)
- ☒ **D** tobacco mosaic virus (TMV)

(iii) Rabies virus is an RNA virus.
How many of the following viruses are RNA viruses?

- Ebola virus
- human immunodeficiency virus (HIV)
- lambda phage (λ phage)
- tobacco mosaic virus (TMV)

(1)

- ☒ **A** 1

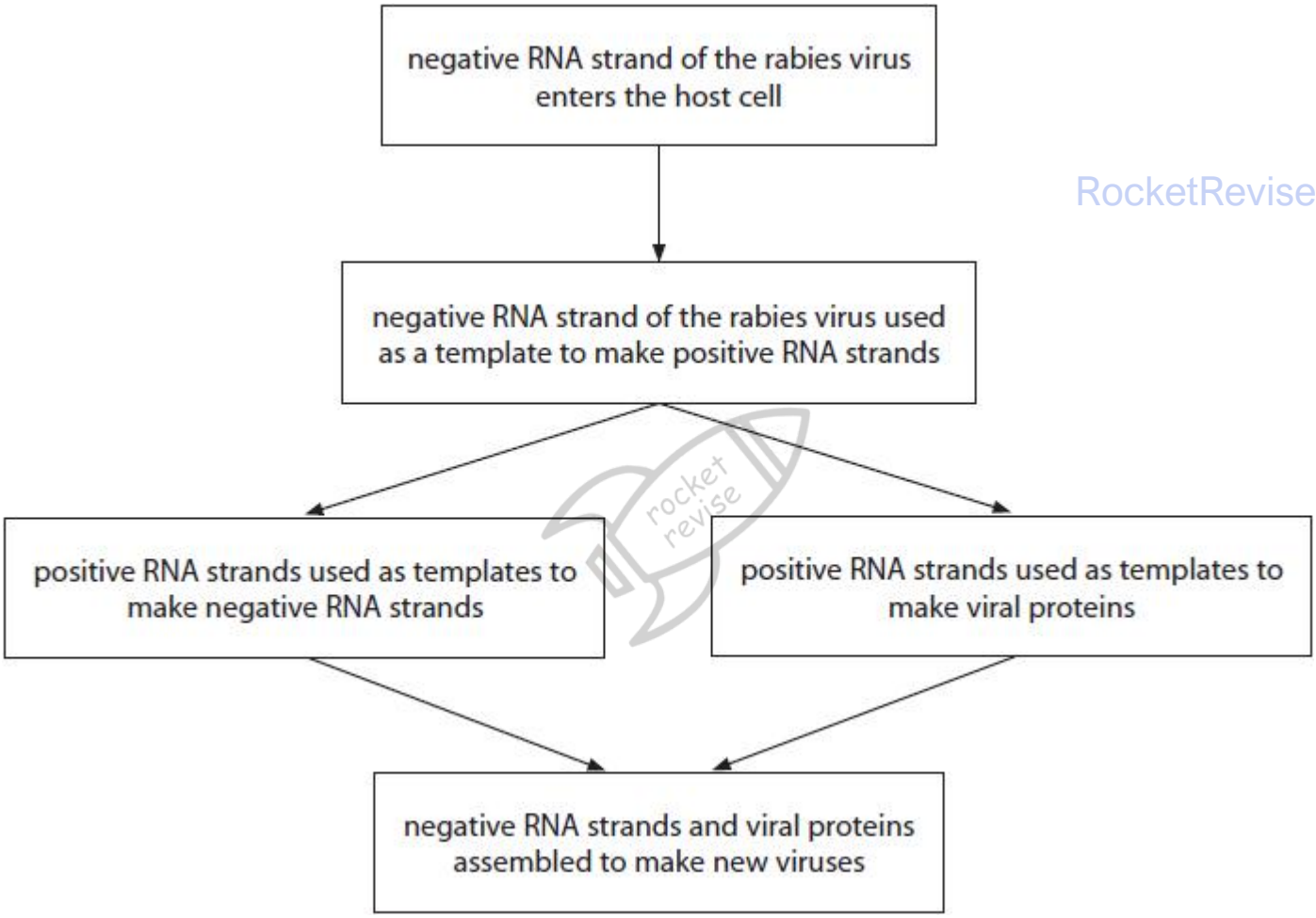
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- **B** 2
- **C** 3
- **D** 4

(b) The rabies virus replicates in a lytic cycle.

The RNA of the rabies virus is a negative RNA strand.
The diagram shows how the negative RNA strand of the rabies virus is used to make positive RNA and proteins.



(i) The diagram shows part of the base sequence in the negative RNA strand.
Complete the diagram to show the corresponding base sequence in the positive RNA strand.

(1)

Negative RNA strand	A	C	C	A	A	G	G	C	G
Positive RNA strand									

(ii) Explain why a positive RNA strand has to be made.

(2)

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(c) Lemurs are found in Madagascar. It is thought that they might carry rabies viruses.

The photograph shows a lemur.



(Source: Caroline Wilcox)

A person was bitten by a lemur.
This person did not receive any treatment for rabies until 18 days after being bitten.
Explain why doctors were worried that this person had left it too long for the treatment to be successful.

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(Total for question = 10 marks)

Q30.

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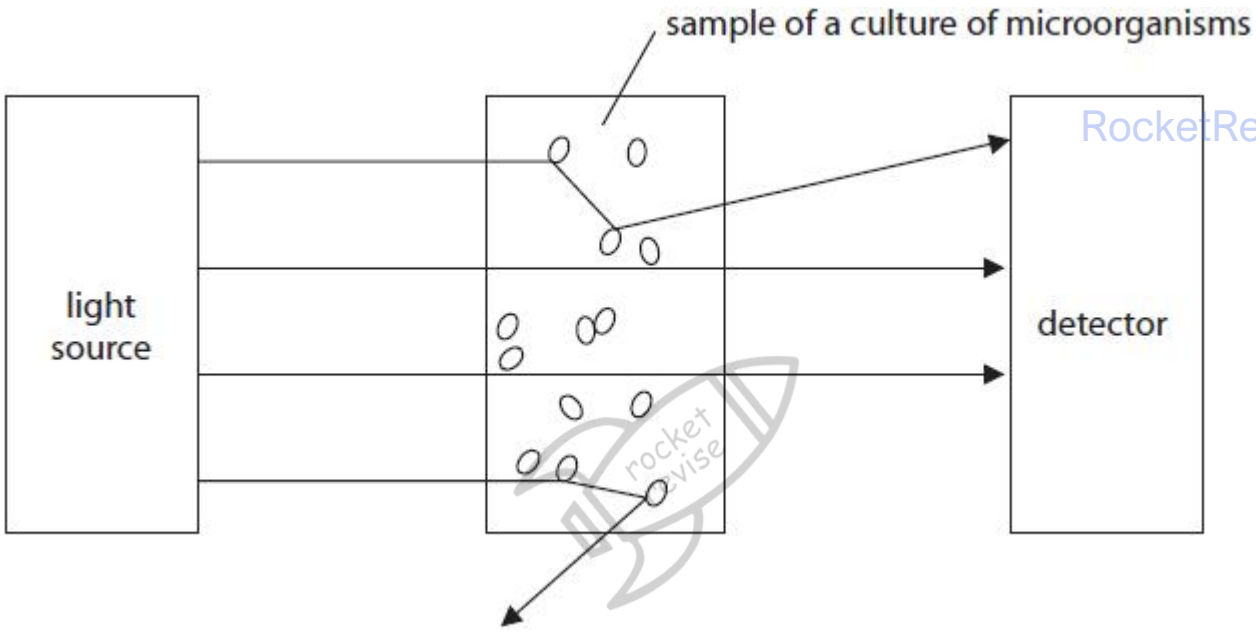
The growth of microorganisms can be studied using optical methods (turbidity).

The number of cells in 1 cm^3 of a culture can be estimated by using a photometer to measure the optical density of the culture.

Light is shone through a sample of the culture and a detector records the optical density.

The more light absorbed by the culture, the higher the optical density.

The diagram shows what happens to light shone at a culture of microorganisms.

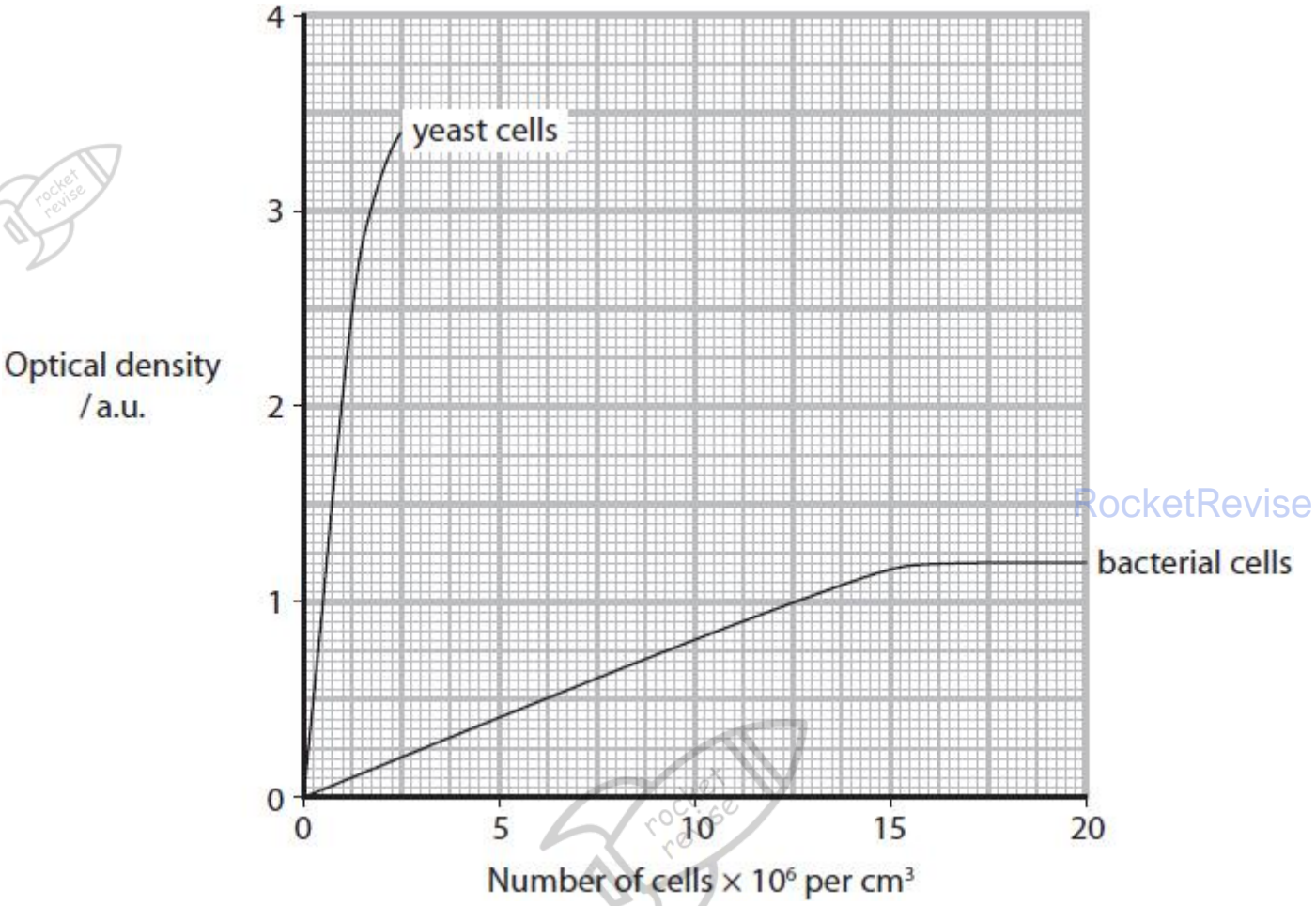


The number of cells in the sample is determined using a calibration curve.

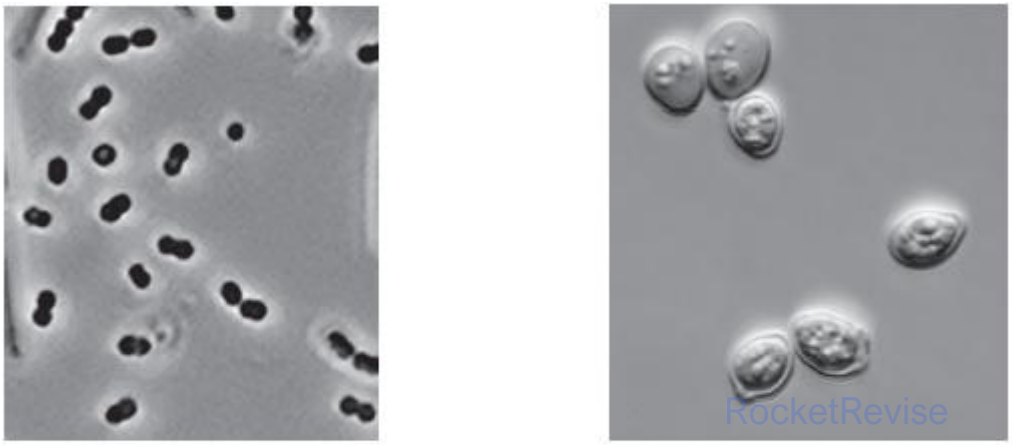
The graph shows a calibration curve for bacterial cells and yeast cells.

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The photographs show bacterial cells and yeast cells, as seen using a light microscope.



bacterial cells

yeast cells

scale: 5 µm

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(a) State the relationship between optical density and the size of the microorganisms.

(1)

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(b) Calculate how many times more bacterial cells there are than yeast cells if both samples have an optical density of 1.0 a.u.

(2)

Answer .....

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(c) (i) State the relationship between the number of bacterial cells and optical density.

(1)

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(ii) Suggest why using optical density to measure the concentration of microorganisms is more accurate at lower concentrations of cells.
Use the information in the diagram to support your answer.

(2)

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(d) A liquid medium was inoculated with 6×10^3 bacterial cells and the culture was incubated for a period of time.

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At the end of this incubation period, there were 1.2×10^7 bacterial cells.
Calculate the time (t) of this incubation period.
Use a value of 0.963 for the growth rate constant (k) and the equation

$$k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$$

(3)

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Answer .....

*(e) Explain why one calibration curve cannot be used to determine the growth of all microorganisms.

Use all the information in this question and your own knowledge to support your answer.

(6)

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(Total for question = 15 marks)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The resistance of bacteria to antibiotics can be studied using a mast ring.

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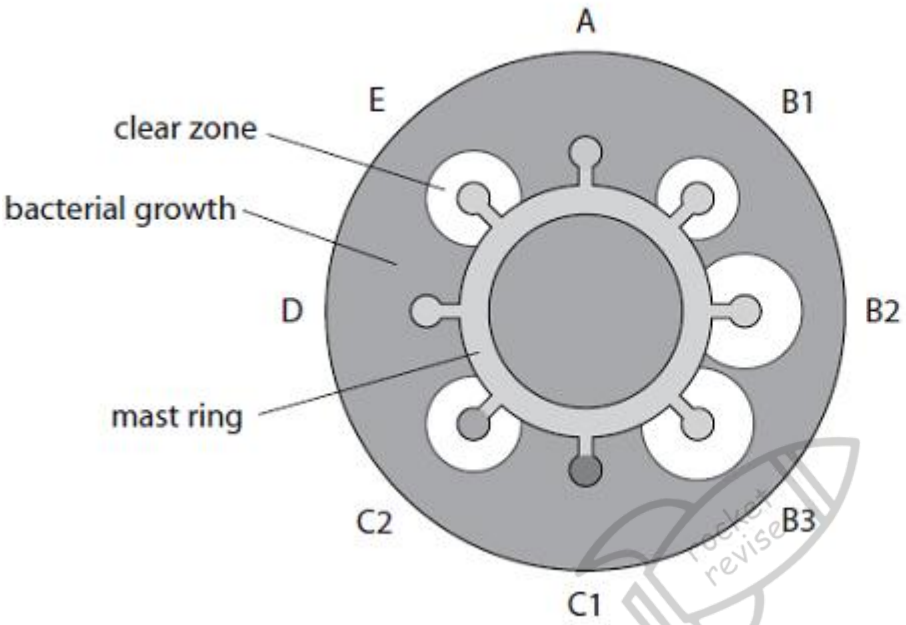
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A mast ring is made of filter paper.

Each part of the mast ring contains a different type of antibiotic or different mass of the same antibiotic.

The diagram shows a mast ring placed on agar in a Petri dish that has been spread with bacteria.

The table shows the mass of each antibiotic in the mast ring.



Part of mast ring	Antibiotic and its mass
A	100 μg antibiotic A
B1	1 μg antibiotic B
B2	10 μg antibiotic B
B3	0.5 mg antibiotic B
C1	20 μg antibiotic C
C2	200 μg antibiotic C
D	100 μg antibiotic D
E	100 μg antibiotic E

Clear zones around the mast ring indicate that the growth of the bacteria has been inhibited.

The greater the diameter of the zone of inhibition, the more effective the antibiotic.

(a) State **two** precautions that should be taken to ensure that only one type of bacteria is grown in this culture.

(2)

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(b) How much more antibiotic is in part B3 than in part B2?

(1)

- ☒ **A** 0.5 times
- ☒ **B** 5.0 times
- ☒ **C** 50.0 times
- ☒ **D** 500.0 times

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(c) Explain how the results shown in the diagram can help a doctor decide how to prescribe antibiotics to a patient who is infected with this type of bacteria.

(3)

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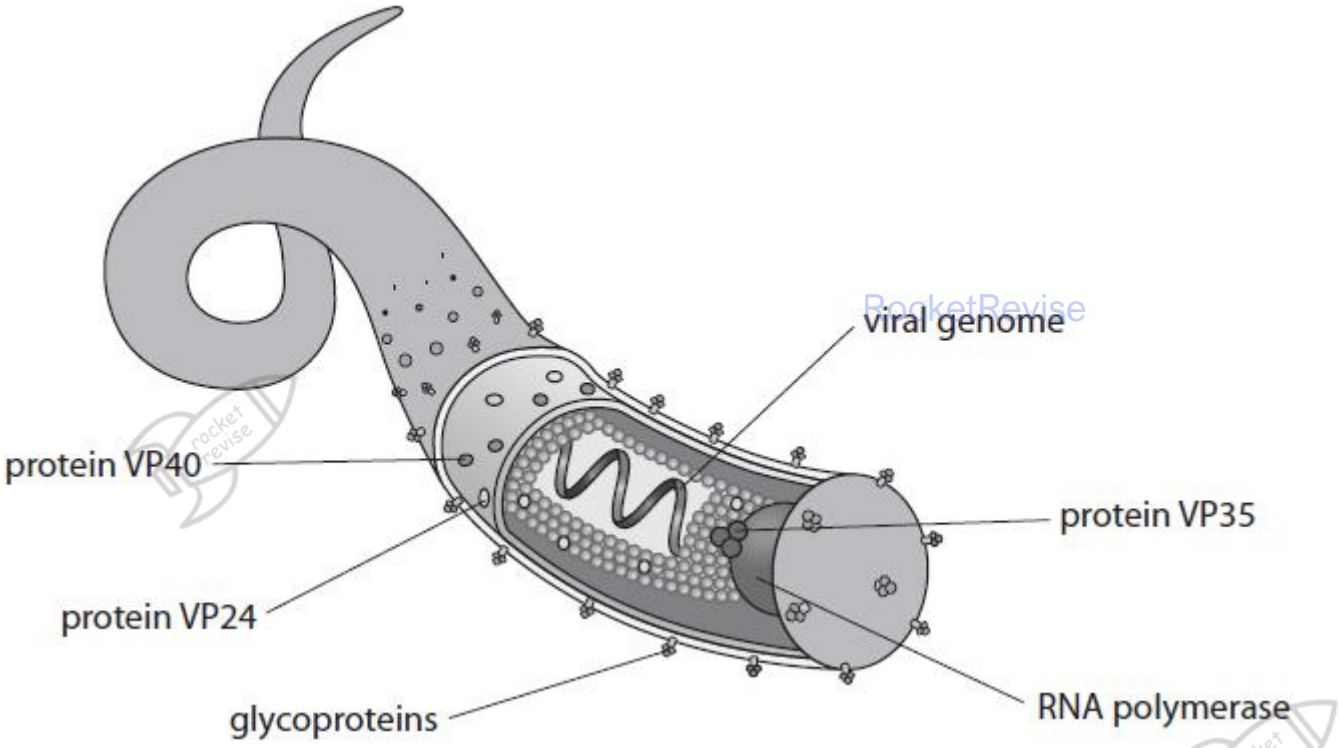
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(Total for question = 6 marks)

Q32.

In December 2018 there was an outbreak of Ebola, with 500 cases being reported.

The diagram shows the structure of an Ebola virus.



(a) Describe the role of each of the following.

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(i) The viral genome

(2)

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(ii) The glycoproteins

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(b) The enzyme RNA polymerase is involved in RNA transcription.

Explain why the structure of the Ebola virus includes RNA polymerase.

(2)

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(c) The proteins VP24 and VP40 are involved in virus assembly.

State what is meant by the term virus assembly, using Ebola as an example.

(2)

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(d) The protein VP35 inhibits the production of interferon by host cells.

Explain why the structure of the Ebola virus includes VP35.

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(Total for question = 10 marks)

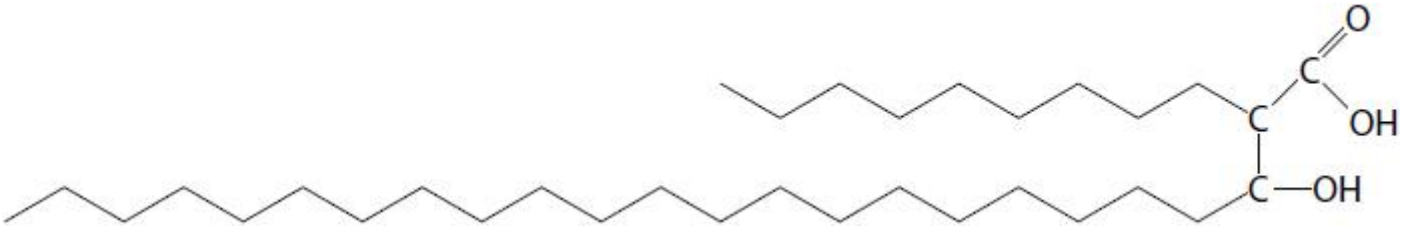
Q33.

Tuberculosis is a disease caused by infection with *Mycobacterium tuberculosis*.

The cell wall of *M. tuberculosis* is different from most other types of bacteria as it contains different types of mycolic acid.

Mycolic acid protects the bacteria from lysozyme action, dehydration and polar (hydrophilic) antibiotics.

The diagram shows part of one type of mycolic acid.



(a) What type of molecule is mycolic acid?

(1)

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- ☐ **A** carbohydrate
- ☐ **B** fatty acid
- ☐ **C** nucleic acid
- ☐ **D** polypeptide

(b) Treatment for tuberculosis involves the use of different types of antibiotic, taken over several months.

The table gives details of four antibiotics used to treat tuberculosis.

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Antibiotic	Mode of action
E	disrupts the formation of the cell wall
J	inhibits the synthesis of mycolic acids
P	activates an enzyme that inhibits the synthesis of fatty acids
R	binds to the active site of RNA polymerase

(i) Suggest why antibiotic E is effective only when *M. tuberculosis* bacteria are dividing.

(1)

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(ii) Explain why antibiotic J could result in an increase in antigen presentation by macrophages.

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(iii) Explain how antibiotic P could affect *M. tuberculosis*.

(3)

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(iv) Explain how antibiotic R could affect *M. tuberculosis*.

(3)

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(v) Explain why taking a combination of antibiotics for several months could increase the resistance of *M. tuberculosis* to these antibiotics.

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(Total for question = 12 marks)

Q34.

Infection with *Mycobacterium tuberculosis* results in tuberculosis (TB), which is one of the top ten causes of death in the world.

(a) In 2019, an estimated 1.2 million children, 3.2 million women and 5.6 million men developed TB.

Calculate the ratio of children to women to men who developed TB.

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(1)

Answer .....

(b) Explain how infection with *Mycobacterium tuberculosis* may result in death.

(3)

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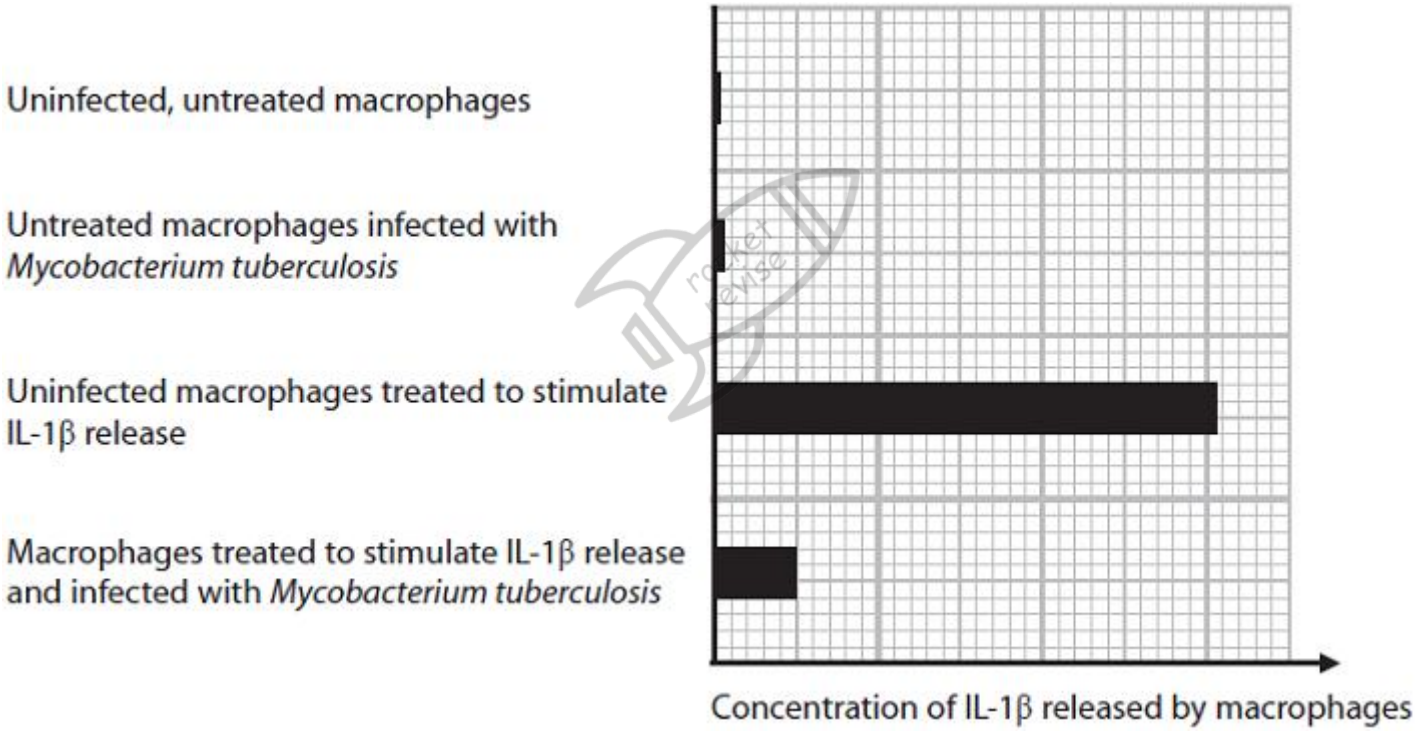
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(c) *Mycobacterium tuberculosis* has evolved to evade host immunity by interfering with macrophage function.

In one investigation, scientists showed that *Mycobacterium tuberculosis* can inhibit the activation of a process in macrophages. This inhibition results in less of the chemical IL-1β being released.

(i) The graph shows some of the results of this investigation.



Explain why the macrophages had to be treated to stimulate IL-1β release in this investigation. Use the information in the graph to support your answer.

(2)

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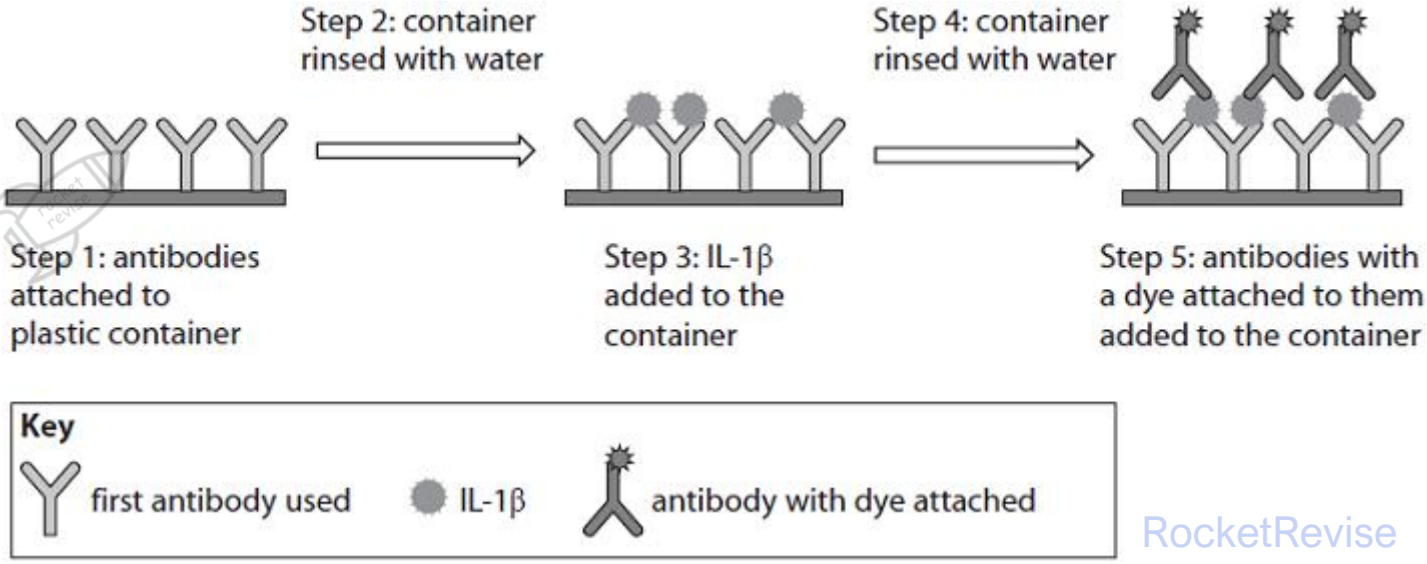
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(ii) An ELISA is used to detect a specific antigen.
The IL - 1β was detected using ELISA.
The diagram shows an ELISA.

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Explain how the steps in the ELISA result in the identification of IL-1β.
Use the information in the diagram to support your answer.

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(iii) The scientists identified a protein that could be responsible for inhibiting the activation of the macrophages.

They used *Mycobacterium tuberculosis* with a mutation in the gene coding for this protein as part of their investigation.

This involved comparing the growth rate of these mutant bacteria with the unmutated bacteria.

Describe how optical methods (turbidity) can be used to compare the growth rates of these two populations of bacteria.

(4)

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(iv) Suggest how this protein, released by the *Mycobacterium tuberculosis*, could be used to develop a treatment for TB.

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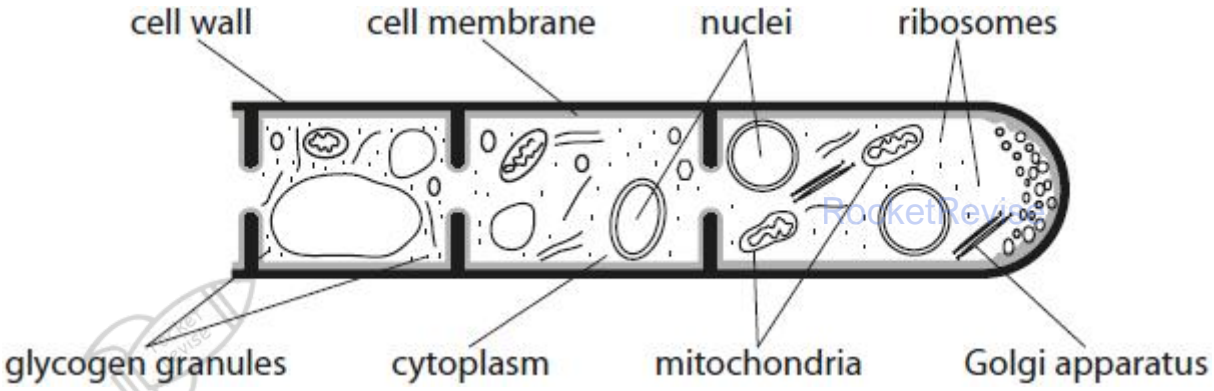
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(Total for question = 15 marks)

Q35.

Bacteria and fungi can both cause infections.

(a) The diagram shows part of the structure of a fungus.



(i) How many of the organelles labelled in this diagram have membranes?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 7

(ii) Explain why fungi are not classified as bacteria, plants or animals.
Use the information in the diagram and your knowledge of cell structure to support your answer.

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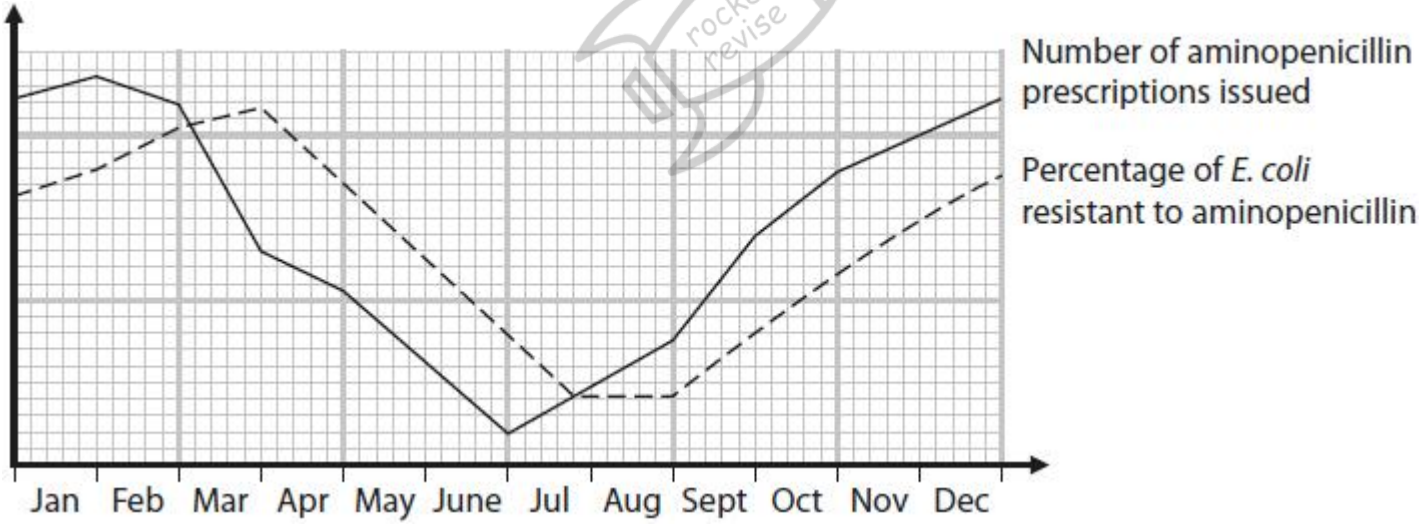
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(b) The graph shows the number of prescriptions of the antibiotic, aminopenicillin, issued in one year.

The graph also shows the percentage of *E. coli* bacteria that were resistant to aminopenicillin during the same year.



Explain the relationship between the number of aminopenicillin prescriptions issued and the percentage of *E. coli* resistant to aminopenicillin during this one-year period.
Use the information in the graph to support your answer.

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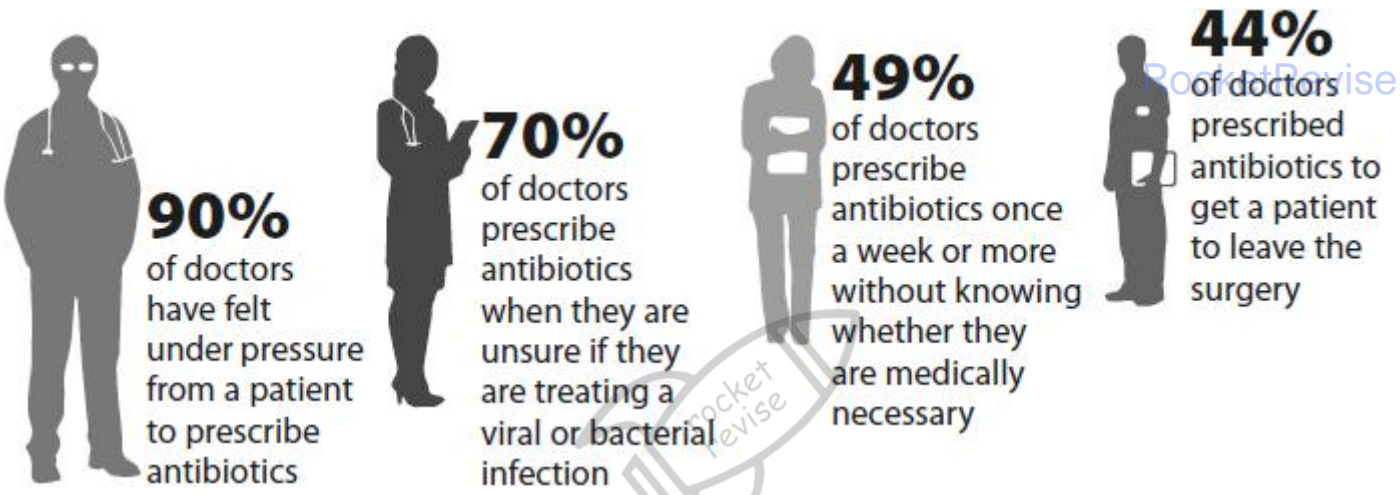
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(c) The results of a survey are shown in the diagram.



Explain why the results of this survey are a cause for concern.

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(Total for question = 11 marks)

Q36.

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In 2018, an organisation carried out a global survey into the attitudes of people towards vaccines.

Vaccines provide artificial active immunity.

(a) Describe how a vaccine results in active immunity.

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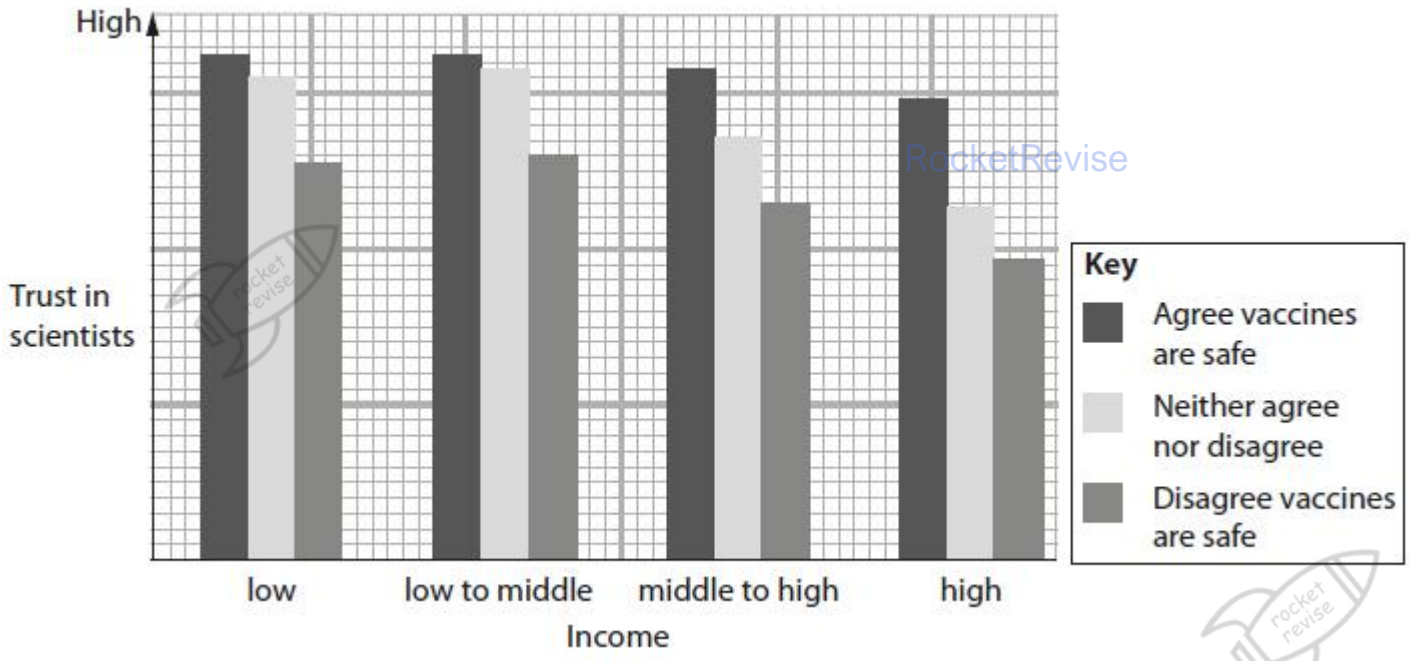
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(b) This survey included three aspects:

- the trust that people have in scientists
- their thoughts on the safety of vaccines
- their income.

The graph shows some of the results of this survey.



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(i) Identify **three** conclusions that can be drawn from the results of this survey.

(3)

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(ii) Suggest **one** reason for the different attitudes of these people.

(1)

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(c) The more people in a population who are vaccinated against a disease, the less likely it is for non-vaccinated people to become infected.

Suggest why vaccination is more successful when a greater proportion of people are vaccinated.

(2)

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(Total for question = 10 marks)

Q37.

Viruses can infect bacteria.

(a) Which virus can infect bacteria?

(1)

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☒ **A** Ebola virus

☒ **B** human immunodeficiency virus (HIV)

☒ **C** lambda phage (λ phage)

☒ **D** tobacco mosaic virus (TMV)

(b) Some viruses that infect bacteria cause the production of molecules called holins.

Holins form protein channels in the cell membranes of bacteria. This allows polar molecules called lysins to reach the cell wall by facilitated diffusion.
The DNA of these viruses codes for lysins.

(i) Describe the role of channel proteins in the facilitated diffusion of lysins.

(2)

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(ii) Explain how the primary structure and the tertiary structure of holins determine the properties of these channel proteins.

(3)

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(iii) Lysins break down the cell wall of bacteria.
Explain the role of lysins in the lytic cycle.

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(Total for question = 8 marks)

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Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The following features of viruses can be used to classify them:

- the type of nucleic acid they contain
- the arrangement of the proteins in their capsid
- the presence or absence of an envelope
- the enzymes they contain.

(a) For each structure, put one cross ☒ in the appropriate box, in each row, to show which viruses have that structure.

(3)

Structure	Virus			
	Ebola only	HIV only	both Ebola and HIV	neither Ebola nor HIV
DNA	☐	☐	☐	☐
helical capsid	☐	☐	☐	☐
envelope	☐	☐	☐	☐

(b) Human cells contain several types of DNA polymerase.

One type of DNA polymerase, found in some cancer cells, has been shown to synthesise DNA from an RNA template.

(i) Name the enzyme, found in some types of virus, that can synthesise DNA from an RNA template.

(1)

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(ii) Suggest why this DNA polymerase may be a target for drugs used to treat cancer.

(1)

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(Total for question = 5 marks)

Q39.

Two students, student P and student Q, each cultured bacteria in a liquid medium for 20 hours.

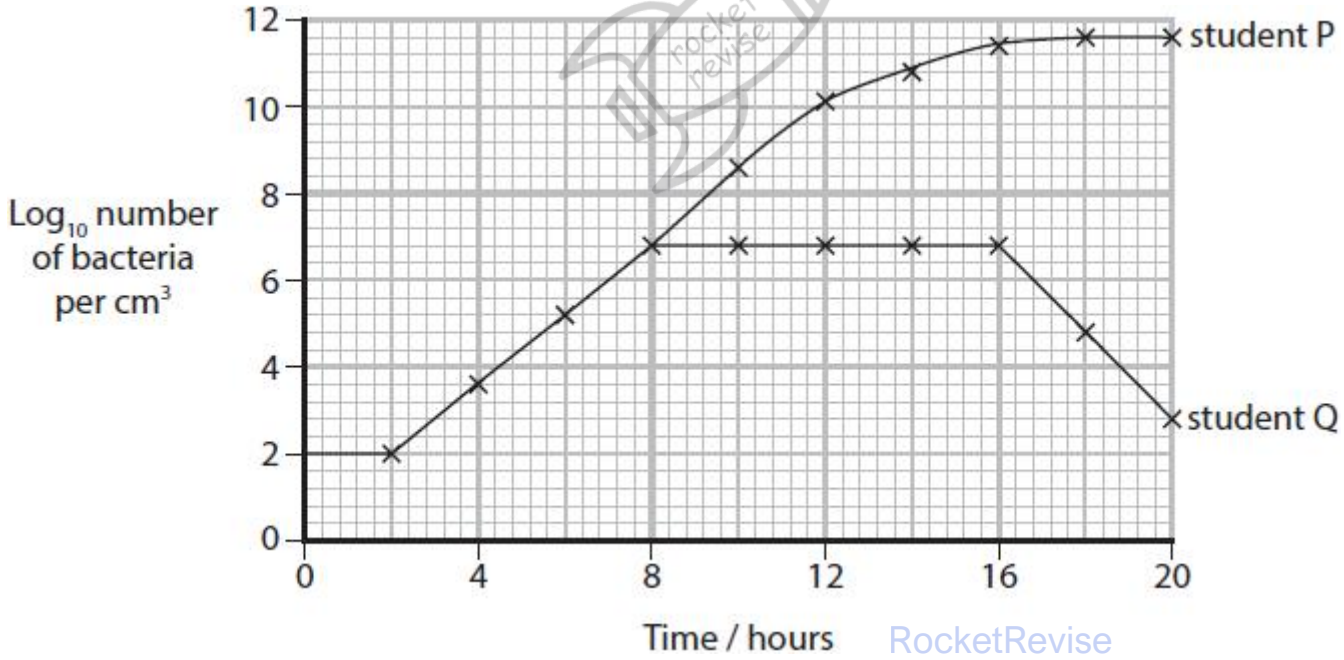
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The culture conditions used by both students were identical.

Every two hours, each student removed a sample of the culture to count the number of bacteria.

Each student used a different method to count the number of bacteria in the samples.

The graph shows the bacterial growth curve that each student drew using their results.



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(a) Both students used aseptic techniques to culture the bacteria.

(i) Explain the importance of using aseptic techniques to culture bacteria.

(2)

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(ii) Explain **one** aseptic technique that should be used when culturing bacteria.

(2)

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(b) Calculate the growth rate constant (k) of these bacteria between 2 hours and 6 hours of culture.

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(2)

Use the equation:

$$k = \frac{\log_{10}N_t - \log_{10}N_0}{0.301 \times t}$$

where:
$\log_{10}N_t$ = $\log_{10}$ number of cells at 6 hours
$\log_{10}N_0$ = $\log_{10}$ number of cells at 2 hours.

Answer .....

(c) The number of bacteria in the samples taken after 10 hours were:

398 107 171 in the culture of student P
6 309 573 in the culture of student Q.

(i) Calculate the ratio of the number of bacteria in these two cultures.

(1)

Answer ..... :1

(ii) Student P and student Q used different methods for determining the number of bacteria.
State which method each student used.
Give a reason for your answer.

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(2)

Student P .....

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Student Q .....

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(Total for question = 9 marks)

Q40.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Herpes simplex is a virus that causes sores around the nose and mouth.

A modified form of this virus has been developed to infect and kill cancer cells in humans.

(a) Explain how a cell can become a cancer cell.

(2)

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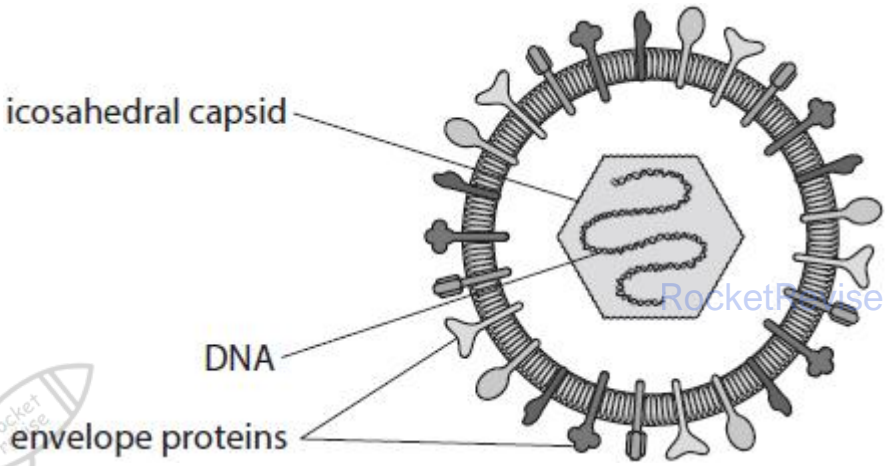
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(b) The diagram shows a herpes simplex virus.



(i) Which of the following statements are true?

(1)

1 Ebola virus has the same type of genetic material as herpes simplex virus
2 Both Ebola virus and human immunodeficiency virus (HIV) have envelope proteins
3 HIV has an icosahedral capsid

☒ **A** Statements 1 and 2 only
☒ **B** Statements 1 and 3 only
☒ **C** Statements 2 and 3 only
☒ **D** Statements 1, 2 and 3

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(ii) State the function of the envelope proteins.

(1)

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(iii) Explain why the herpes simplex virus has to be modified before it can be used to infect cancer cells.

(2)

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(iv) Explain how the infection of a cancer cell with herpes simplex virus can result in the death of this cell.

(2)

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(c) Using a modified herpes simplex virus activates the immune system.

Explain why activation of the immune system would result in destruction of the cancer cells.

(4)

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(Total for question = 12 marks)

Q41.

Individuals develop artificial immunity either actively or passively.

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(a) The table gives some statements about immunity.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show whether these statements are true for these types of immunity.

(2)

Statement	Type of artificial immunity			
	both active and passive	active only	passive only	neither active nor passive
Antigens are injected into the person	☐	☐	☐	☐
Immunity is long term	☐	☐	☐	☐

(b) Vaccination against a virus results in the production of antibodies.

Describe how antibodies are produced following activation of T cells.

(2)

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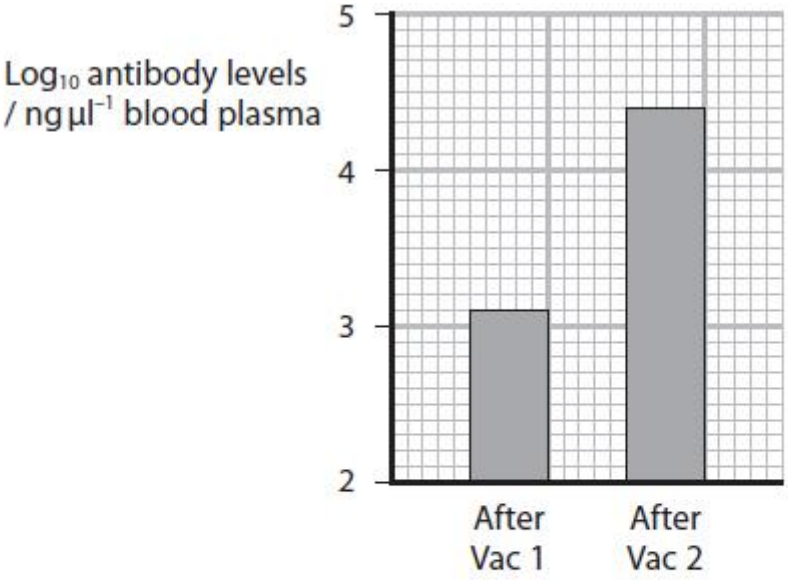
(c) The levels of antibodies following vaccinations were investigated.

There were two groups of people. Group 1 **had not** been infected with the virus, before the vaccine was given. Group 2 **had** been infected with the virus before the vaccine was given.
Both groups of people received two vaccinations, Vac 1 and Vac 2.

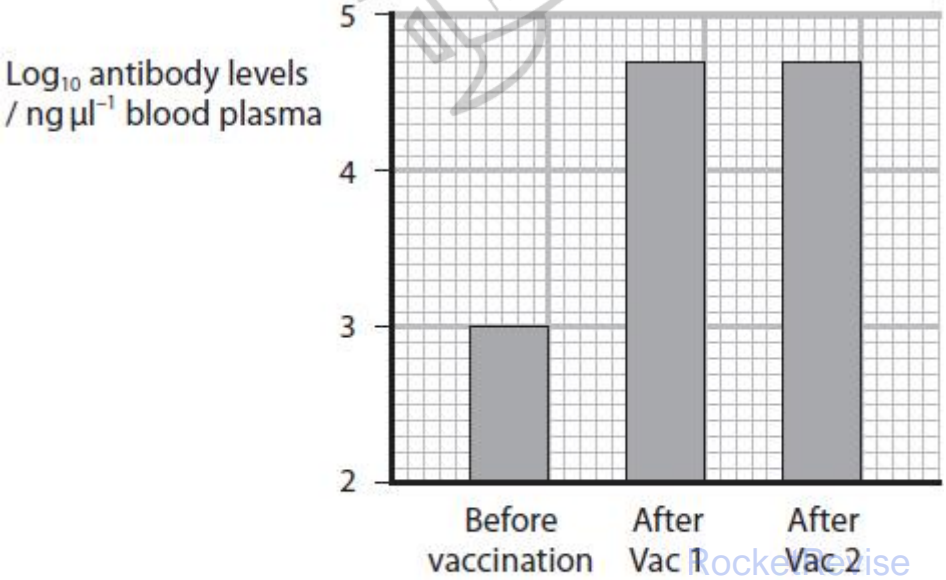
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The graphs show the results of this investigation.



Group 1:
People who had not been infected before the vaccine was given



Group 2:
People who had been infected with the virus before the vaccine was given

(i) Explain **two** ways in which this investigation would have to be controlled to provide valid data.

(2)

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(ii) In group 1, Vac 2 increased the levels of antibody more than Vac 1.
Calculate how many times greater this increase was.

(2)

Answer .....

(iii) Suggest why there is no data on the graph for group 1 before vaccination.

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(1)

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*(iv) Explain the results of this investigation.

(6)

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(Total for question = 15 marks)

Q42.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Skin flora provide a barrier, protecting the body from infection by bacteria.

(a) Which of the following also act as barriers, protecting the body from **infection** by bacteria?

(1)

- ☐ **A** antigens and stomach acid
- ☐ **B** antigens and interferon
- ☐ **C** interferon and skin
- ☐ **D** skin and stomach acid

(b) The diagrams show the effects of different methods of washing hands on skin flora.

The dark areas show the presence of microorganisms and the light areas show clean areas.

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1. before washing



2. "rinse and shake"



3. six seconds, no soap



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4. six seconds with soap



5. fifteen seconds with soap



6. thirty seconds with soap



(i) Describe **two** conclusions that can be made about the different methods of washing hands.

(2)

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(ii) Explain the differences in the effects that these methods of washing hands have on the skin flora.

(2)

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(Total for question = 5 marks)

Q43.

Mycobacterium tuberculosis (Mtb) and human immunodeficiency virus (HIV) are two pathogens that infect people.

When people infected with either pathogen become infected with the other pathogen the disease is worse.

(a) Explain why HIV infection causes tuberculosis (TB) to be worse.

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(4)

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(b) In one year, it was estimated that 33.4 million people were infected with HIV.

It was estimated that 30% of these people were also infected with *Mtb*.

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(i) Suggest **two** reasons why these figures are only estimates.

(2)

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(ii) Calculate the number of people with both infections.
Give your answer in standard form.

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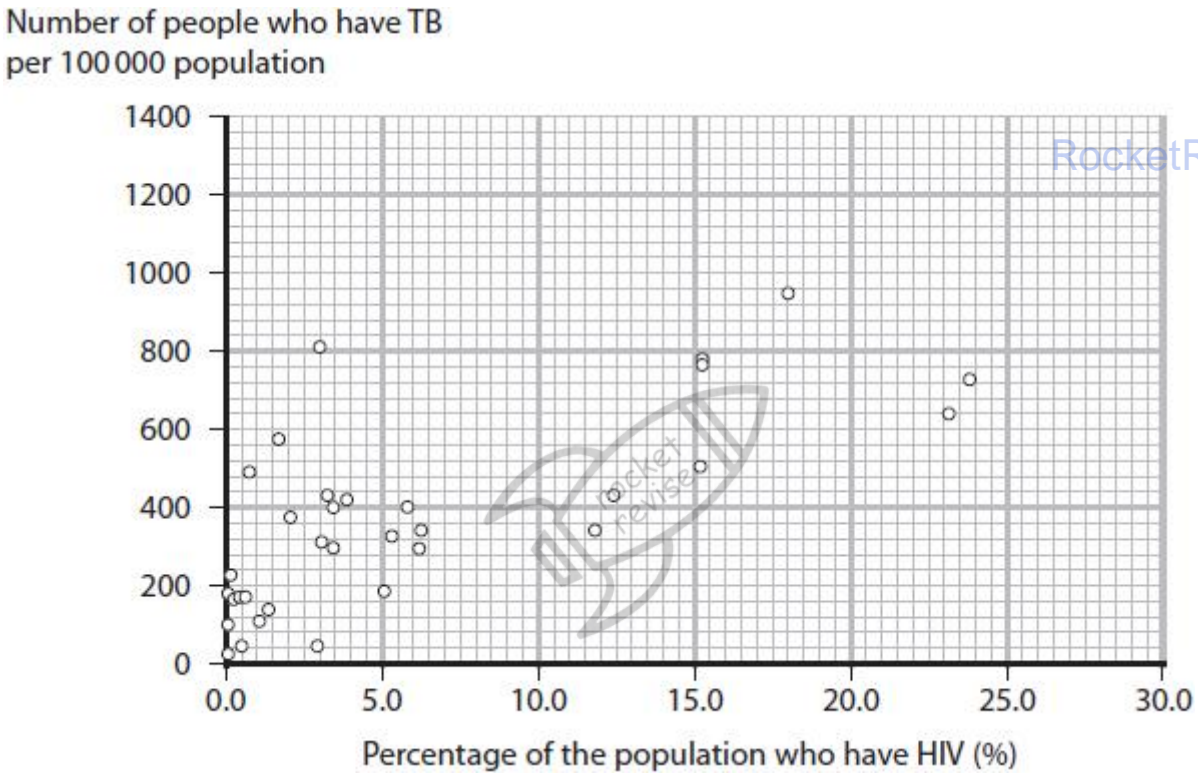
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(1)

Answer .....

(c) The graph shows the number of people who have TB and the percentage of the population who have HIV.

Each plotted point represents data from a different country.



(i) Comment on the relationship between the number of people who have TB and the percentage of people who have HIV.

(2)

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(ii) Describe how the strength of the relationship between the number of people who have TB and the percentage of people who have HIV could be determined.

(3)

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(Total for question = 12 marks)

Q44.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Aquaculture is the controlled cultivation of aquatic organisms such as fish, shellfish and plants. It is an important source of food in many countries of the world.

The photograph shows freshwater aquaculture ponds in Africa.



(Source: © E.Westmacott/Alamy Stock Photo)

Vibrio are bacteria that are common in aquatic habitats. They cause diarrhoea and cholera and are also pathogenic to shellfish.

The resistance of *Vibrio* to antibiotics and to heavy metals in aquaculture ponds in Nigeria was investigated.

(a) Samples of water were taken from 120 ponds. Of these, 74 of the ponds had been treated previously with antibiotics.

Which of the following is the percentage of **untreated** ponds?

(1)

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- **A** 38
- **B** 61
- **C** 62
- **D** 163

(b) *Vibrio* were grown on agar containing different heavy metals at a range of concentrations.

The highest concentration of heavy metal tested was 3.2 mg cm^{-3} and the lowest concentration was 6.25 µg cm^{-3}.

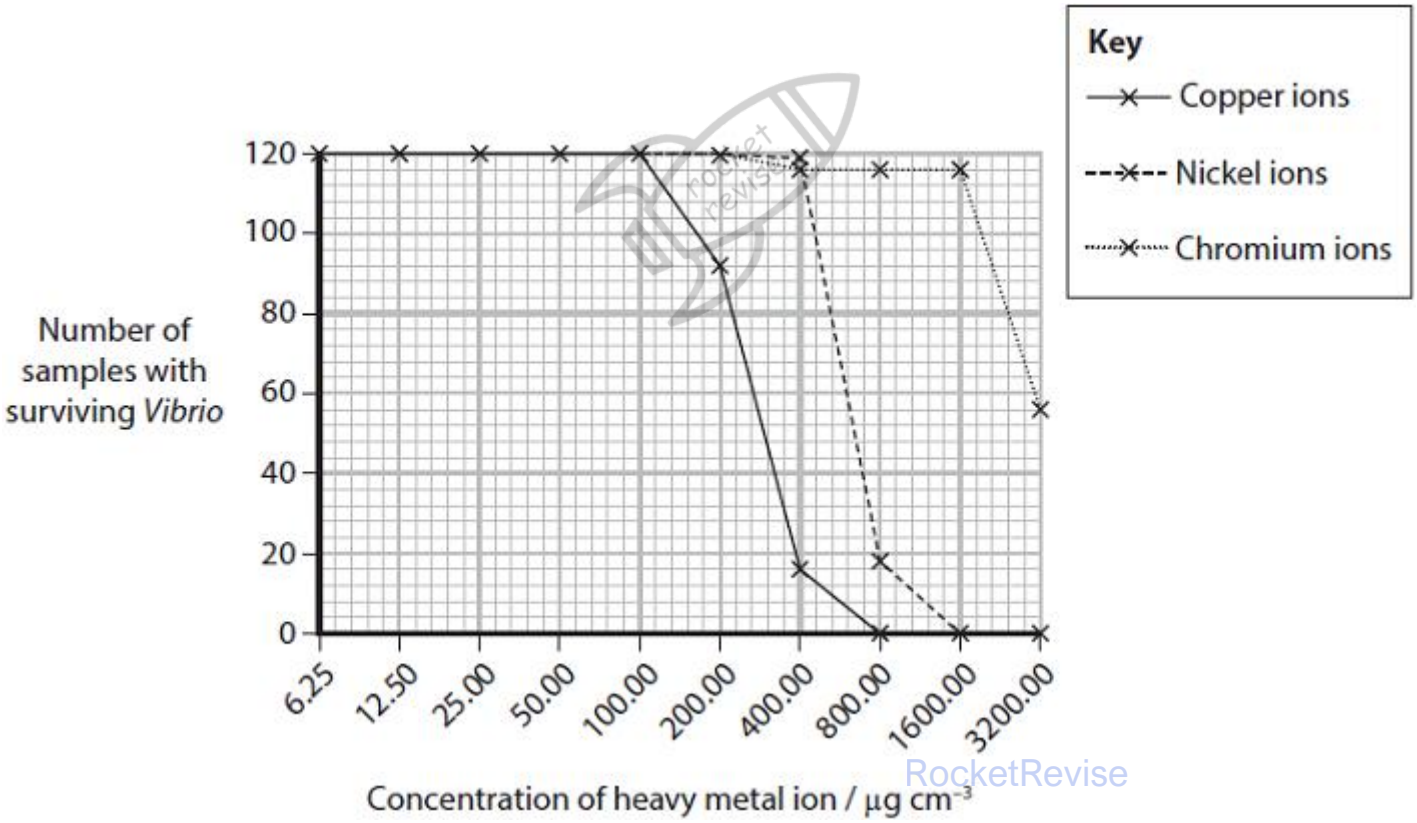
(i) How many times does the highest concentration need to be diluted to make the lowest concentration?

(1)

- **A** 5.12
- **B** 51.2
- **C** 512
- **D** 5120

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(ii) The graph shows the effect of different concentrations of heavy metals on the survival of the *Vibrio* collected from each of the 121 ponds.



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Describe the conclusions that can be made about the effects of heavy metal ions on the survival of *Vibrio*.

(3)

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(c) The samples of *Vibrio* taken from each pond were spread on agar containing different antibiotics.

(i) The scientists expected the percentage of resistant bacteria to be higher in the samples of *Vibrio* taken from the ponds previously treated with antibiotics.
Explain why the scientists expected this difference.

(3)

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(ii) Some of the antibiotics used were bacteriostatic and some were bactericidal.
State the difference between a bacteriostatic antibiotic and a bactericidal antibiotic.

(1)

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(iii) The method did not distinguish between bacteriostatic and bactericidal antibiotics because no colonies grew.
Suggest how the method would need to be modified to show which antibiotics were bacteriostatic and which were bactericidal.
Give a reason for your answer.

(2)

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(Total for question = 11 marks)

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Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Respiratory syncytial virus (RSV) usually causes mild, cold-like symptoms from which most people recover.

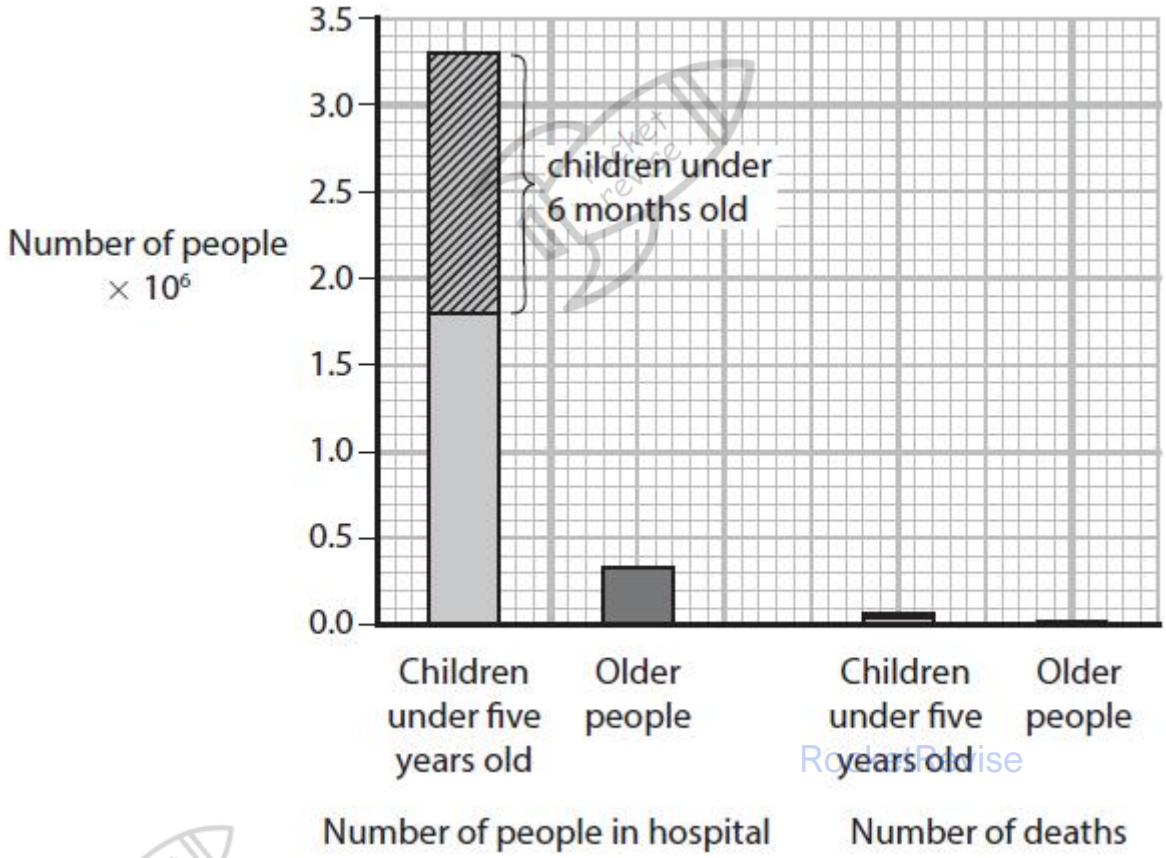
Infection with RSV can be serious in young children and older people.

Currently, there are no approved vaccines against RSV, although there are some in clinical trials.

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(a) The graph shows the number of people who had to go to hospital in one country in one year.

The graph also shows the number of deaths in children under five years old and in older people in the same country in that year.



(i) Which is the proportion of the children under five in hospital who are under 6 months old?

(1)

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☒ **A** $\frac{5}{11}$

☒ **B** $\frac{6}{11}$

☒ **C** $\frac{5}{6}$

☒ **D** $\frac{6}{5}$

(ii) Which would be a more appropriate scale to use on the y-axis so that the range of values for the number of deaths can be seen clearly?

(1)

☒ **A** linear
☒ **B** logarithmic
☒ **C** standard deviation
☒ **D** standard form

(b) In one trial, people received a new vaccine against RSV.

The results showed that this vaccine protected against two subtypes of RSV.
Give **two** reasons why one vaccine can protect against two subtypes of RSV.

(2)

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(c) In another trial, pregnant women were given a vaccine against RSV.

This vaccine did not cross from the mother into the developing fetus.
(i) As a result of passive immunity, the babies of these pregnant women were born with high levels of antibodies against RSV.
Explain how passive immunity developed in these babies before they were born.

(5)

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(ii) The scientists running this trial claimed that the vaccine was 85 % successful at protecting babies against RSV infections that required medical attention.

Describe how the data could have been collected to support this claim.

(3)

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(Total for question = 12 marks)

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Q46.

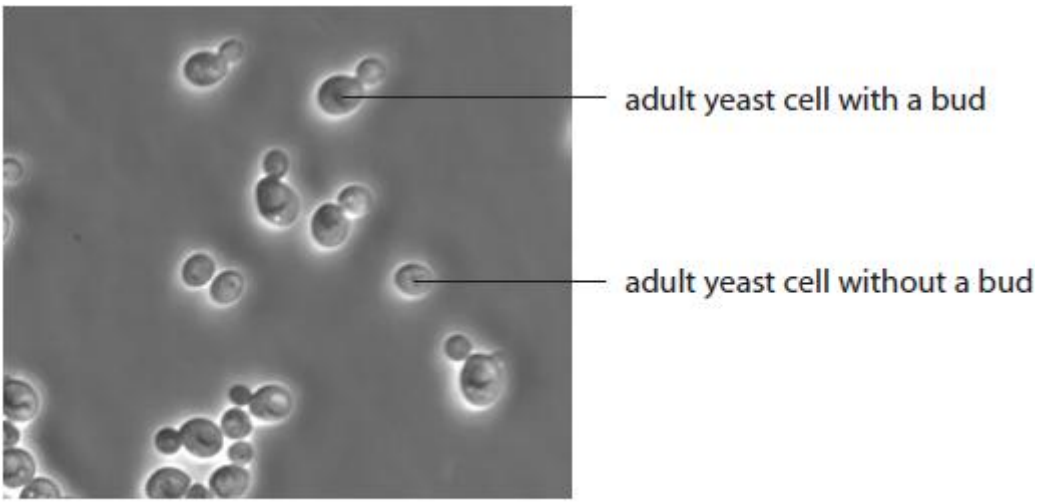
Yeast are single-celled eukaryotic microorganisms.

They reproduce asexually by budding. The new yeast cell (bud) grows out of the cell of an adult yeast cell following mitosis.

The photograph shows some yeast cells.

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(Source: © Andre Nantel/Shutterstock)

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(a) One species of yeast, *S. cerevisiae*, has a diameter from 5 μm to 10 μm.

One type of bacteria, *S. aureus*, has a diameter from 500 nm to 1500 nm.

(i) Calculate the smallest difference between the diameters of *S. cerevisiae* and *S. aureus*.
Give suitable units with your answer.

(1)

Answer .....

(ii) The table gives some statements about cells.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show whether these statements are true for the types of microorganisms shown.

(2)

Statement about cells	Type of microorganism			
	both *S. cerevisiae* and *S. aureus*	*S. cerevisiae* only	*S. aureus* only	neither *S. cerevisiae* nor *S. aureus*
Contain both DNA and RNA	☐	☐	☐	☐
Have membranes around the cytoplasm and around the nucleus	☐	☐	☐	☐

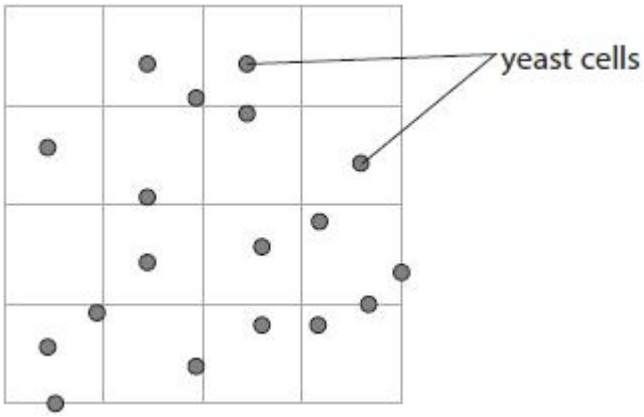
(b) Yeast cells can be cultured and the rate of growth determined using a haemocytometer (counting chamber / Neubauer chamber).

(i) A student had a suspension of yeast cells in a liquid culture and wanted to determine the concentration of yeast cells in this culture.
A sample of this culture was added to the haemocytometer and observed under a light microscope.

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The diagram shows yeast cells in part of the haemocytometer.



The volume of the sample covering this part of the haemocytometer is 0.00625 µl.

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1 µl = 10^{-6}dm^3

Explain how to calculate the concentration of yeast cells, in cells per cm^3, in the culture. Include in your answer an explanation of how the number of cells was determined from the diagram of a haemocytometer.

(3)

Explanation .....

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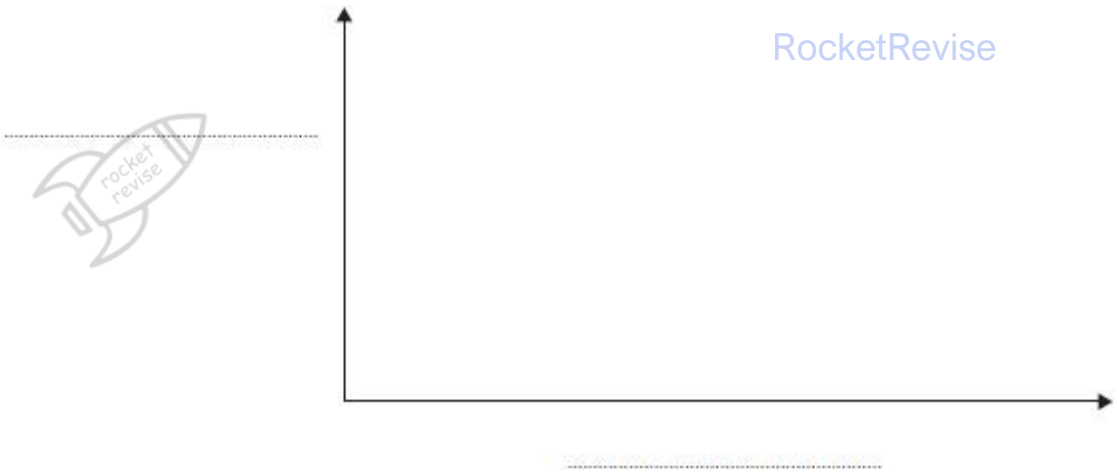
Answer ..... cells per cm^3

(ii) The growth curve for yeast grown in culture has four phases, similar to bacteria grown in culture.

Sketch a growth curve on the axes drawn below.
Label the axes and the four phases of the growth curve.

(4)

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(iii) Describe how the student could use a growth curve to determine the exponential growth rate constant for the yeast.

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(Total for question = 14 marks)

Q47.

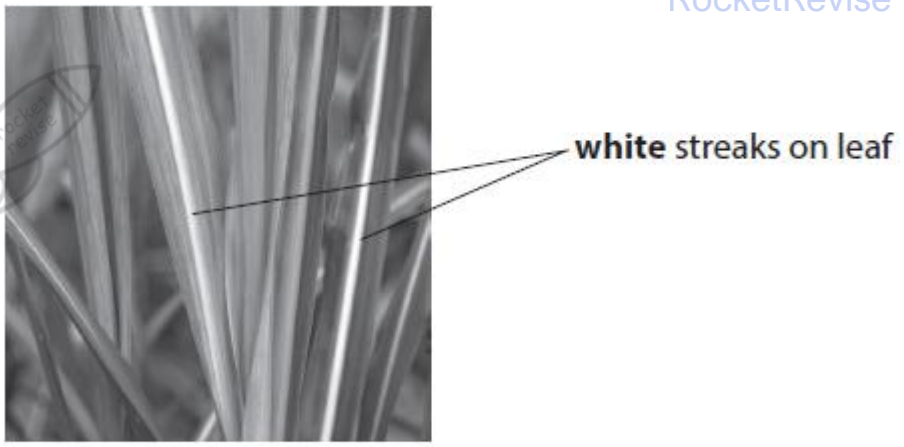
Leaf scald is a disease of sugarcane caused by bacteria.

These bacteria are found in the **xylem** and release chemicals, one of which is called albicidin.

Some of these chemicals stimulate the xylem vessels to produce a sticky gum.

Albicidin inhibits chloroplast development.

The photograph shows the appearance of part of a sugarcane leaf infected with these bacteria.



(Source: © Nigel Cattlin / Alamy Stock Photo)

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(a) Explain how infection with these bacteria results in the death of the leaf cells and eventually the whole plant.

(4)

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(b) Albicidin is an inhibitor of the enzyme DNA gyrase.

This enzyme is present in prokaryotic cells.
It binds to DNA and causes the DNA to supercoil, a process needed for cells to function properly.

(i) Describe **two** differences in the structure of DNA found in prokaryotic cells and the structure of DNA found in human cells.

(2)

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(ii) Albicidin is being developed as an antibiotic to human pathogens.
Suggest why albicidin can be used as an antibiotic.
Use the information in the question to support your answer.

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(iii) Explain the importance of developing new antibiotics, such as albicidin.

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(Total for question = 11 marks)

Q48.

A person's immune response to viral infections involves T killer cells.

Cells infected with viruses present viral antigens to the T killer cells.

(a) Peptide fragments bind to MHC molecules to form an antigen-MHC complex. This takes place in the rough endoplasmic reticulum (rER).

The complex is then transported to the surface of the infected cell.
Describe how the antigen-MHC complex is transported from the rER onto the outside of the surface membrane of the infected cell.

(2)

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(b) An electron microscope with a resolution of 3.7 Å can be used to study antigen-MHC complexes.

1 µm = 1 × 104 Å

(i) Calculate the resolution of this electron microscope in nanometres.

(1)

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Answer ..... nm

(ii) State what is meant by a resolution of 3.7 Å

(1)

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(c) Explain why antigens need to be presented to T killer cells in a person's immune response to viral infections.

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(d) Explain how the action of T killer cells results in a person recovering from a viral infection.

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(Total for question = 9 marks)

Q49.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The development of antibiotic resistance in bacteria is a global concern.

A new type of drug is being developed against bacteria.

This drug has a number of effects on bacteria including making the bacteria more susceptible to antibiotics.

This drug is not an antibiotic.

(a) State the meaning of the term **antibiotic**.

(1)

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(b) In 2019, 1.27 million deaths were due to antibiotic resistance.

It is estimated that this will rise to 10 million per year in 2050.
Which is the percentage increase in the number of deaths due to antibiotic resistance?

(1)

- ☐ **A** 13%
- ☐ **B** 87%
- ☐ **C** 687%
- ☐ **D** 787%

(c) Explain why it is important that these new types of drugs are developed.

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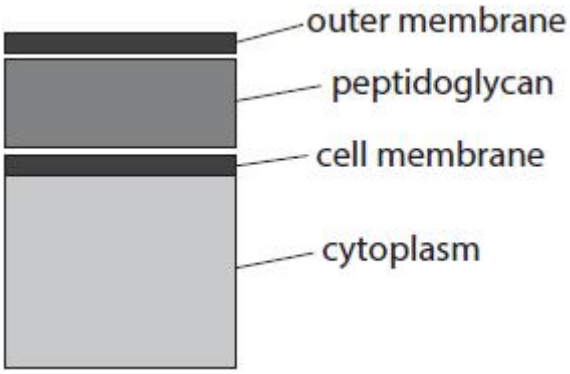
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(d) The diagram shows the structure of the cell wall of a bacterial cell. It also shows the cell membrane.

The molecules in the outer membrane vary in different types of bacteria.

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When bacteria were treated with the new drug, together with antibiotics, the antibiotics were effective at lower doses.
The new drug attached to the outer membrane.
Suggest why the new drug resulted in the antibiotics being effective at lower doses.

(2)

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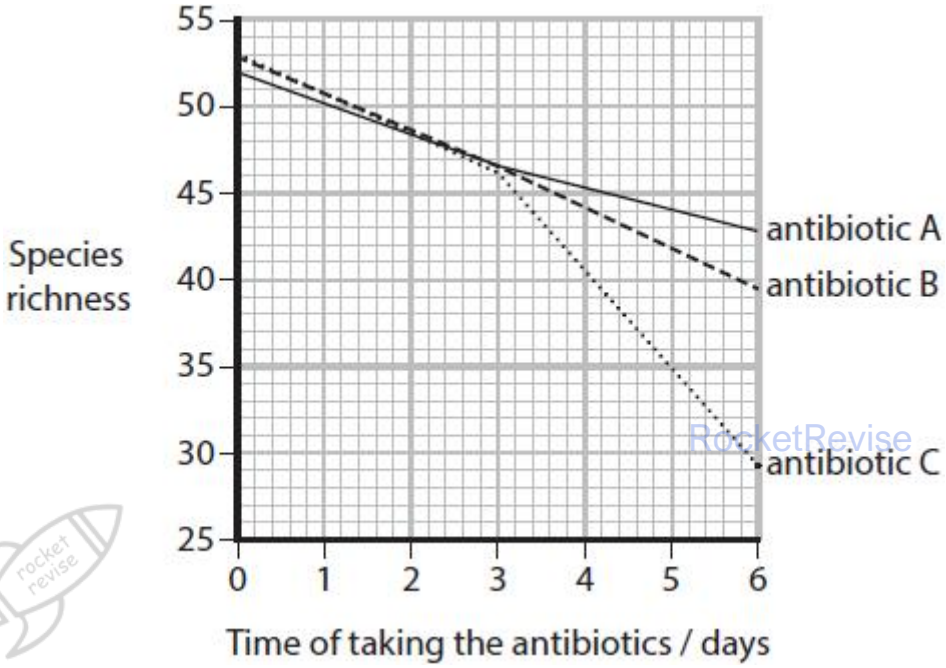
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(e) The graph shows the effects of three antibiotics on the species richness of gut flora.



(i) Describe **two** conclusions that can be made about the effects of these three antibiotics on the species richness of gut flora.

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(ii) Explain the advantages of adding this new drug when treating a bacterial infection with antibiotic C.

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(Total for question = 11 marks)

Q50.

Since the early 2020s, the number of cholera cases and deaths due to this disease are high as the world is facing its seventh pandemic of cholera.

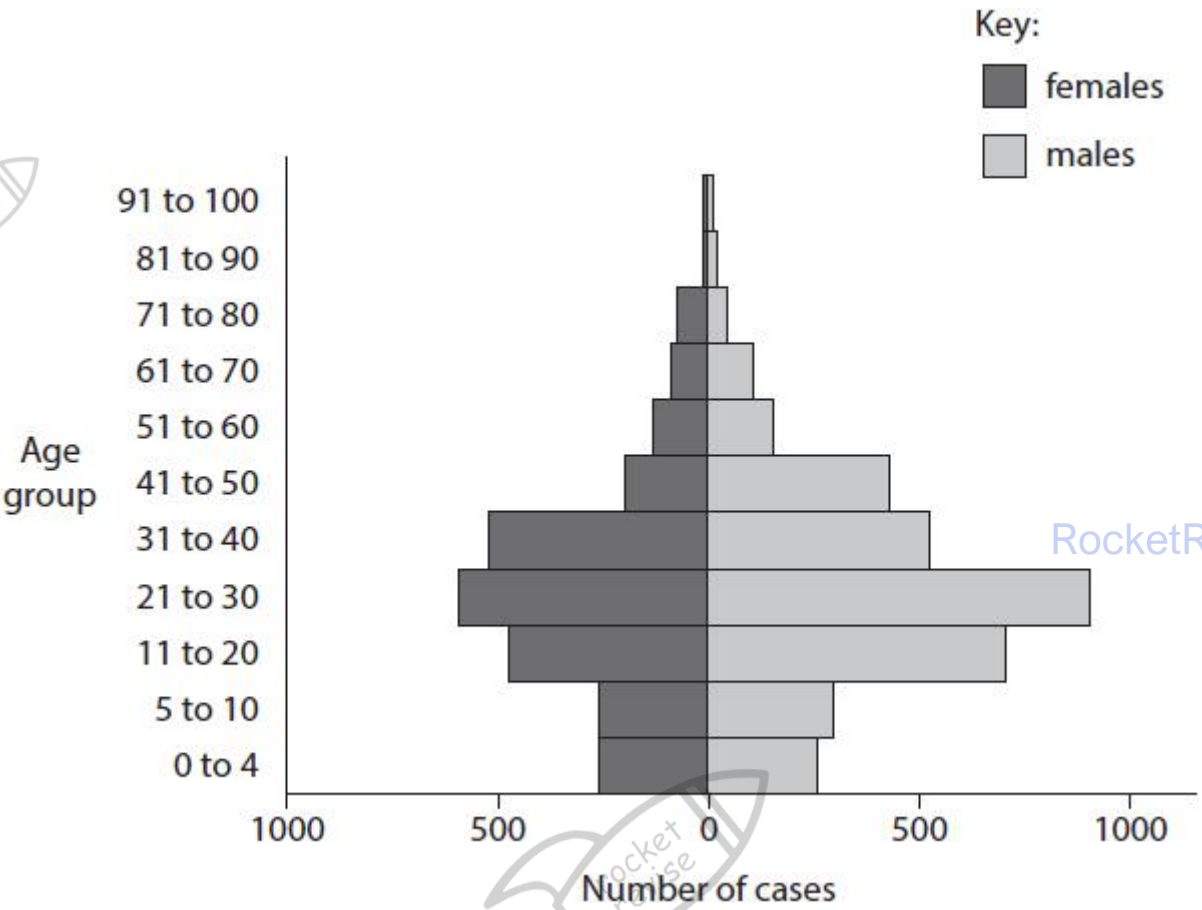
Cholera is caused by ingestion of food or water contaminated with the bacterium *Vibrio cholerae*.

(a) The diagram shows the distribution of cholera cases by age group and sex at birth in Malawi, in the first nine months of 2022.

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Describe **two** conclusions that can be made about the distribution of cholera cases by age group and sex.

(2)

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(b) Vibrio cholerae produce a toxin made up of several subunits.

Genetically modified rice plants are grown that produce rice grains that contain one of these subunits.
This subunit acts as a vaccine.
The rice grains were harvested and ground into a powder that can be eaten.
In a clinical trial, three groups of people were given one of three doses of this rice powder vaccine.
Another group of people were given a placebo.
The change in the levels of antibody against the subunit in their blood was measured over an eight-week period.
The graphs show the data obtained.

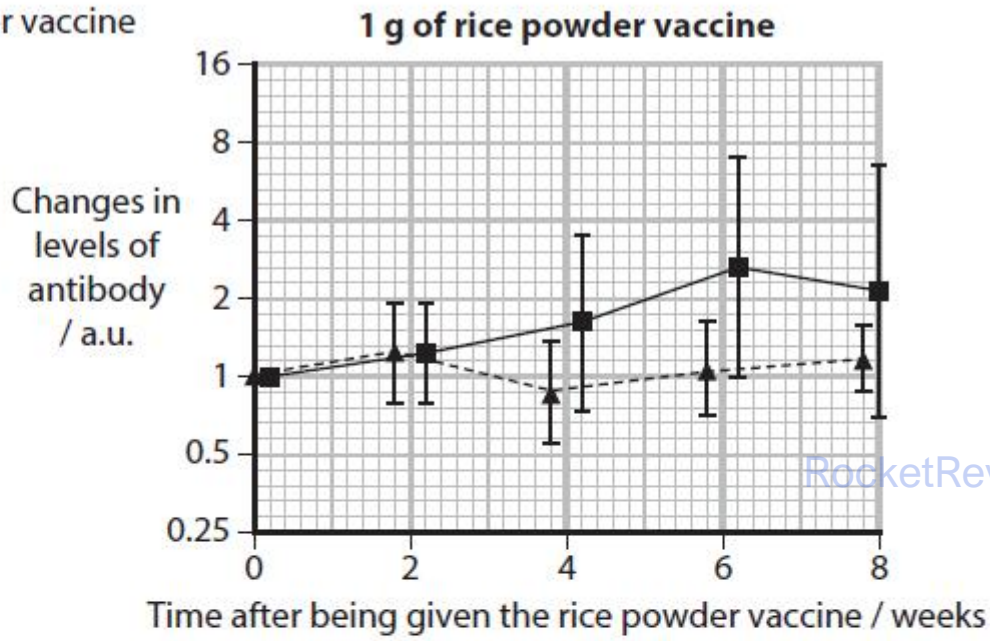
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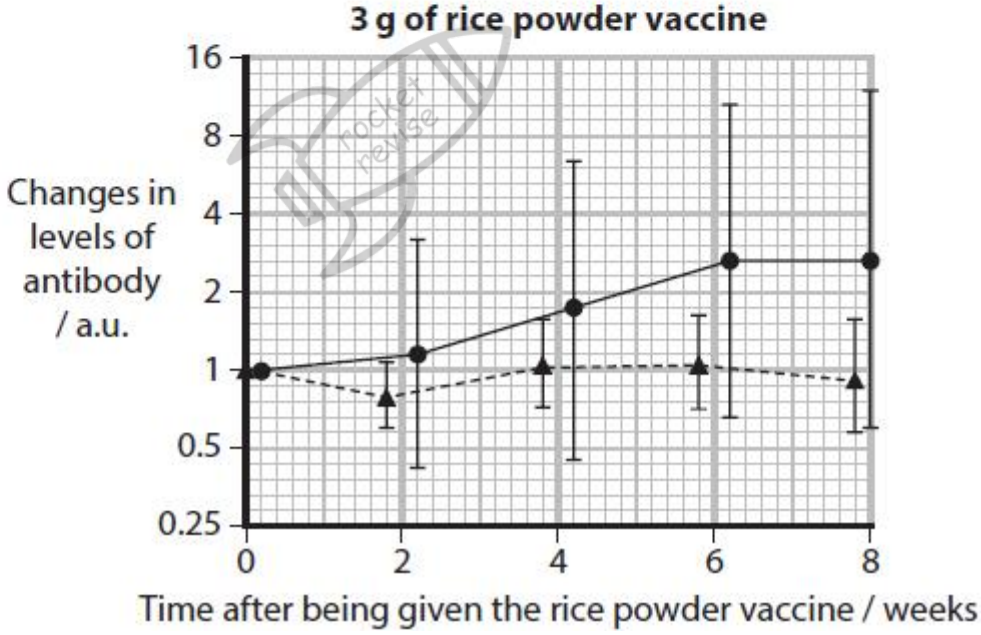
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Key:

——— rice powder vaccine

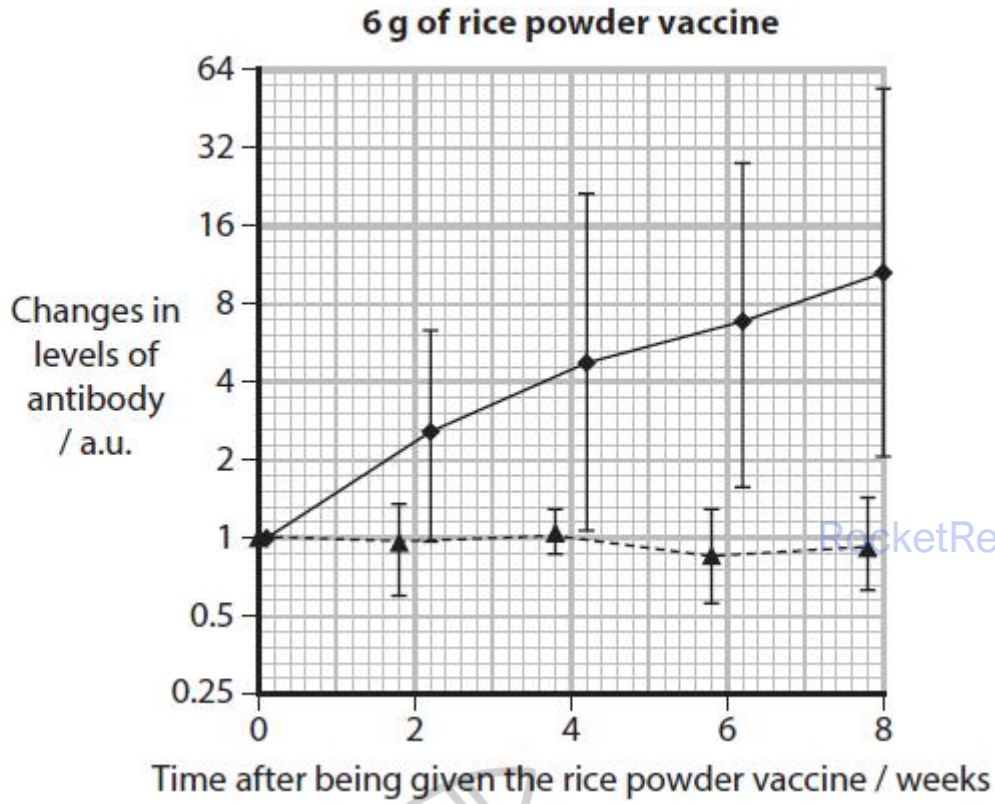
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(i) Describe suitable placebos for use in this clinical trial.
Use the information in the question to support your answer.

(2)

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(ii) The scientists running the trial concluded that a dose of 6 g of rice powder vaccine was needed to stimulate an immune response.
State why they came to this conclusion.

(1)

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(iii) The scientists found that these antibodies also bind to different toxins produced by another type of bacteria.
Explain why these antibodies, produced as a result of the rice powder vaccine, could bind to different toxins.

(2)

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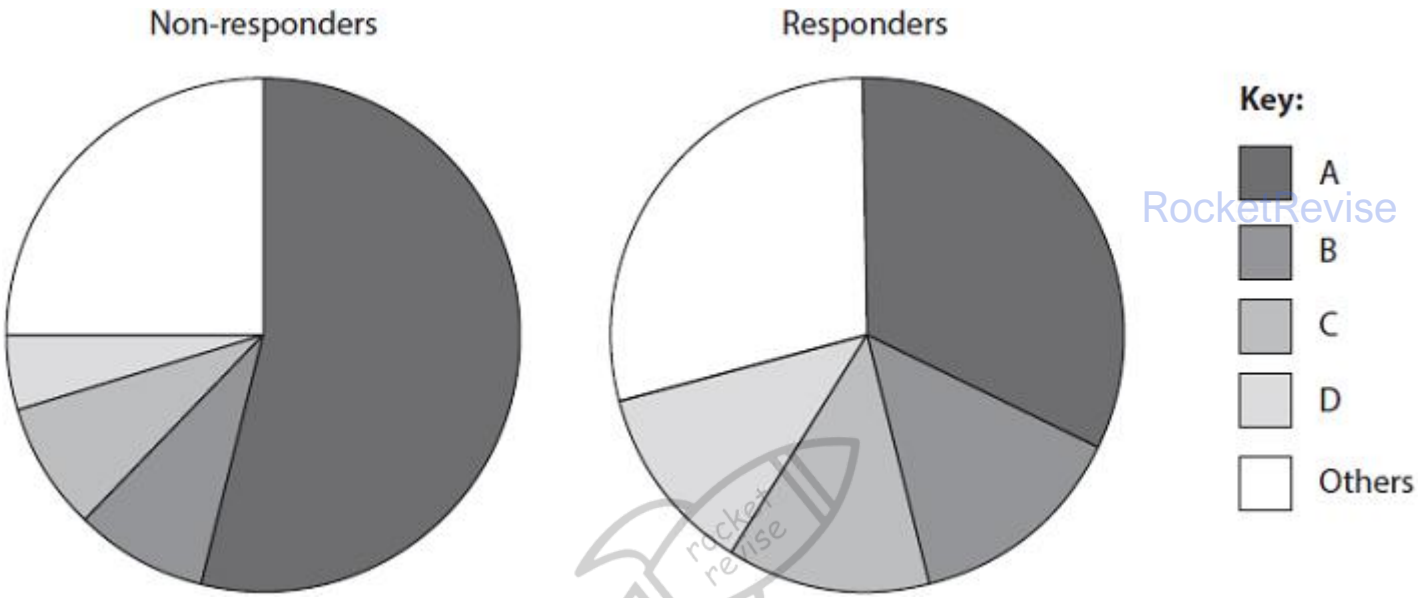
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(iv) Some people in the clinical trial did not respond to the rice powder vaccine.
The scientists determined the proportions of different types of bacteria (A, B, C, D and others) in the gut flora of these non-responders with those of the responders.
The charts show the results.



Compare and contrast the results for the non-responders and the responders.

(3)

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(v) Give **two** limitations when using this data to determine if the type of gut flora could cause a person to be a non-responder to the vaccine.

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(Total for question = 12 marks)

Q51.

Carbon dioxide, methane and nitrous oxide are greenhouse gases.

Nitrous oxide is produced by bacteria in the soil.

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(a) State **one** anthropogenic source of carbon dioxide and **one** different anthropogenic source of methane.

(1)

Carbon dioxide

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Methane

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(b) Describe how greenhouse gases cause global warming.

(3)

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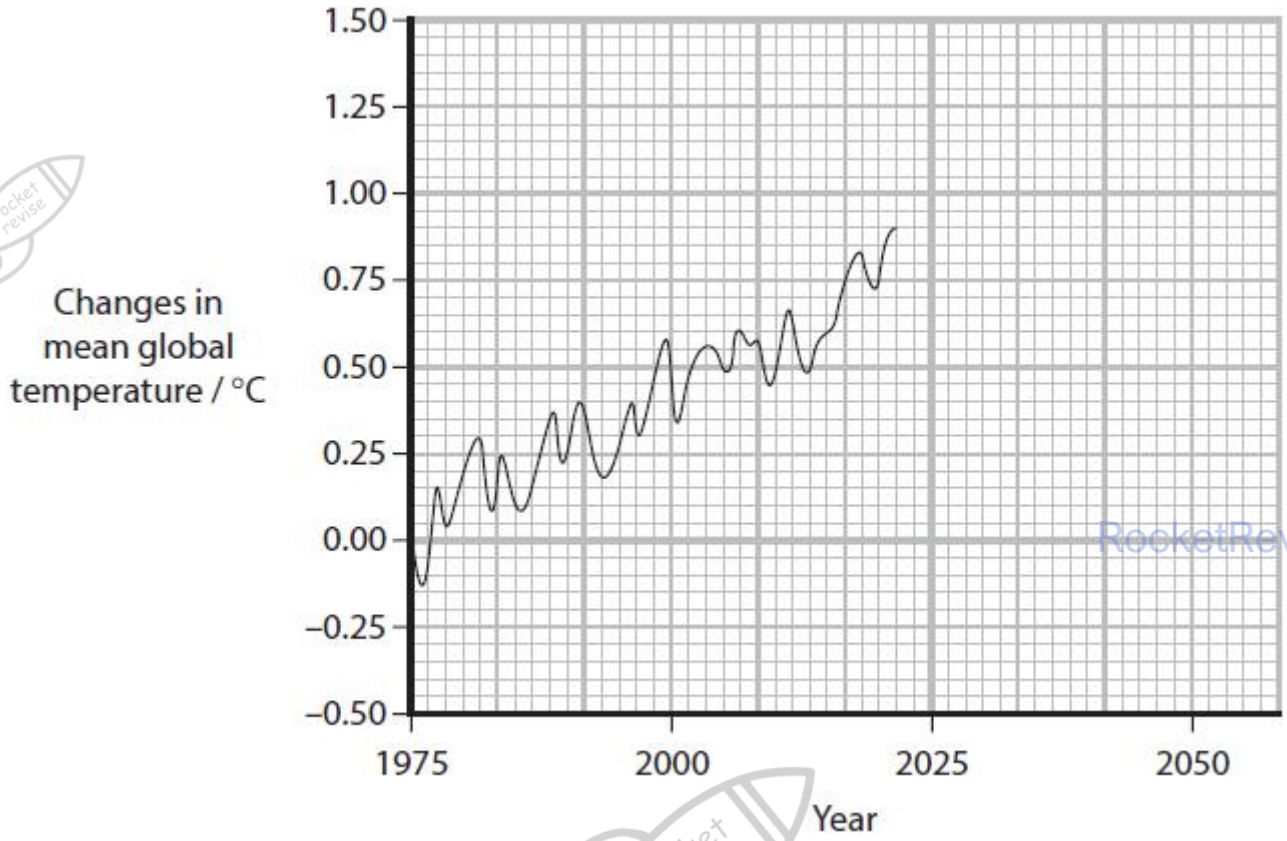
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(c) The graph shows the changes in mean global temperature from 1975 to 2023.

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Predict the change in the mean global temperature in 2050 from the data in this graph.

(1)

Answer ..... °C

(d) The greenhouse gases have different warming potentials (WP).

Over a 100-year timescale:

- one tonne of methane would generate 28 times the WP of one tonne of carbon dioxide
- one tonne of nitrous oxide would generate 265 times the WP of one tonne of carbon dioxide.

(i) Calculate the ratio of the WP of nitrous oxide to the WP of methane.
Give your answer to **two** significant figures.

(1)

Answer ..... : 1

(ii) Carbon dioxide can last in the atmosphere for thousands of years. Methane lasts for about 12 years and nitrous oxide for about 121 years.
Discuss the implications of this data and the WPs when scientists are deciding how to tackle climate change.

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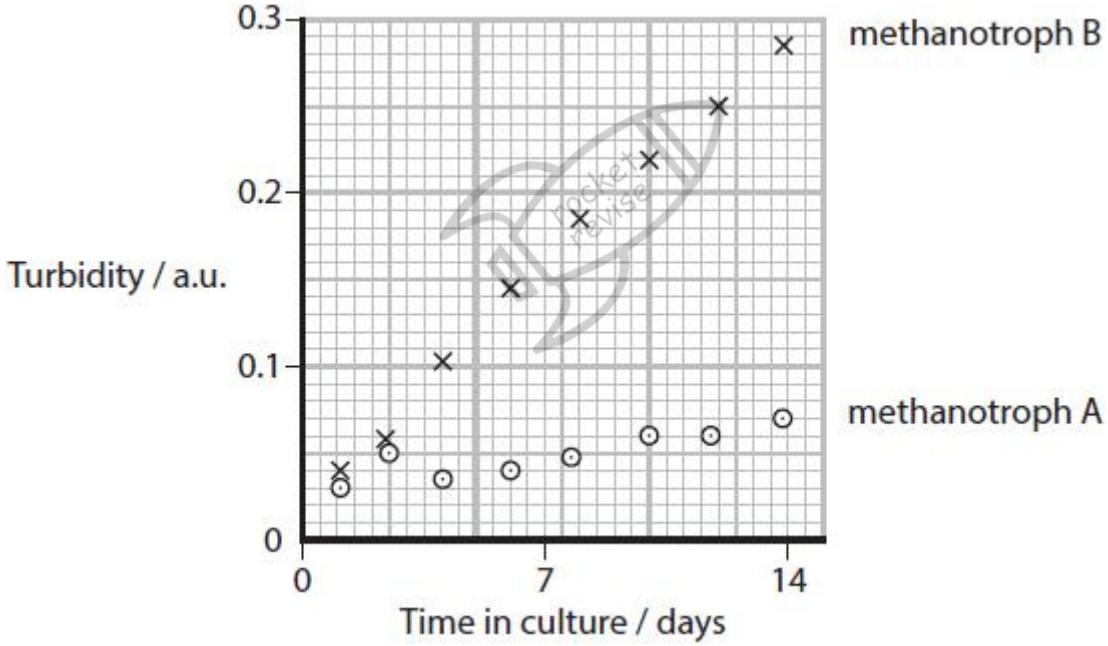
(e) Methanotrophs are a group of bacteria that can absorb methane and use it as a source of energy and as a material to convert into biomass.

Most methanotrophs grow best in methane concentrations of 5 000 a.u. to 10 000 a.u. This is greater than the methane concentration in the atmosphere.
Scientists wanted to identify the methanotrophs that could grow at lower methane concentrations, as these could help reduce the levels of methane in the atmosphere.

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(i) The graph shows the growth curve for two types of methanotrophs grown in a methane concentration of 500 a.u.

The increase in turbidity of the culture was used to assess the growth of the methanotrophs.



Explain how this graph can be used to compare the growth rate of these two types of methanotrophs.

(2)

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(ii) Suggest why low methane concentrations are not able to support the growth of some methanotrophs, such as methanotroph A.

(1)

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(iii) Suggest why the scientists analysed the cultures of methanotroph B for the production of nitrous oxide and carbon dioxide.

(2)

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(Total for question = 14 marks)

Q52.

The human genome contains sequences of DNA called endogenous retroviruses (ERVs).

These ERVs result from infection with retroviruses.

Retroviruses are a group of RNA viruses that have a replication cycle similar to that of human immunodeficiency virus (HIV).

This replication cycle results in latency.

Some of the ERVs in the human genome have been present for millions of years, because the ERVs are passed down through many generations.

*(a) Explain how retroviruses introduce ERVs into human cells and how the ancient ERVs can be transmitted to offspring in families.

(6)

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(b) One group of ERVs code for viral envelope proteins.

In normal cells, these proteins are not synthesised due to epigenetic regulation.
Explain how epigenetic regulation prevents the synthesis of these viral envelope proteins.

(3)

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(c) In cancer cells, the viral envelope proteins are synthesised and incorporated into the cancer cell membrane.

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Explain why these viral envelope proteins could be used in a vaccine against cancer cells.
Use the information in the question to support your answer.

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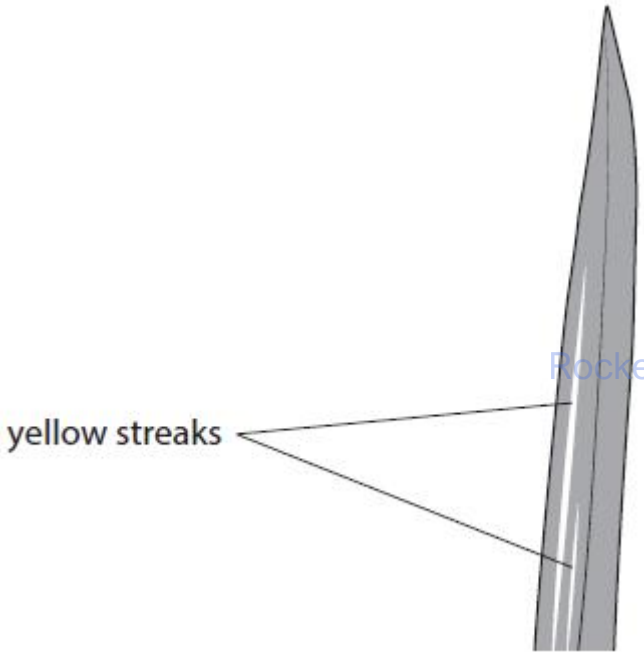
(Total for question = 14 marks)

Q53.

The rice yellow mottle virus (RYMV) infects species of rice plants grown in Africa.

The leaves of infected rice plants become discoloured with yellow or orange streaks.

The diagram shows a leaf of a rice plant infected with RYMV.



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(a) One country in Africa can produce 1.7 million tonnes of rice per year.

Between 20% and 100% of the harvest may be lost due to infection with RYMV.

(i) Calculate how many tonnes of rice are lost if 20% of the harvest is lost.
Give your answer in standard form.

(1)

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Answer ..... tonnes

(ii) Explain why it is important to identify rice plants infected with RYMV as early as possible.

(2)

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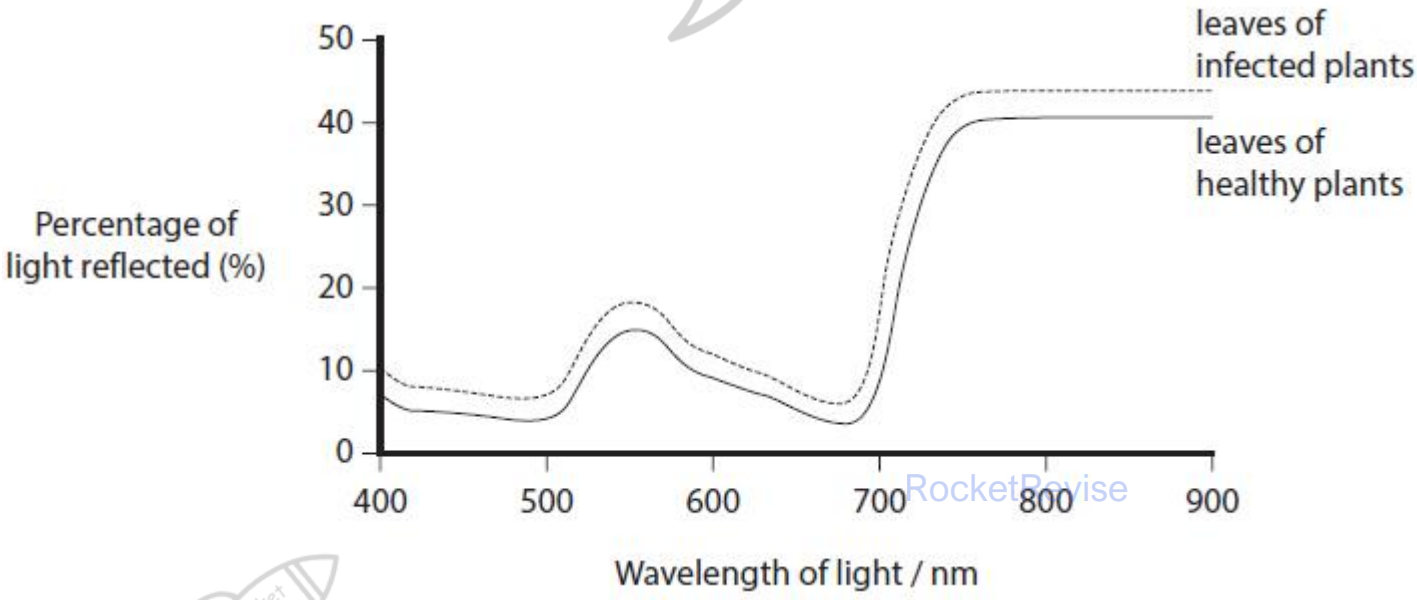
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(b) Scientists are trying to develop ways to identify rice plants recently infected with RYMV.

One investigation measured the reflection spectrum of the leaves of infected rice plants and compared this with healthy plants.
Different wavelengths of light were shone onto the leaves of rice plants and the percentage of light reflected at each wavelength was measured.
The graph shows the results of this investigation for healthy plants and plants that had been infected for two days with RYMV.



(i) Suggest why wavelengths of light between 400 nm and 900 nm were used in this investigation.

(1)

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(ii) Explain why the scientists thought that there would be a difference in the reflection spectra for healthy leaves and leaves infected with RYMV.

(2)

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(iii) Discuss the extent to which the results shown in the graph could be useful to farmers in identifying infected plants.

(3)

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(c) The types of rice plants most susceptible to infection with RYMV are those imported from Asia into Africa.

Explain why rice plants from Africa are less susceptible to the virus than those imported from Asia.

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(Total for question = 13 marks)

Q54.

Treatment of patients with antibiotics can affect the composition of their gut flora.

(a) Gut flora are important as they help protect the body from pathogens.

State the meaning of the term **pathogen.**

(1)

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(b) State the meaning of the term **antibiotic.**

(1)

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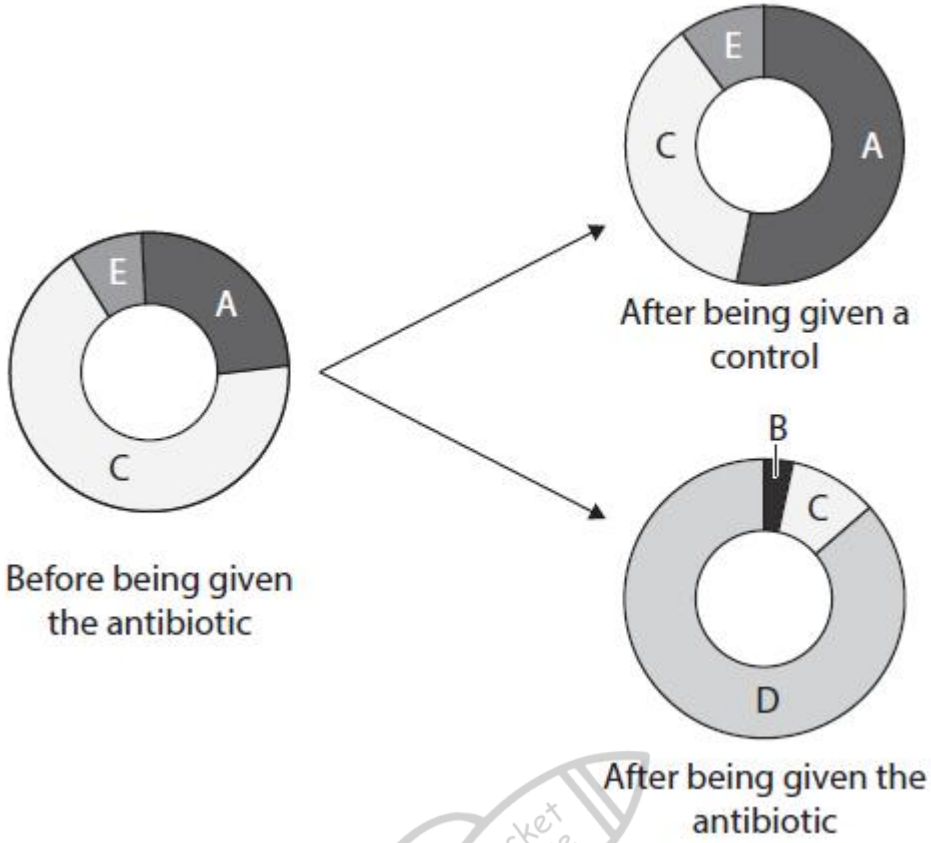
(c) The effect of an antibiotic on the proportion of different types of bacteria, A, B, C, D and E, in gut flora was investigated:

Step 1 – The types of bacteria in the gut flora of a group of people were determined.
Step 2 – Half the group was then given a dose of the antibiotic and the other half was given a control.
Step 3 – The types of bacteria in the gut flora were again determined.
The diagram shows the proportions of the types of bacteria in the gut flora.

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(i) State **one** conclusion about the effect of the control on the **variety** of bacteria in the gut flora.

(1)

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(ii) Estimate the decrease in the proportion of bacteria type C in the group given the antibiotic.

(1)

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Answer .....

(iii) Explain how treatment with antibiotics can affect the gut flora.

(2)

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(Total for question = 6 marks)

Q55.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Antibodies are produced by the immune system and play an important role in the destruction of pathogens by macrophages.

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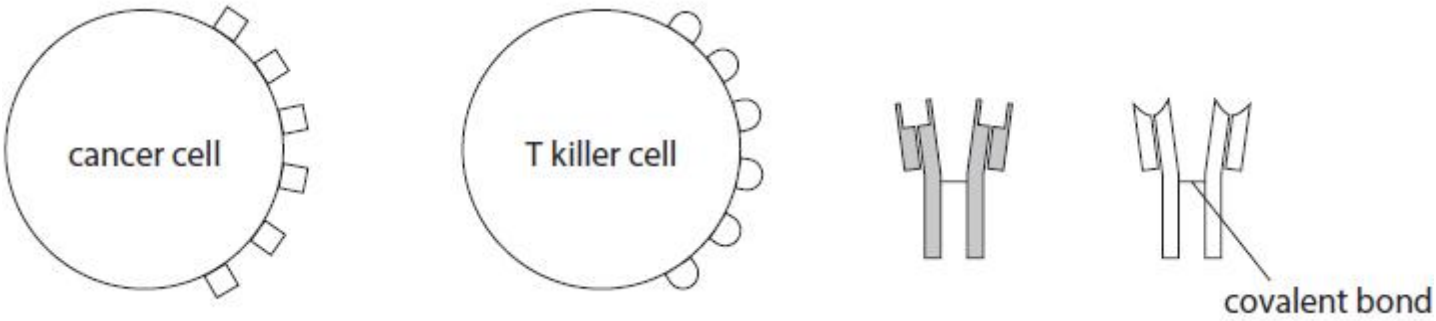
(a) Which types of immunity result in the production of antibodies?

(1)

- ☐ **A** artificial active and natural active
- ☐ **B** artificial active and natural passive
- ☐ **C** artificial passive and natural active
- ☐ **D** artificial passive and natural passive

(b) Cells called T killer cells are involved in destroying cells infected with viruses.

These cells can also destroy cancer cells in a similar way.
Scientists have made a new single antibody that can bind to both cancer cells and T killer cells.
The diagrams show the antigens on a cancer cell, the antigens on a T killer cell and two different antibodies.



(i) Explain why a single antibody had to be made by the scientists.
Use the information in the diagrams to support your answer

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(2)

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(ii) Suggest how the scientists could make this single antibody.
Use the information in the diagram to support your answer.

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(3)

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(ii) Explain how this single antibody helps the body destroy cancer cells.

(2)

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(Total for question = 8 marks)

Q56.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Beta-carotene is a pigment produced by plants.

It can be harvested from plants and used as an antibacterial agent, as an antioxidant and as an ingredient in foods.

Bacteria called *C. parietis* produce beta-carotene.

Scientists experimented to find the culture conditions required for *C. parietis* to produce the maximum yield of beta-carotene.

(a) *C. parietis* growing on nutrient agar were used to inoculate a liquid broth.

The liquid cultures were then shaken in an incubator.
The diagram shows this procedure.

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Colonies of *C. parietis* growing on nutrient agar

C. parietis in a liquid culture

Liquid culture of *C. parietis* shaken in an incubator

(i) Describe how to inoculate the liquid broth using the colonies growing on nutrient agar.

(3)

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(ii) Suggest why these liquid cultures were shaken in an incubator.

(2)

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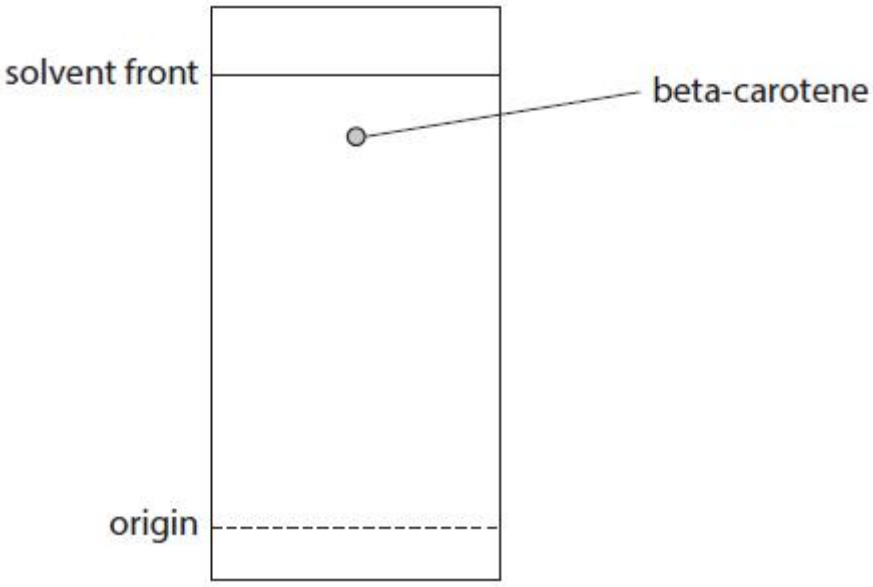
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(b) The beta-carotene could be identified using chromatography.

The diagram shows a chromatogram of beta-carotene.

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(i) Which is the Rf value for beta-carotene in this chromatogram?

(1)

- ☒ **A** 0.75
- ☒ **B** 0.87
- ☒ **C** 1.16
- ☒ **D** 1.33

(ii) Explain why chromatography can be used to identify beta-carotene.

(2)

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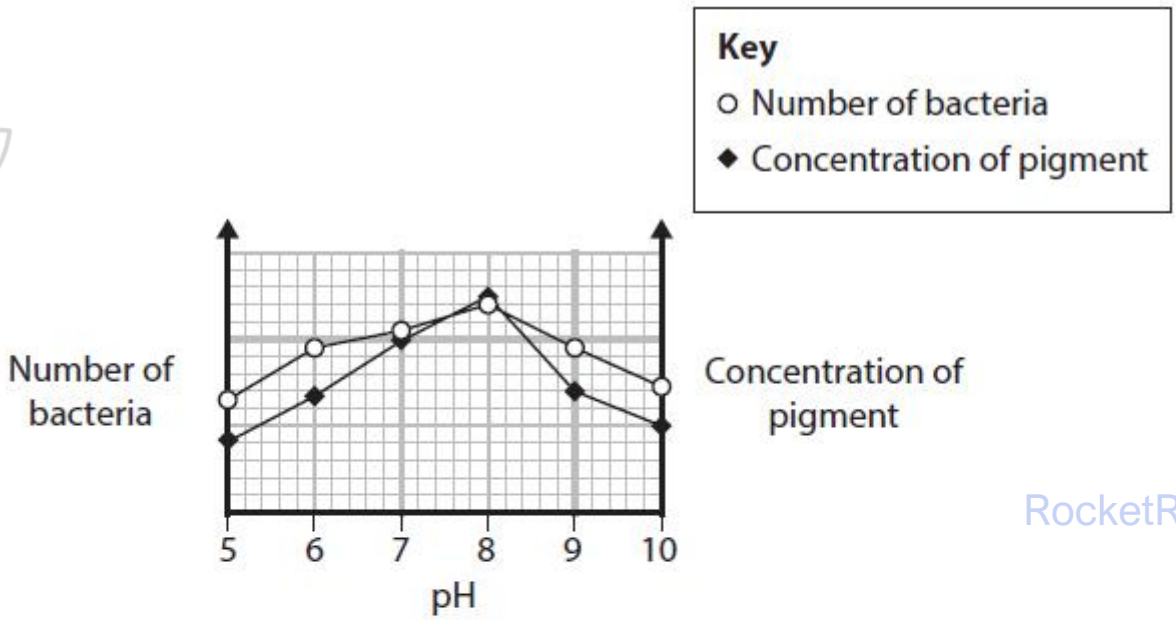
(c) The scientists cultured the bacteria at different pHs and different temperatures.

The graphs show the results.

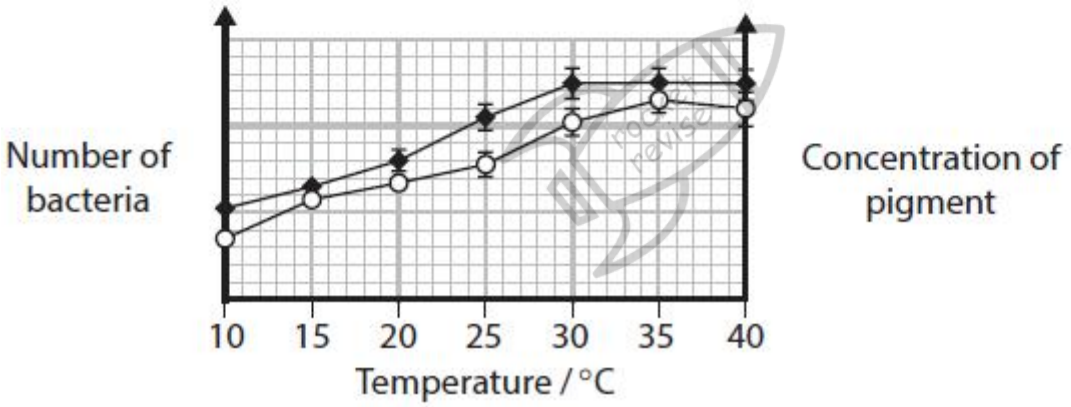
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Describe **three** conclusions that can be made from the data shown in these graphs.

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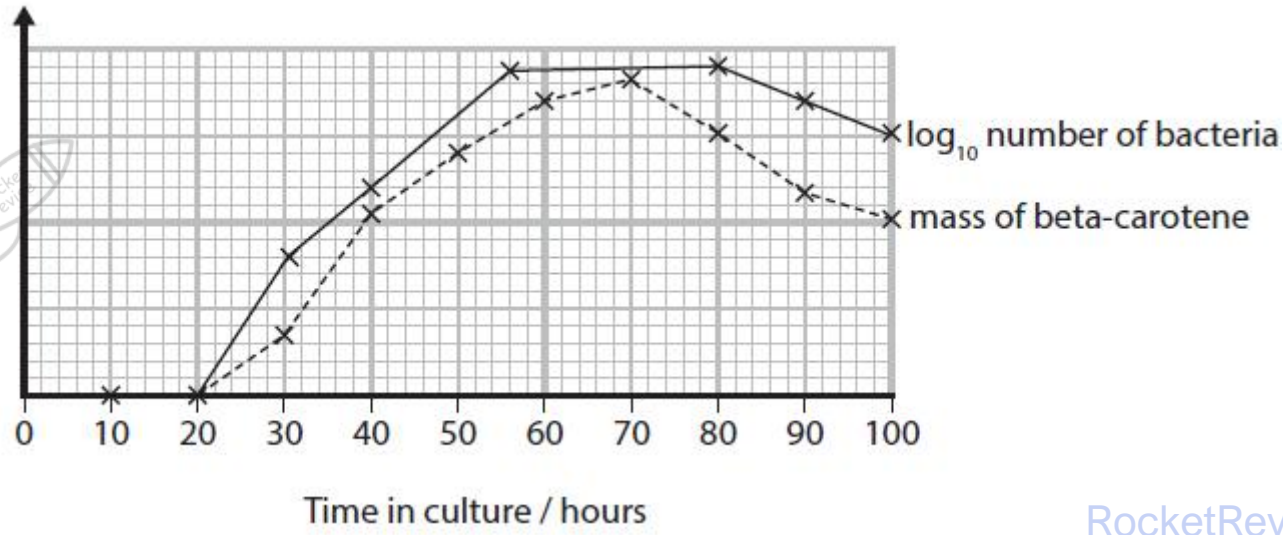
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(d) The graph shows the mass of beta-carotene produced and the number of bacteria in the culture as determined by dilution plating.

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Explain the changes in the mass of beta-carotene produced by this culture of bacteria.
Use the information in the graph to support your answer.

(4)

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(Total for question = 15 marks)

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